

Introduction to Digital Media Engineering Website DigiMedia Guide

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Khon Kaen University

Faculty of Engineering, Digital Media Engineering

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Chapter 1 - Introduction

1.1 Product Information

Product Name	MON And The Kids ; Attendance and Leave Management System
Date	March 17, 2026
Functional & Non-Functional requirement	functional & non-functional
Lark links	Backlog - Lark
Docs links	Member & Contribution %
Figma (Sprint 1)	Sprint 1 - Figma
Figma (Sprint 2,3)	Sprint 2,3 - Figma

1.2 Background of the Project

Digital Media Engineering (DME) is a relatively new and rapidly growing field of study at Khon Kaen University's Faculty of Engineering. Despite its growing relevance in today's technology-driven world, many prospective students and their parents remain unfamiliar with what the program entails, what career paths it leads to, and how to apply. The program's official information is scattered across multiple platforms, written in technical academic language that is difficult for high school students and parents to interpret, and often outdated by the time applicants need it most.

This project was initiated to address these challenges by developing a centralized, user-friendly website — DigiMedia Guide, designed to guide high school students, parents, and teachers through every stage of the university exploration and admission process. The website consolidates program information, admission score calculators, curriculum details, alumni outcomes, and career path guidance into a single accessible platform, reducing confusion and supporting more confident decision-making for all users.

1.3 Objectives of the Project

1. To design and develop a web-based platform that presents Digital Media Engineering program information in a clear, accessible, and visually engaging format for high school students, parents, and academic advisors.
2. To provide an admission score calculator that allows prospective students to input their academic scores and check their eligibility across Portfolio, Quota, and Admission rounds.
3. To deliver curriculum details, instructor profiles, alumni outcomes, and career path information that help users make informed decisions about pursuing the program.
4. To support non-Thai curriculum applicants (GED, GCE/A-Level) by accommodating multiple score input formats within the eligibility checker.
5. To build the system iteratively using Agile Scrum methodology across three sprints, ensuring continuous improvement based on review and feedback.

1.4 Scope of the Study

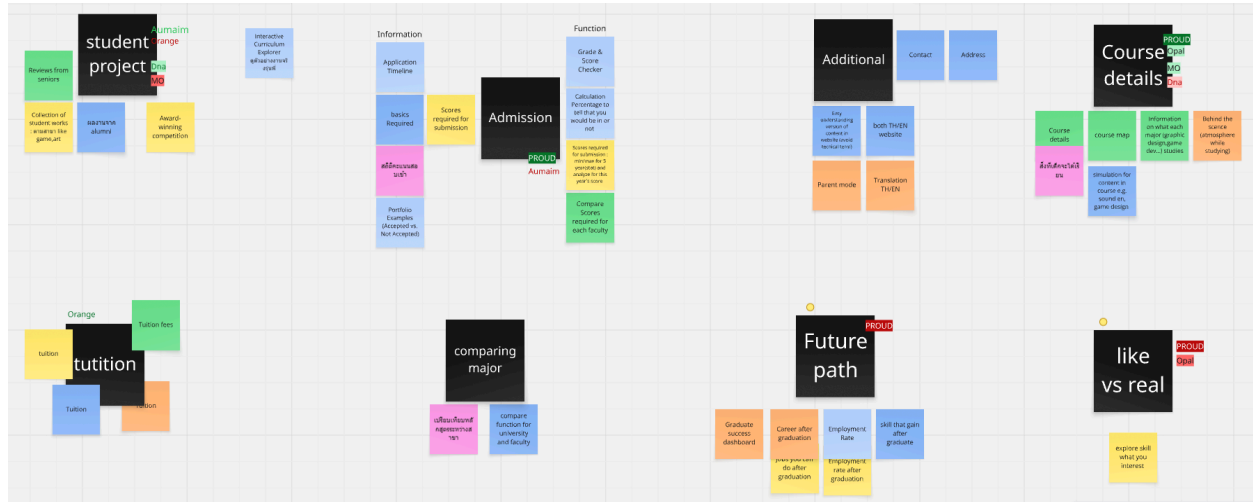
The scope includes the following:

- User Analysis: Development of three user personas (high school student, parent, and teacher/advisor) and user journey maps to guide design decisions.
- Requirements Analysis: Definition of functional and non-functional requirements covering nine system areas including authentication, self-exploration, course information, admission score calculation, document submission, portfolio, career information, and contact/support.
- Sprint 1: UI design and system modeling (Use Case Diagram, DFD, ER Diagram) for the admission score checker feature, supporting Thai Standard Curriculum applicants across Portfolio, Quota, and Admission rounds.
- Sprint 2: Iterative refinement of the score checker UI based on Sprint 1 feedback, including improved input validation, error messaging, and result display. Initial design of the course information feature.
- Sprint 3: Expansion of the score checker to support additional curriculum types (GED, GCE/A-Level, Netsat, A-Level), design of multi-curriculum result pages, and full test case coverage.

Chapter 2 Requirements and User analysis

2.1 Communication sheets

The communication sheets used for brainstorming website features. Team members generated ideas on individual cards, grouped them into thematic categories, and voted to determine which topics should be prioritized for the development of the DME recommendation website.



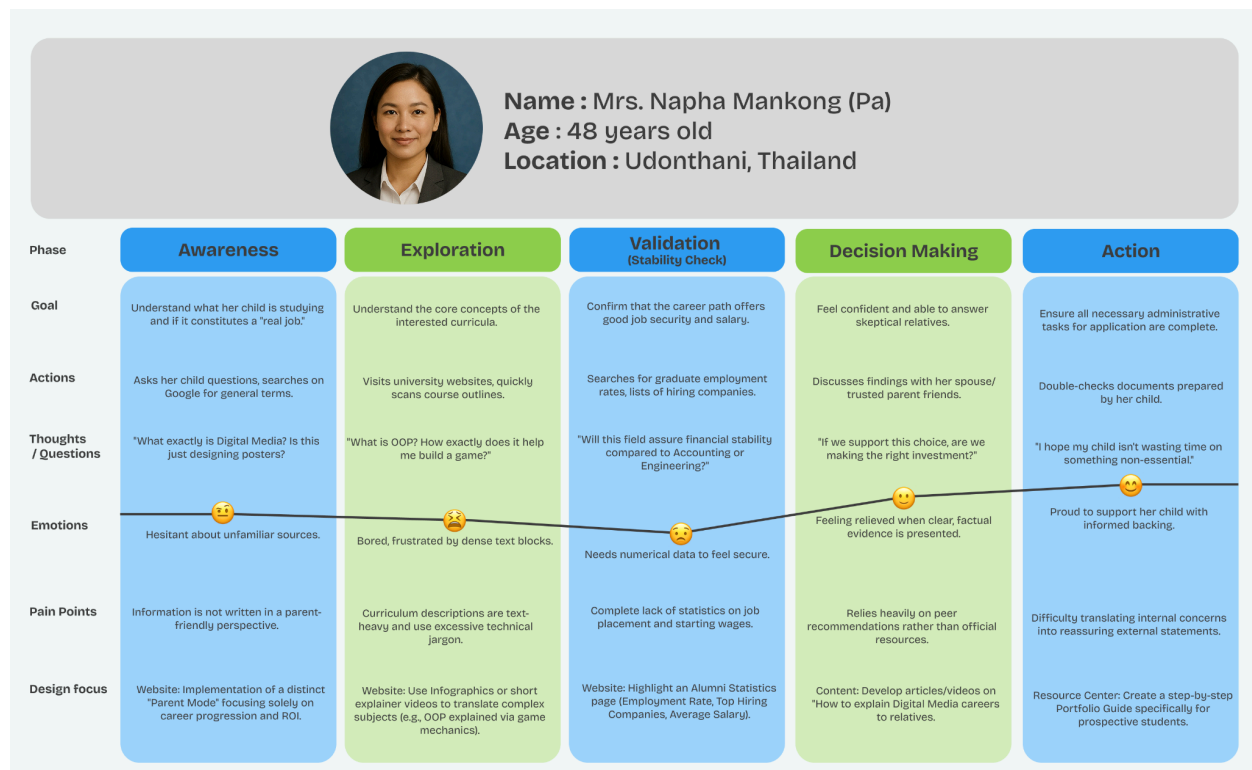
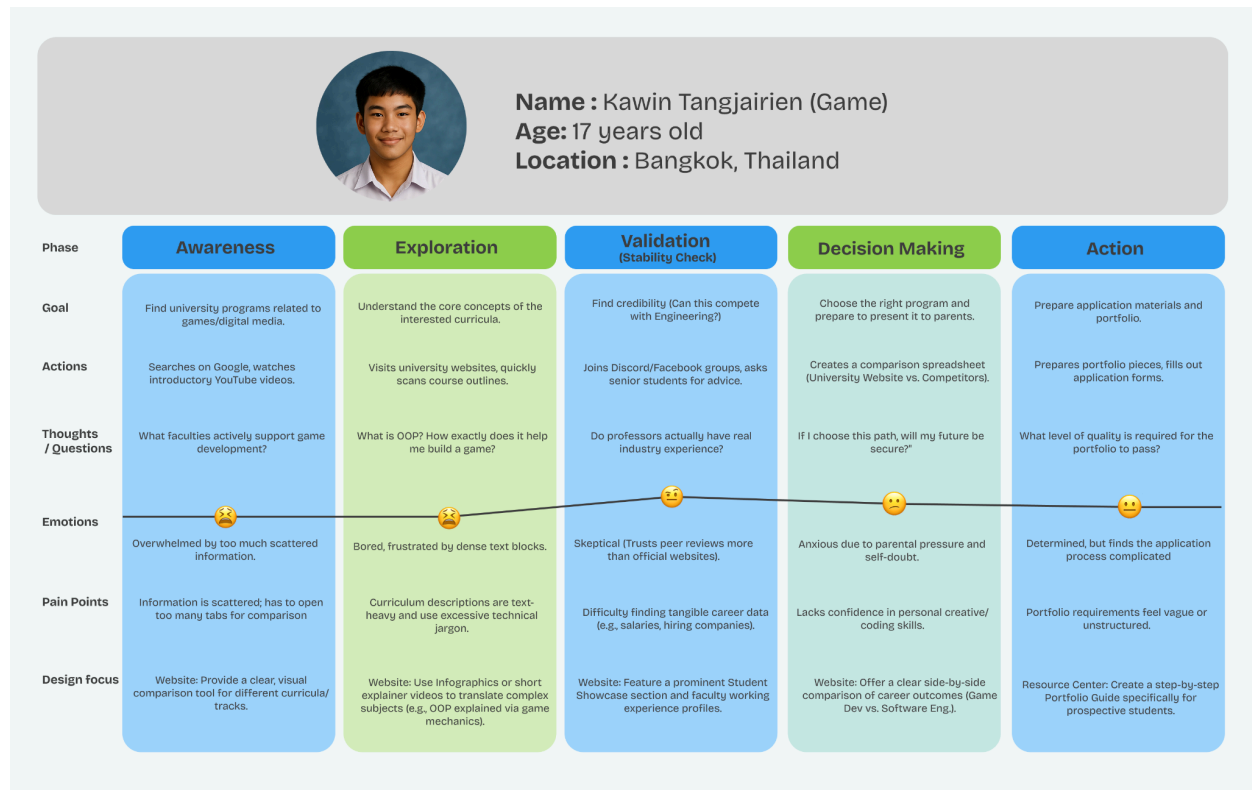
2.2 Persona

Photo			
Name	Kawin Tangjairien (Game)	Mrs. Napha Mankong (Pa)	Arisa Chantararat (Aom)
Age	17	48	27
Gender	Male	Female	Female
Location	Bangkok, Thailand	Udonthani, Thailand	Bangkok, Thailand

Job / Role	Grade 12 (Mathayom 6) Science-Math Program Debsirin School	Private company employee (Accounting) Small business owner	University application advisor for Grade 12 students High School Teacher (ICT / Digital Skills)
Tech Skills	High Gaming Literacy (RPGs/Strategy) Research via YouTube, Reddit, Discord Creative Software: ArtStation/Pinterest Active on Discord & gaming communities	Very active on LINE and Facebook Mobile Banking, Google Search General computer use for office work Finds UX/UI & Coding jargon intimidating	Google Classroom, Kahoot Notion (info organization) Canva (visual content) Blender, Unity (basic) YouTube & online reviews
Device	Desktop PC (high-spec) Smartphone	Smartphone (primary) Laptop (for accounting)	Laptop (primary) Tablet
Goals	Find program combining gaming/art with logic skills Discover aptitude: Programming / Art / Game Design Convince parents that digital media is a stable career	Understand practical curriculum (not marketing phrases) Learn job placement rates & starting salaries Get credible info to support her child confidently	Help students discover their best-fit track Provide clear, trustworthy program guidance Explain that digital media is a growing career Reduce confusion from academic jargon

Pain Points	Info scattered across 10+ outdated websites Jargon-heavy curriculum hard to visualize Family pressure vs. passion for digital media	Overwhelmed by scattered, outdated info Jargon disconnected from real career outcomes Anxious about digital media's job stability	University websites full of technical jargon Students lack self-awareness of their strengths Parents still view digital careers as unstable
Behavior	Trusts YouTubers & senior reviews over official sites Needs to see real portfolios before deciding Compares details meticulously (like PC specs)	Trusts official university websites Shares info in parent LINE groups Browses Facebook parent groups at night	Collects curricula & portfolios in Notion Shares concise summaries with students & parents Runs classroom workshops with Kahoot & Canva
Needs from Website	Visual Roadmap (subjects → real outputs) Alumni showcase & salary/job placement stats Self-assessment quiz (Art vs. Code)	Plain-language career & salary info Career Roadmap infographics Alumni success stories	Comparison Tool: majors & universities Student-fit quiz for classroom use Exportable PDFs / shareable via LINE

2.3 User Journey Map





Name : Miss Arisa Chantararat (Aom)

Age : 27 years old

Location : Bangkok, Thailand

Phase	Awareness	Exploration	Validation (Stability Check)	Decision Making	Action
Goal	To get an overview of updated digital media programs to prepare for advising Grade 12 students.	To gather in-depth program curriculum details (subjects/tuition) from multiple universities for comparison.	To verify "which student fits which major" and find data to address parents' concerns about job security.	To help students choose the right major and communicate with parents using credible information.	To ensure student portfolios meet specific university criteria.
Actions	Asking students in class about their current interests	Visiting individual university websites one by one.	Administering career/self-discovery quizzes with students.	Holding 1-on-1 or small group advising sessions.	Helping students proofread documents.
Thoughts / Questions	Game Design is trending this year, but students still can't distinguish whether they should study Programming or Art.	This website is so hard to navigate. What is 'OOP' subject? How do I explain this to kids? Why does every uni name this major differently?	Do graduates actually get jobs? Will the parents accept it if their child studies Gaming?	I need nice visual graphics to show parents; otherwise, they won't believe this career is stable.	Did the Portfolio criteria change again this year? I hope the kids don't miss anything.
Emotions					
Pain Points	Students have misconceptions about majors	University websites use complex jargon, have outdated info, or the curriculum download button is hidden.	Lack of clear statistics or easy-to-understand infographics to prove career stability to parents.	No "Comparison Tool" for curriculums; has to open multiple tabs to compare manually.	TCAS admission criteria are complex, and good Portfolio examples are hard to find.
Design focus	Student Fit Quiz A simple quiz for teachers to run with the whole class to categorize interests (Dev vs. Art vs. Design).	Teacher Mode / Plain Language A summary mode for teachers that removes jargon or has pop-ups explaining difficult	Parent-Friendly Infographics Shareable/printable graphics showing career paths and salary ranges for parents.	Comparison Tool A feature to compare highlights, tuition, and subjects of 2-3 universities on a single screen.	Portfolio Guideline Hub A collection of successful portfolio examples with a checklist of requirements.

Chapter 3 Product Planning

3.1 User Stories

No.	Persona	User Requirement	User Story	Acceptance Criteria
1	High School Student	US-001 : Score Check	As a high school student, I want to check their required scores so that I can know whether my scores pass the admission criteria.	- Display required scores per program clearly - Allow input of student scores for comparison - Show pass/fail result against criteria
2	High School Student	US-002 : Course Content Review	As a high school student, I would like to review the content and teaching approach of each course, so that I can determine whether the program aligns with my interests.	- Show course content and teaching method per program - Content is accessible without login
3	High School Student	US-003 : Instructor Profiles	As a high school student, I want to see instructors' profiles, so that I can gain confidence in the teaching quality of the program.	- Display instructor name, photo, and credentials - Show expertise and teaching subjects

4	High School Student	US-004 : Admission Info & Timeline	As a high school student, I want to see clear admission information and timelines, so that I do not miss important dates or make mistakes during the application process.	- Display admission steps and deadlines clearly - Highlight upcoming important dates - Information is up-to-date each academic year
5	High School Student	US-005 : Curriculum Structure	As a high school student, I want to see the curriculum structure, so that I know what subjects I will study and whether the program fits me.	- Show full list of subjects by year/semester - Indicate core vs. elective courses
6	High School Student	US-006 : Senior Reviews	As a high school student, I want to read senior reviews and real experiences, so that I can understand the learning environment and social life at the university.	- Display verified reviews from current students or alumni - Include rating or feedback category (e.g., academics, campus life)

7	High School Student	US-007 : Alumni Outcomes	As a high school student, I want to see alumni outcomes, so that I know what kind of careers I can pursue after graduation.	- Show example job titles and companies of alumni - Include employment rate or graduate statistics
8	High School Student	US-008 : Language Switch	As a high school student, I want to switch the website language between Thai and English, so that I can understand the information more easily.	- Language toggle (TH/EN) available on all pages - All content switches language without page reload
9	High School Student	US-009 : Search Function	As a high school student, I want to search for courses, news, or projects, so that I can quickly find the information I need.	- Search bar accessible from any page - Returns relevant results with keyword matching - Supports Thai and English keywords
10	High School Student	US-010 : Career Paths	As a high school student, I want to see possible career paths, so that I can understand job opportunities and future prospects.	- Display career path diagram per program

				- Include job titles, industries, and growth potential
11	High School Student	US-011 : Career Matching Quiz	As a high school student, I want to take a career matching quiz, so that I can get guidance on which program may suit me best.	- Quiz presents relevant questions about interests/skills - Result recommends one or more matching programs - User can retake the quiz
12	High School Student	US-012 : Score Calculator	As a high school student, I want to calculate my admission score, so that I can know whether I meet the admission criteria.	- Input fields match actual admission score components - Calculates total score and compares with cutoff - Shows whether student is likely eligible

13	High School Student	US-013 : Admission Notifications	As a high school student, I want to receive admission notifications, so that I do not miss important announcements.	- User can subscribe to notifications (email or in-app) - Notifications sent for key admission events - User can manage notification preferences
14	High School Student	US-014 : Admission Status Check	As a high school student, I want to check my admission status, so that I know whether my application is complete and successful.	- Applicant can log in to view application status - Status clearly shows: Pending / Complete / Accepted / Rejected - Timestamp of last status update is shown

3.1.1 Sprint 1

US-001 Score Check : As a high school student, I want to check their required scores so that I can know whether my scores pass the admission criteria.

3.1.2 Sprint 2

US-001 Score Check : As a high school student, I want to check their required scores so that I can know whether my scores pass the admission criteria.

3.1.3 Sprint 3

US-001 Score Check : As a high school student, I want to check their required scores so that I can know whether my scores pass the admission criteria.

3.2 Product Backlog (Lark)

ID	Backlog Item	Type	Priority	Estimate	Planned Sprint
PB-01	Define System Requirements and User Needs	Analysis	High	3	Sprint 1
PB-02	Create Communication Sheets and Feature Categories	Analysis	High	2	Sprint 1
PB-03	Design ERD for System Data Structure	System Design	High	3	Sprint 1
PB-04	Context Diagram	System Modeling	High	2	Sprint 1
PB-05	Design DFD Level 1	System Modeling	High	2	Sprint 1
PB-06	Design DFD Level 2	System Modeling	High	2	Sprint 1
PB-07	UI Design: Homepage	UI Design	High	3	Sprint 1
PB-08	UI Design: Admission Information Page	UI Design	High	2	Sprint 1
PB-09	UI Design: Course Details Page	UI Design	Medium	3	Sprint 2

PB-10	UI Design: Tuition Information Page	UI Design	High	2	Sprint 2
PB-11	Develop Score Input Form	Development	High	3	Sprint 2
PB-12	Implement Eligibility Calculation Logic	Development	High	5	Sprint 2
PB-13	Implement Input Validation	Development	High	3	Sprint 2
PB-14	Develop Result Display Page	Development	High	3	Sprint 3
PB-15	Improve UI Consistency and Styling	UI Improvement	Medium	2	Sprint 3
PB-16	Revise DFD and System Diagrams	System Design	Medium	2	Sprint 3
PB-17	Refine Test Cases Based on User Story	Testing	Medium	2	Sprint 3
PB-18	System Testing and Bug Fixing	Testing	High	3	Sprint 3

Chapter 4 Sprint 1 Implementation

4.1 Sprint 1 Planning

4.1.1 Sprint Backlog (Lark)

☐ Task	Notes	Status	Task leader	Task start ti...	Task due d...	Expecte...	Task start a...	Task due da...	Time us...
▼ As a high school student, I...									
1 > Design UI: Admission criteria ...	UX/UI	Done	Opal	02/10/2026	02/10/2026	30.0	2026/02/10	2026/02/10	26.0
2 Design DFD : select round	Backend	Done	Opal	02/12/2026	02/12/2026	30.0	2026/02/12	2026/02/12	28.0
3 > Design UI : criteria portfolio r...	UX/UI	Done	Deena	02/10/2026	02/10/2026	30.0	2026/02/10	2026/02/10	29.0
4 > Design UI criteria quota round	UX/UI	Done	Orange	02/10/2026	02/10/2026	30.0	2026/02/10	2026/02/10	27.0
5 > Design UI : criteria admission ...	UX/UI	Done	Mo	02/10/2026	02/10/2026	30.0	2026/02/10	2026/02/10	28.0
6 > Design UI : calculator portfoli...	UX/UI	Done	Pround	02/10/2026	02/10/2026	30.0	2026/02/10	2026/02/10	32.0
7 > Design UI : calculator portfoli...	UX/UI	Done	Pround	02/12/2026	02/12/2026	30.0	2026/02/12	2026/02/12	62.0
8 > Design UI : calculator portfoli...	UX/UI	Done	Deena	02/13/2026	02/13/2026	30.0	2026/02/14	2026/02/14	35.0
9 > Design UI : calculator portfoli...	UX/UI	Done	Deena	02/15/2026	02/15/2026	30.0	2026/02/15	2026/02/15	37.0
10 > Design UI : calculator portfoli...	UX/UI	Done	Pround	02/14/2026	02/14/2026	30.0	2026/02/14	2026/02/14	27.0
11 > Design UI : calculator portfoli...	UX/UI	Done	Orange	02/13/2026	02/13/2026	30.0	2026/02/14	2026/02/14	35.0
12 > Design UI : calculator portfoli...	UX/UI	Done	Opal	02/14/2026	02/14/2026	30.0	2026/02/14	2026/02/14	28.0
13 > Design UI : calculator quota r...	UX/UI	Done	Pround	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	22.0
14 Design UI : calculator quota round ...	UX/UI	Done	Pround	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	25.0
15 Design UI : calculator portfolio rou...	UX/UI	Done	Pround	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	28.0
16 Design UI : calculator admission ro...	UX/UI	Done	Pround	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	22.0
17 Design UI : calculator admission ro...	UX/UI	Done	Pround	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	25.0
18 Design DFD : select calculator button	Backend	Done	Mo	02/12/2026	02/12/2026	30.0	2026/02/12	2026/02/12	28.0
19 Design DFD : fill in detail portfolio r...	Backend	Done	Aumaim	02/12/2026	02/12/2026	30.0	2026/02/12	2026/02/12	30.0
20 Design DFD : fill in detail quota rou...	Backend	Done	Aumaim	02/14/2026	02/14/2026	30.0	2026/02/14	2026/02/14	27.0
21 Design DFD : fill in detail admission...	Backend	Done	Aumaim	02/15/2026	02/15/2026	30.0	2026/02/15	2026/02/15	25.0
22 Design ER for admission criteria	Backend	Done	Orange	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	28.0
23 Design DFD : level 0	Backend	Done	Aumaim	02/10/2026	02/10/2026	30.0	2026/02/10	2026/02/10	25.0
24 Design DFD : fill in detail portfolio l...	Backend	Done	Aumaim	02/12/2026	02/10/2026	30.0	2026/02/10	2026/02/10	30.0
25 > Design UI : calculator portfoli...	UX/UI	Done	Deena	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	28.0
26 > Design UI : calculator portfoli...	UX/UI	Done	Opal	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	29.0
27 > Design UI : calculator portfoli...	UX/UI	Done	Orange	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	26.0
28 > Design UI : calculator portfoli...	UX/UI	Done	Orange	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	18.0
29 > Design UI : calculator portfoli...	UX/UI	Done	Orange	02/16/2026	02/16/2026	30.0	2026/02/16	2026/02/16	16.0

[Sprint Backlog 1](#)

4.1.2 Sprint Goal

The goal of Sprint 1 is to design and develop the core information-browsing experience for high school students visiting the website.

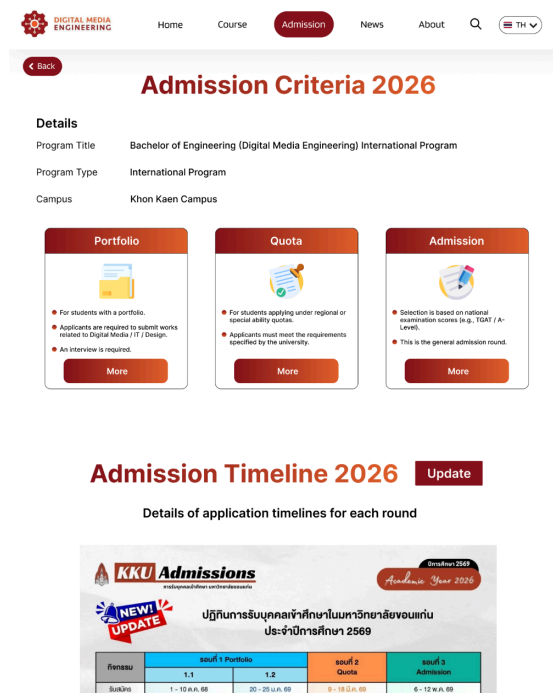
This sprint focuses on delivering a functional admission score checker that allows students to input their subject scores and instantly see whether they meet the program's admission criteria — addressing the most immediate need of prospective applicants.

By the end of this sprint, the system will support score input , calculate totals for Portfolio, Quota and Admission rounds, and display a clear pass/fail result against the program's historical cutoff data. The feature will be accessible without login, ensuring that all visitors can use it without barriers.

This sprint establishes the foundation for the self-service admission tools that will be expanded in future sprints.

4.2 Sprint 1 Development (Increment)

4.2.1 User Interface (UI) Design



Picture 4.1 Main Page

DIGITAL MEDIA
ENGINEERING

Home
Course
Admission
News
About

TH

Back
Portfolio
Calculate Score

High-Potential School Student Admission Program
receive 4 person

High-Potential Student Program
receive 20 person

DIGITAL MEDIA
ENGINEERING

Home
Course
Admission
News
About

TH

Back
Portfolio
Calculate Score

High-Potential School Student Admission Program
receive 4 person

High-Potential Student Program
receive 20 person

Qualifications

- Applicants from **Thai standard curriculum**
- Does not accept **international curriculum**
- Does not accept **vocational curriculum**
- Does not accept **Non-Formal Education (NFE) curriculum**
- Does not accept **GED applicants**
- Minimum GPAX : 3.25
- Must graduate in the current academic year only

Details

- Applicants who are currently studying in Grade 12 (M.6) from High Potential Schools that have an academic cooperation agreement with the Faculty of Engineering, Khon Kaen University (29 schools), including:
 - 1. Khon Kaen Wittayayon School
 - 2. Kaen Nakhon Wittayalai School
 - 3. Demonstration School of Khon Kaen University (Secondary) – Modindaeng Campus
 - 4. Demonstration School of Khon Kaen University (Education Faculty)
 - 5. Loei Pittayakom School
 - 6. Suranaree Wittaya School
 - 7. Buriram Pittayakom School
 - 8. Ubon Ratchathani School
 - 9. Benchama Maharat School
 - 10. Yasothorn Pittayakom School
 - 11. Srisaket Wittayalai School
 - 12. Udon Pittayanukul School
 - 13. Sakon Nakhon School
 - 14. Kalasin Pittayakom School
 - 15. Maha Sarakham Pittayakom School
 - 16. Roi Et Wittayalai School
 - 17. Buriram Rajabhat University Demonstration School
 - 18. Sisaket School
 - 19. Nong Bua Pittayakom School
 - 20. Anant Charoen School
 - 21. Sunn School
 - 22. Maha Sarakham Demonstration School (Secondary)
 - 23. Loei Wittayakom School
 - 24. Phra Pittayakom School
 - 25. Bueng Kan School
 - 26. Pathumrat Pittayakom School
 - 27. Loei Rajabhat University Demonstration School
 - 28. Nong Khai Wittayakom School
 - 29. Demonstration School of Khon Kaen University, Nong Khai Campus
- Must have a cumulative GPA (4 semesters of upper secondary level) of not less than 3.25
- Must have GPA in the subject groups of Mathematics, Chemistry, Physics, and English not less than 3.25 in each group.
- Must receive at least grade B in the following preparatory engineering courses :

EN 001 203	Computer Programming
EN 001 202	Engineering Drawing
- For applicants of the international program. Applicants must have English language test scores in order to be able to apply. However, if any of the English language test scores are lower than the following criteria, applicants must attend an English language preparatory course as announced by the Faculty of Engineering. The scores must be **no more than 2 years old** from the date of application.

TOEFL (IBT)	≥ 59 points
TOEFL (ITP)	≥ 470 points
IELTS (Academic Module)	≥ Band 5
TOEIC	≥ 500 points
KEPT	≥ 68 points
KKU-AELT (Overall score)	≥ Band 4 (7-band system)
- has academic achievements or has participated in academic projects in science, mathematics or computer science conducted at the high school level at the provincial, regional, national or international level, or other achievements, or has foreign language test results with scores as per the attached document (downloadable at the link <https://kku.world/porten69>)

Acceptance conditions

Portfolio	40%
Essay	10%
Interview	50%
Total	100%

considering answers to questions in English, abilities related to digital media engineering, quick wit, personality, attitude, and social etiquette.

High-Potential Student Program
receive 20 person

DIGITAL MEDIA
ENGINEERING

Home
Course
Admission
News
About

TH

Back
Portfolio
Calculate Score

High-Potential School Student Admission Program
receive 4 person

High-Potential Student Program
receive 20 person

Qualifications

- Accepts applicants from **Thai standard curriculum**
- Accepts applicants from **international curriculum**
- Does not accept **vocational curriculum**
- Does not accept **Non-Formal Education (NFE) curriculum**
- Accepts **GED applicants**
- Minimum GPAX : 2.75

Details

- Must be a student currently studying in the 6th year of secondary school with an average GPA of at least 2.75 over 4 semesters, and have an average GPA in the subjects of Mathematics, Chemistry, and Physics.
- Must be a person with GED scores in 4 subjects, with each subject having a score of at least 145, and the applicant is at least 16 years old or older; or
- Must be a person with General Certificate of Education (Advanced Level) GCE A-Level scores, with each subject having a score of at least C in 3 subjects, from which 2 subjects are chosen from the 1st-4th ranking and 1 subject from the 5th-11th ranking as follows: 1) Math 2) Physics 3) Chemistry 4) Computer Science 5) Digital Media and Design 6) Mechanics 7) Statistics 8) Geography 9) Design and Technology 10) Information Technology 11) Economics; or
- Must be a person who has completed secondary education (Grade 12) with an average GPA of at least 2.75 and an average GPA in the Mathematics, Chemistry, and Physics groups of at least 2.75; or
- Must be a person who has graduated from a foreign university.
- In the case of applicants who have completed their education abroad or according to the education system of a foreign country and have received equivalency from the Ministry of Education of Thailand, or
- applicants must have English language test scores in order to be able to apply. However, if any English language test score is lower than the following criteria, they must undergo English language preparatory training as announced by the Faculty of Engineering. The test score must be no more than 2 years old from the date of application

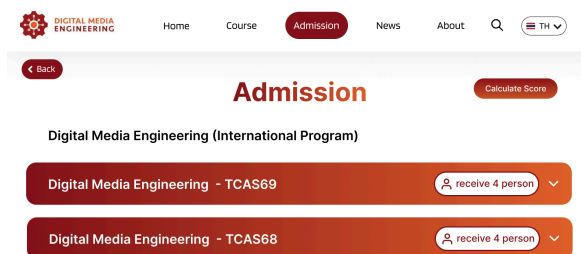
TOEFL (IBT)	≥ 59 points
TOEFL (ITP)	≥ 470 points
IELTS (Academic Module)	≥ Band 5
TOEIC	≥ 500 points
KEPT	≥ 68 points
KKU-AELT (Overall score)	≥ Band 4 (7-band system)
- has academic achievements or has participated in academic projects in science, mathematics or computer science conducted at the high school level at the provincial, regional, national or international level, or other achievements, or has foreign language test results with scores as per the attached document (downloadable at the link <https://kku.world/porten69>)

Acceptance conditions

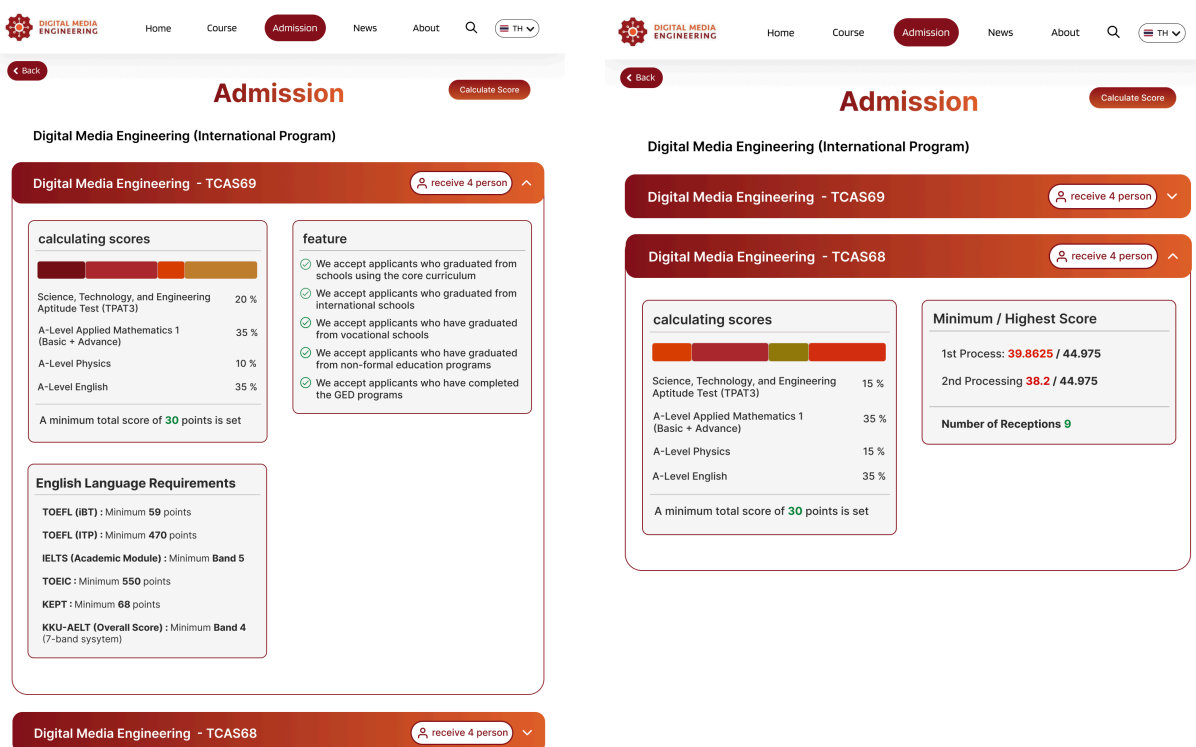
Portfolio	40%
Essay	10%
Interview	50%
Total	100%

considering answers to questions in English, abilities related to digital media engineering, quick wit, personality, attitude, and social etiquette.

Picture 4.3 Dropdown Portfolio criteria Page



Picture 4.4 Admission criteria Page



Picture 4.5 Dropdown Admission criteria Page


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ENGINEERING**

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Quota

[Calculate Score](#)

NETSAT is an assessment center that evaluates the intelligence and learning ability of upper secondary school students to determine their readiness for higher education.

Eligibility Criteria for NETSAT Applicants

- Be a student currently studying in:
 - Upper secondary level (Grade 11-12), or
 - Vocational education level (Vocational Certificate Year 2-3)
- Be studying at a school located in the Northeastern region of Thailand (Isan)
 - House registration in the Northeastern region is not required

Digital Media Engineering (International Program)
receive 12 person

Subjects Used for Selection

English Language	40%
Mathematics	30%
Physics	15%
Engineering Aptitude	15%
Total	100%

**Total score > 30 **

Score Statistics

Max - Min score quota 68

Maximum Score	67.12
Minimum Score	47.57
Total Score	100

Picture 4.6 Quota criteria Page


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Portfolio Round Eligibility Check

High-Potential School Student Admission Program receive 4 person

Educational Information

☐ Currently studying in Grade 12

Graduated Program

☐ Thai standard curriculum
School/graduated from:

☐ Others

Academic Performance

GPA (in semesters) [Enter your GPA](#)

Combined GPA (Mathematics, Chemistry, Physics, English) [Enter your GPA](#)

preparatory engineering courses

EN 001 203 Computer Programming [Enter your GPA](#)

EN 001 202 Engineering Drawing [Enter your GPA](#)

English Proficiency Score

TOEIC Score [Enter your Score](#)

☐ Academic Achievements

[Check Eligibility](#)

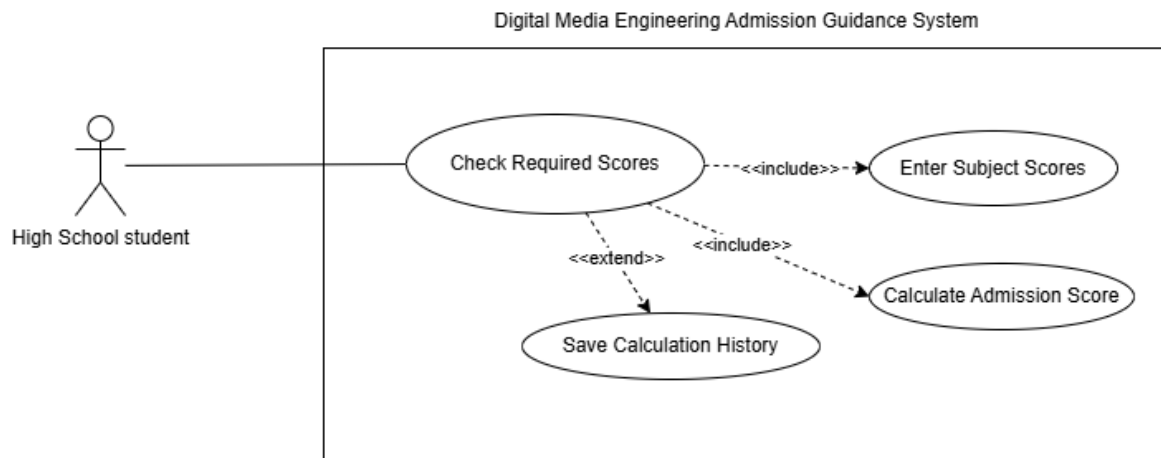
Result

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Educational Information
Graduated Program
Academic Performance
Result

Picture 4.7 Portfolio Eligibility Check Page

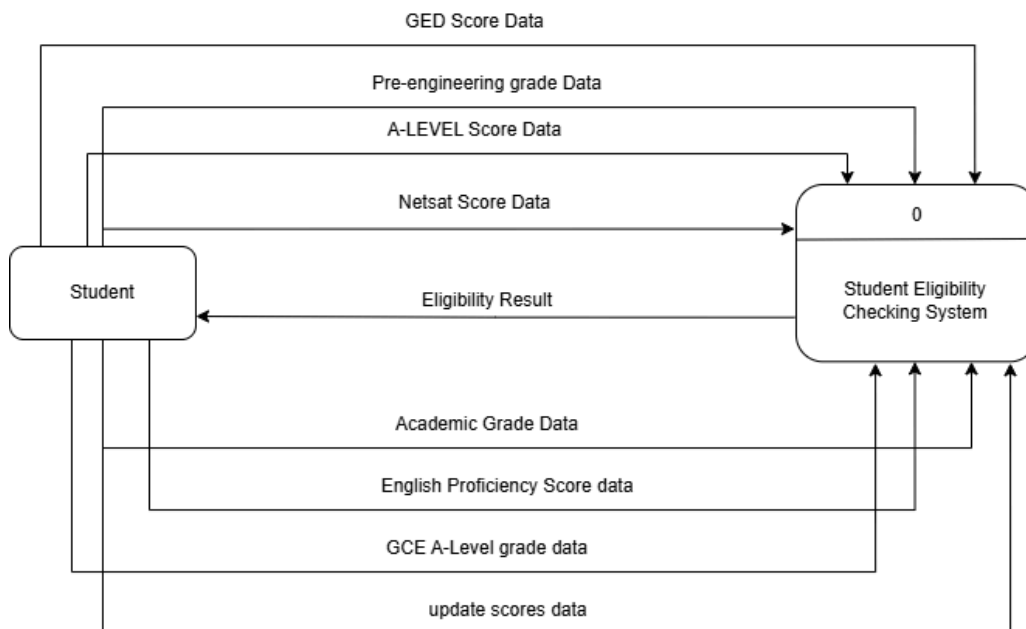
4.2.2 Use Case Diagram



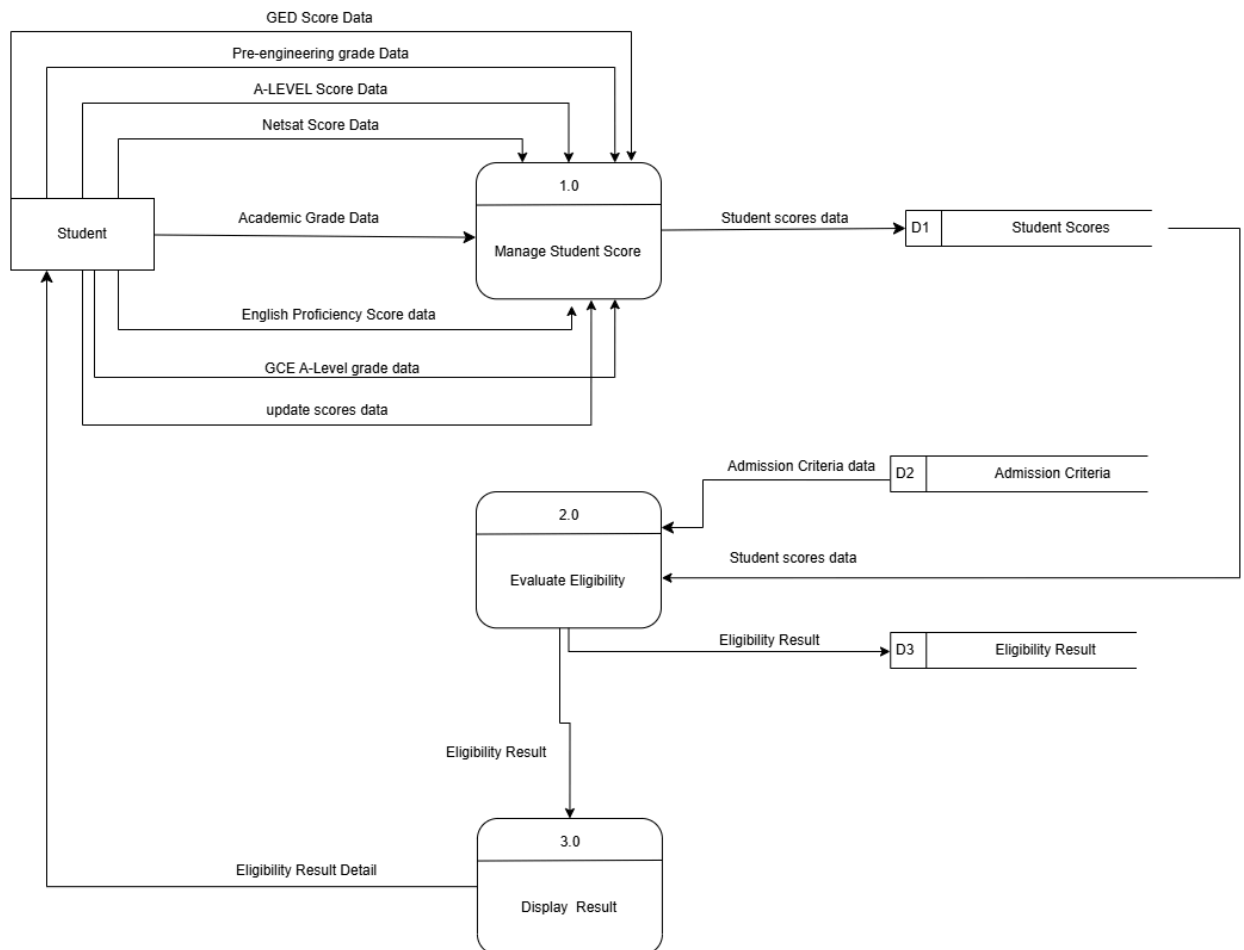
This use case diagram represents the process of checking admission score requirements in the Digital Media Engineering Admission Guidance System. A high school student can enter subject scores and calculate the admission score to determine whether their results meet the admission criteria. The system also provides an optional feature to save the calculation history.

4.2.3 Data Flow Diagram (DFD)

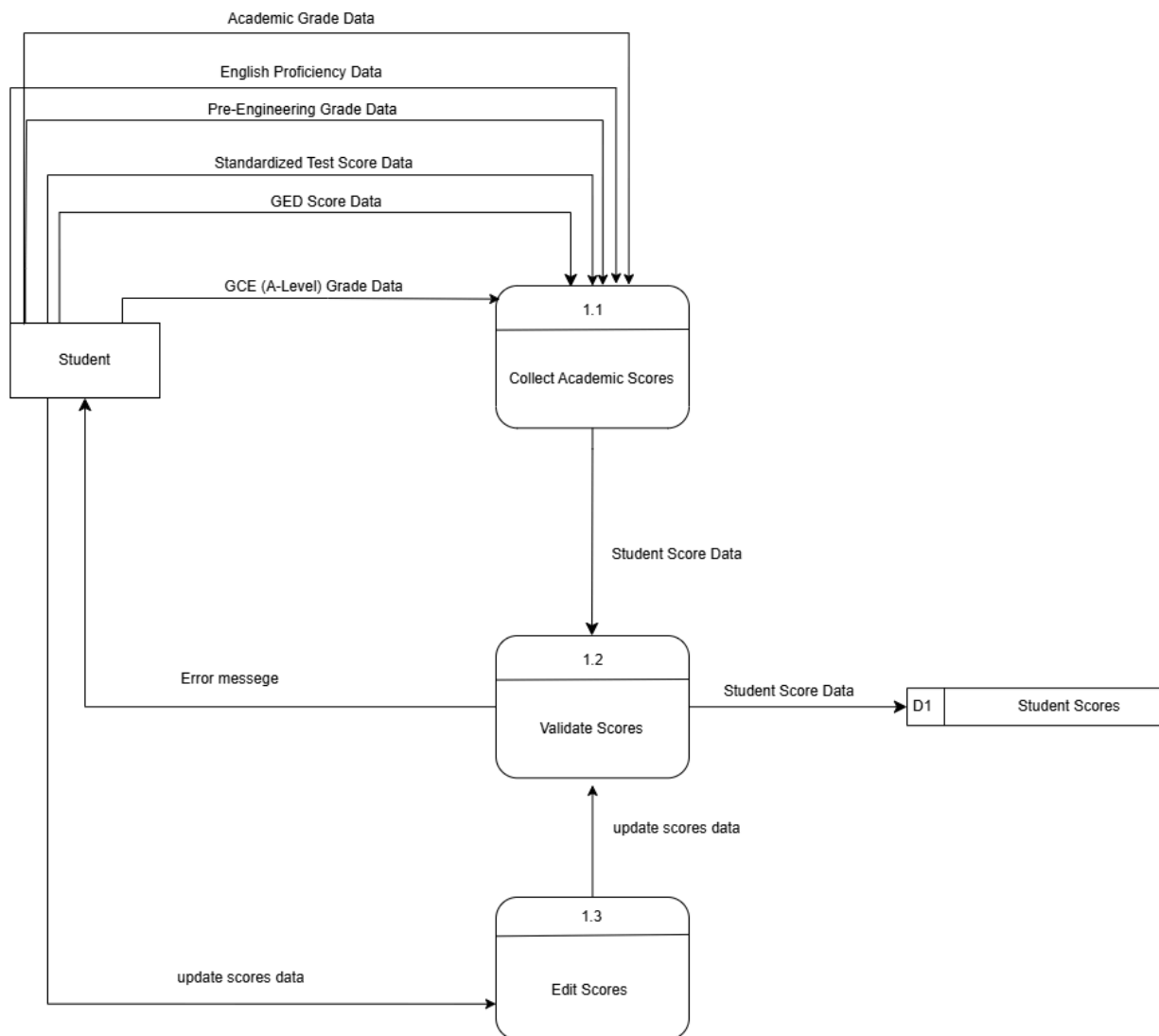
- Context DFD



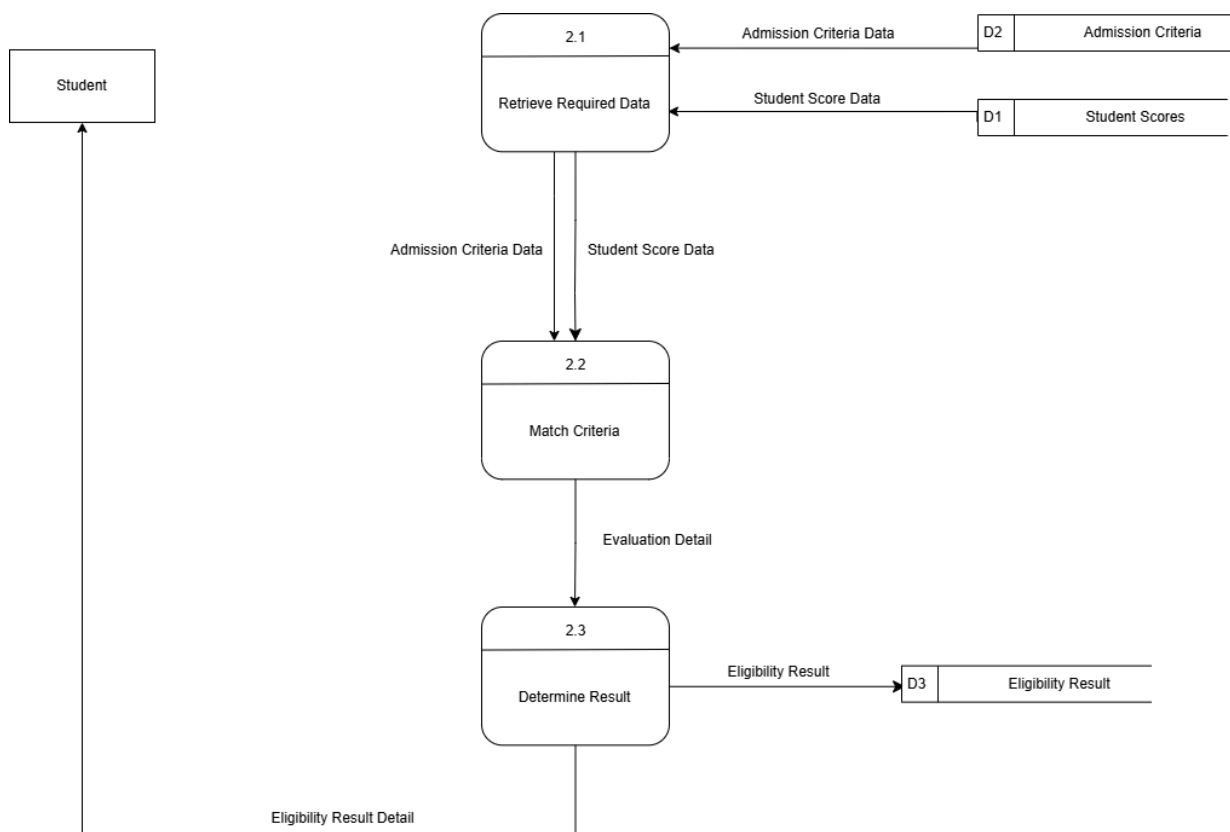
- DFD Level 1



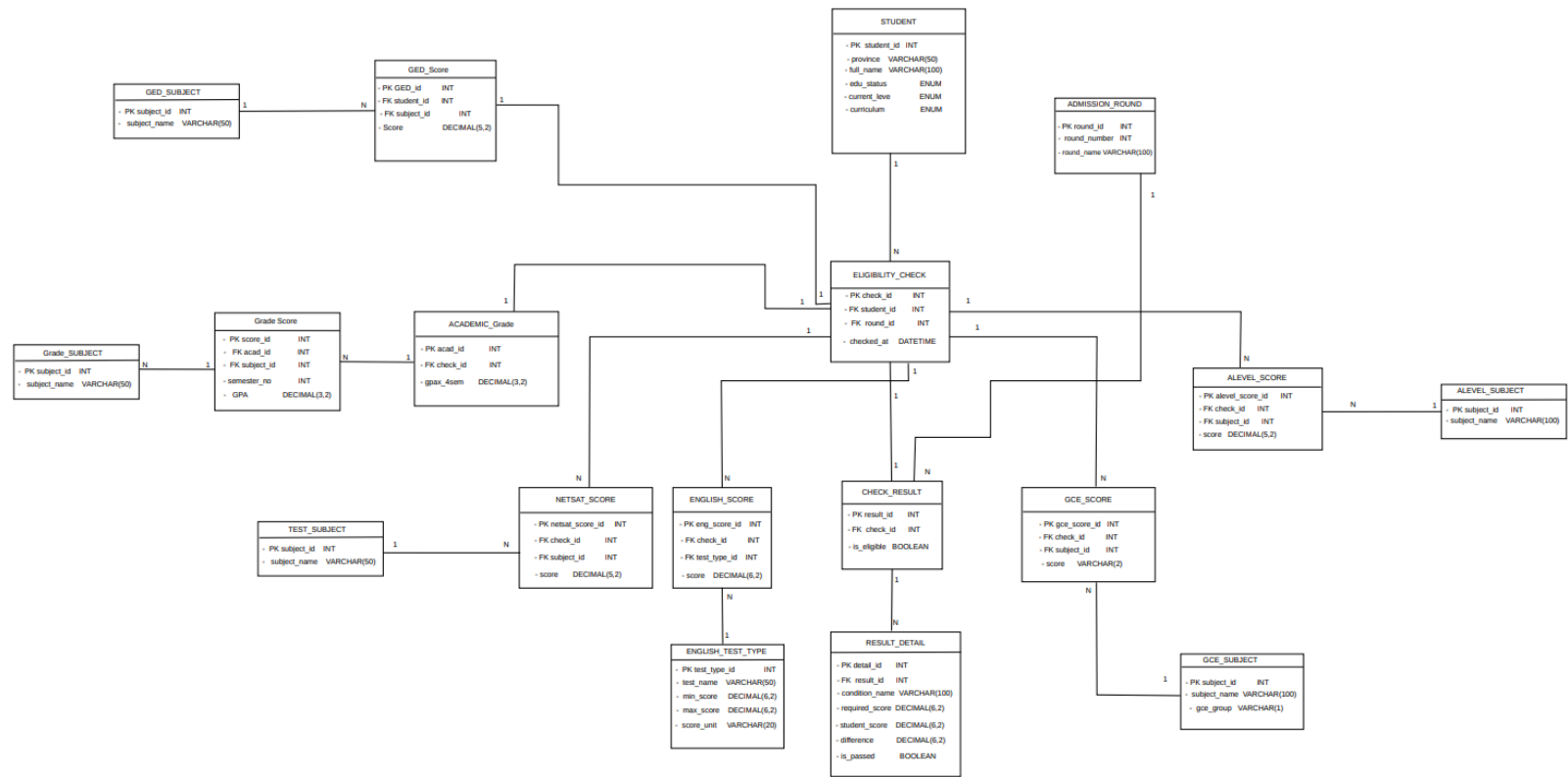
- DFD Level 2
- DFD Level 2 1.0



- DFD Level 2 2.0



4.2.4 Entity-Relationship Diagram (ER)



4.2.5 Test Cases

4.2.5.1 Test Scenarios

User story ID	Test scenario ID	Test scenario	Number of test cases	Priority
US0001	TS0001	Verify whether the system displays admission criteria information and navigates to the correct page when clicking More	4	High
US0001	TS0002	Verify user can access and use all functions in Portfolio Round Criteria page	7	High
US0001	TS0003	Verify user can check eligibility for admission rounds (Portfolio, Quota, Admission)	8	High

4.2.5.2 Test Cases

Test Scenario	Test Case ID	Test Case	Pre-Condition	Test Step	Test Data	Expect Result	Post Condition	Actual Result	Status (Pass/Fail)
TS0001 : Verify whether the system displays admission criteria information and navigates to the correct page when clicking More	TC01	Verify that admission criteria information is displayed correctly on the page	User opens the admission criteria page	1.Navigate to the Admission Criteria section on the page. 2.Observe the admission criteria information displayed on the page.	-	The system displays admission criteria information correctly such as round name, description, and related details.	User remains on the homepage and can continue interacting with the page.	The admission criteria information is displayed correctly on the page.	Pass
TS0001 : Verify whether the system displays	TC02	Verify that clicking More on Portfolio navigates	User opens the Portfolio page	1.Navigate to the Portfolio page from the Admission Criteria page. 2.Observe the	-	The system displays Portfolio program information	User remains on the Portfolio page and	The system successfully navigates to the Portfolio	Pass

admission criteria information and navigates to the correct page when clicking More		to Portfolio page correctly		Portfolio program information displayed on the page.		such as qualifications, details, and acceptance conditions correctly.	can continue interacting with the page.	page.	
TS0001 : Verify whether the system displays admission criteria information and navigates to the correct page when clicking More	TC03	Verify that clicking More on Quota navigates to Quota page correctly	User opens the Admission Criteria page.	1.Navigate to the Admission Criteria page. 2.Click More on the Quota section.	-	The system navigates to the Quota page.	User can view admission score statistics.	The system successfully navigates to the Portfolio page.	Pass
TS0001 : Verify whether the system displays admission criteria information and navigates to the correct page when clicking More	TC04	Verify that clicking More on Admission navigates to Admission page correctly	User opens the Admission Criteria page.	1.Navigate to the Admission Criteria page. 2.Click More on the Quota section.	-	The system navigates to the Quota page.	User can view admission score statistics.	The system successfully navigates to the Quota page.	Pass
TS0002 : Verify user can view and interact with criteria round pages (Portfolio, Quota, Admission)	TC01	Verify user can view portfolio criteria page and download file from High-Potential School Student	User opens the Portfolio criteria page.	1.Navigate to the Portfolio criteria page. 2.Click High-Potential School Student Admission Program. 3.Click the Download File button.	-	The system displays the program details and allows the user to download the document.	The file is downloaded and the user remains on the Portfolio page.	The program details are displayed and the file can be downloaded successfully.	Pass

TS0002 : Verify user can view and interact with criteria round pages (Portfolio, Quota, Admission)	TC02	Verify user can view portfolio criteria page and download file from High-Potential Student	User opens the Portfolio criteria page.	1.Navigate to the Portfolio criteria page. 2.Click High-Potential Student Admission Program. 3.Click the Download File button.	-	The system displays the program details and allows the user to download the document.	The file is downloaded and the user remains on the Portfolio page.	The program details are displayed and the file can be downloaded successfully.	Pass
TS0002 : Verify user can view and interact with criteria round pages (Portfolio, Quota, Admission)	TC03	Verify user can select High-Potential School Student and calculate score	User opens the Portfolio criteria page.	1.Navigate to the Portfolio criteria page. 2.Select High-Potential School Student Admission Program. 3.Select Educational Information — Currently studying in Grade 12. 4.Select Graduated Program — Thai standard curriculum, School: School Ladprao. 5.Enter Academic Performance — GPA (4 semesters). 6.Enter Combined GPA (Mathematics, Chemistry, Physics, English). 7.Enter Preparatory Engineering courses — EN 001203 Computer Programming and EN 001202 Engineering Drawing grades. 9.Select English Proficiency Test and enter score. 10.Enter Academic Achievements (if any). 11.Click Check Eligibility button.	1.Program = High-Potential School Student Admission Program 2.Educational Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School = School Ladprao 4.GPA (4 semesters) = 3.50 5.Combined GPA (Mathematics, Chemistry, Physics, English) = 3.20 6.EN 001203 Computer Programming = B+ 7.EN 001202 Engineering Drawing = B 8.English Proficiency Test = TOEIC,	The system calculates the score based on the entered information and displays the eligibility result in the Result panel.	User remains on the Portfolio Round Eligibility Check page. Result panel shows the eligibility result.	The system successfully calculates the score and displays whether the user is eligible for the High-Potential School Student Admission Program.	Pass

					Score = 700 9.Academic Achievements = -				
TS0002 : Verify user can view and interact with criteria round pages (Portfolio, Quota, Admission)	TC04	Verify user can select High-Potential Student and calculate score	User opens the Portfolio criteria page.	1.Navigate to the Portfolio criteria page. 2.Select High-Potential Student Admission Program. 3.Select Educational Information — Currently studying in Grade 12. 4.Select Graduated Program — Thai standard curriculum. 5.Enter GPA (4 semesters). 6.Enter GPA in Mathematics, Chemistry, and Physics. 7.Select English Proficiency Test and enter score. 8.Enter Academic Achievements (if any). 9.Click Check Eligibility button.	1.Program = High-Potential Student Admission Program 2.Educational Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum 4.GPA (4 semesters) = 2.90 5.GPA Mathematics = 3.00, Chemistry = 2.80, Physics = 2.75 6.English Proficiency Test = TOEIC, Score = 520 7.Academic Achievements = participated in science project at provincial level	The system successfully calculates the score and displays whether the user is eligible for the High-Potential Student Admission Program.	User remains on the Portfolio Round Eligibility Check page. Result panel shows the eligibility result.	The system successfully displays the eligibility result based on the entered information.	Fail
TS0002 : Verify user can view and interact with criteria round pages (Portfolio, Quota,	TC05	Verify user can view quota criteria page and calculate score	User opens the Quota criteria page.	1.Navigate to the Quota criteria page. 2.Click Calculate Score button. 3.Select Educational Background. 4.Select Graduation Requirement. 5.Select English	1.Round = Quota 2.Educational Background = Thai standard curriculum 3.Graduation Requirement	The system successfully calculates the score and displays whether the user is eligible for the Quota round.	User remains on the Portfolio Round Eligibility Check page. Result panel	The system successfully displays the eligibility result based on the entered information.	Pass

Admission)				Proficiency Test and enter score. 6.Enter Quota Round Test Score — English, Mathematics, Physics, Engineering Aptitude. 7.Click Check button.	= Graduate in the current academic year 4.English Proficiency Requirement = English test score (valid within 2 years), TOEIC Score = 550 5.English = 75 6.Mathematics = 80 7.Physics = 70 8.Engineering Aptitude = 72		shows the eligibility result.		
TS0002 : Verify user can view and interact with criteria round pages (Portfolio, Quota, Admission)	TC06	Verify user can select TCAS69 and calculate score on admission criteria page	User opens the Admission criteria page.	1.Navigate to the Admission criteria page. 2.Select TCAS69 program. 3.Enter A-Level scores — Science, Technology, and Engineering 4.Aptitude Test (TPAT3), A-Level Applied Mathematics 1, A-Level Physics, A-Level English. 5.Select English Proficiency Test and enter score. 6.Click Check Eligibility button.	1.Program = Digital Media Engineering — TCAS69 2.TPAT3 = 20 3.A-Level Applied Mathematics 1 (Basic + Advance) = 35 4.A-Level Physics = 10 5.A-Level English = 35 6.English Proficiency Test = TOEIC, Score = 550	The system successfully calculates the score and displays whether the user is eligible for TCAS69.	User remains on the Admission Round Eligibility Check page. Result panel shows the eligibility result.	The feature is not yet available. The system does not display the score calculation page for TCAS69.	Fail
TS0002 : Verify user can view and interact with criteria round pages	TC07	Verify user can select TCAS68 and calculate score on admission criteria	User opens the Admission criteria page.	1.Navigate to the Admission criteria page. 2.Select TCAS68 program. 3.Enter A-Level scores — Science, Technology, and	1.Program = Digital Media Engineering — TCAS68 2.TPAT3 = 15 3.A-Level	The system successfully calculates the score and displays whether the user is eligible for	User remains on the Admission Round Eligibility Check page.	The feature is not yet available. The system does not display the score calculation	Fail

(Portfolio, Quota, Admission)		page		Engineering Aptitude Test (TPAT3), A-Level Applied Mathematics 1, A-Level Physics, A-Level English. 4.Click Check Eligibility button.	Applied Mathematics 1 (Basic + Advance) = 35 4.A-Level Physics = 15 5.A-Level English = 35	TCAS68.	Result panel shows the eligibility result.	page for TCAS68.	
TS0003 : Verify user can check eligibility for admission rounds (Portfolio, Quota, Admission)	TC01	Verify user can view eligibility check page	User opens the Portfolio criteria page.	1.Navigate to the Portfolio Round Eligibility Check page. 2.Select Educational Information — Currently studying in Grade 12. 3.Select Graduated Program — Thai standard curriculum and select School. 4.Enter GPA (4 semesters). 5.Enter Combined GPA (Mathematics, Chemistry, Physics, English). 6.Enter Preparatory Engineering courses — EN 001203 Computer Programming and EN 001202 Engineering Drawing grades. 7.Select English Proficiency Test and enter score. 8.Select Academic Achievements (if any). 9.Click Check Eligibility button.	1.Program = High-Potential School Student Admission Program 2.Educational Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School = School Ladprao 4.GPA (4 semesters) = 3.50 5.Combined GPA (Mathematics, Chemistry, Physics, English) = 3.20 6.EN 001203 Computer Programming = B+ 7.EN 001202 Engineering Drawing = B 8.English Proficiency Test = TOEIC, Score = 700 9.Academic	The system successfully displays the eligibility result for the High-Potential School Student Admission Program.	User remains on the Portfolio Round Eligibility Check page. Result panel shows the eligibility result.	The system successfully displays the eligibility result based on the entered information.	Pass

					Achievements = participated in science project at provincial level				
TS0003 : Verify user can check eligibility for admission rounds (Portfolio, Quota, Admission)	TC02	Verify eligible applicant in Portfolio round can proceed to calculate score	1.System is running 2.User is on the Score Entry (Fill Information) page	1.Navigate to the Portfolio Round Eligibility Check page 2.Select Educational Information 3.Select Graduated Program 4.Enter GPA (4 semesters) 5.Enter Combined GPA (Mathematics, Chemistry, Physics, English) 6.Enter Combined GPA (Mathematics, Chemistry, Physics) 7.Enter Preparatory Engineering courses 8.Select English Proficiency Test and enter score 9.Select Netsat and enter scores 10.Select A-Level and enter scores 11.Select GCE A-Level Group A (2 subjects) and Group B (1 subject) and enter grades 12.Select Academic Achievements (if any) 13.Click Check Eligibility button 14.Verify that the Eligibility Result page is displayed with Eligible status for Portfolio round 15.Click Calculate	1.Program = High-Potential Student Admission Program 2.Educational Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School = School Ladprao 4.GPA (4 semesters) = 3.50 5.Combined GPA (Mathematics, Chemistry, Physics, English) = 3.20 6.Combined GPA (Mathematics, Chemistry, Physics) = 3.10 7.EN 001203 Computer Programming = B+, EN 001202 Engineering Drawing = B 8.English Proficiency Test =	The system successfully displays the Eligibility Result page. Portfolio round shows Eligible status with Your Score $\geq$ Requirement for all conditions. Calculate Score button is enabled and user can proceed to the Calculate Score page.	User is navigated to the Calculate Score page for the Portfolio round. The entered score information is retained and pre-filled for score calculation.	System displays Eligible result for Portfolio round and allows user to proceed to Calculate Score page	Pass

				Score button	TOEIC, Score = 700 9.Netsat Mathematics = 80, Physics = 75, English = 70, T-PATs = 78 10.A-Level Mathematics = 100, Physics = 90, English = 80 GCE A-Level Group A = Mathematics (Grade A), Chemistry (Grade B) / Group B = Digital Media and Design (Grade B) 11.Academi c Achievemen ts = Participated in science project at provincial level				
TS0003 : Verify user can check eligibility for admission rounds (Portfolio, Quota, Admission)	TC03	Verify system shows not eligible message for Portfolio round	1.System is running 2.User is on Eligibility Check page	1.Navigate to the Admission Round Eligibility Check page 2.Select Educational Information 3.Select Graduated Program 4.Enter GPA (4 semesters) 5.Enter Combined GPA (Mathematics, Chemistry, Physics, English) 6.Enter Combined GPA (Mathematics, Chemistry,	1.Program = High-Potenti al School Student Admission Program 2.Educationa l Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School =	The system successfully displays the Eligibility Result page. Portfolio round shows Not Eligible status for all conditions with Your Score < Requirement. Difference column displays negative values for	User remains on the Eligibility Result page. No navigation to Calculate Score page occurs. User can return to the Score Entry page to re-enter informatio n.	System displays Not Eligible result for Portfolio round with suggestion for improveme nt.	Pass

				Physics) 7.Enter Preparatory Engineering courses 8.Select English Proficiency Test and enter score 9.Select Netsat and enter scores for Mathematics, Physics, English, and T-PATs 10.Select A-Level and enter scores for Mathematics, Physics, and English 11.Select GCE A-Level Group A (2 subjects) and Group B (1 subject) and enter grades 12.Select Academic Achievements (if any) 13.Click Check Eligibility button 14.Verify that the Eligibility Result page is displayed with Not Eligible status for Admission round	School Ladprao 4.GPA (4 semesters) = 2.50 5.Combined GPA (Mathematics, Chemistry, Physics, English) = 2.00 6.Combined GPA (Mathematics, Chemistry, Physics) = 1.90 7.EN 001203 Computer Programming = D, EN 001202 Engineering Drawing = D 8.English Proficiency Test = TOEIC, Score = 300 9.Netsat Mathematics = 30, Physics = 25, English = 20, T-PATs = 28 10.A-Level Mathematics = 30, Physics = 25, English = 20 11.GCE A-Level Group A = Mathematics (Grade E), Chemistry (Grade E) / Group B = Digital Media and	each condition. Calculate Score button is disabled. Suggestion section is displayed recommending scores that need to be improved.			
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					Design (Grade E) 12.Academi c Achievemen ts = None				
TS0003 : Verify user can check eligibility for admission rounds (Portfolio, Quota, Admission)	TC04	Verify eligible applicant in Quota round can proceed to calculate score	1.System is running 2.User is on Eligibility Check page	1.Navigate to the Quota Round Eligibility Check page 2.Select Educational Information 3.Select Graduated Program 4.Enter GPA (4 semesters) 5.Enter Combined GPA (Mathematics, Chemistry, Physics, English) 6.Enter Combined GPA (Mathematics, Chemistry, Physics) 7.Enter Preparatory Engineering courses 8.Select English Proficiency Test and enter score 9.Select Netsat and enter scores for Mathematics, Physics, English, and T-PATs 10.Select A-Level and enter scores for Mathematics, Physics, and English 11.Select GCE A-Level Group A (2 subjects) and Group B (1 subject) and enter grades 12.Select Academic Achievements (if any) 13.Click Check Eligibility button 14.Verify that the	1.Program = Quota 2.Educationa l Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School = School Ladprao 4.GPA (4 semesters) = 3.50 5.Combined GPA (Mathematic s, Chemistry, Physics, English) = 3.20 6.Combined GPA (Mathematic s, Chemistry, Physics) = 3.10 7.EN 001203 Computer Programmin g = B+, EN 001202 Engineering Drawing = B 8.English Proficiency Test = TOEIC, Score = 700 9.Netsat Mathematics = 80,	The system successfully displays the Eligibility Result page. Quota round shows Eligible status with Your Score $\geq$ Requirement for all conditions. Calculate Score button is enabled and user can proceed to the Calculate Score page.	User is navigated to the Calculate Score page for the Quota round. The entered score informatio n is retained and pre-filled for score calculation .	System displays Eligible result for Quota round and allows user to proceed to Calculate Score page.	Pass

				Eligibility Result page is displayed with Eligible status for Quota round 15. Click Calculate Score button	Physics = 75, English = 70, T-PATs = 78 10. A-Level Mathematics = 100, Physics = 90, English = 80 11. GCE A-Level Group A = Mathematics (Grade A), Chemistry (Grade B) / Group B = Digital Media and Design (Grade B) 12. Academic Achievements = Participated in science project at provincial level				
TS0003 : Verify user can check eligibility for admission rounds (Portfolio, Quota, Admission)	TC05	Verify system shows not eligible message for Quota round	1. System is running 2. User is on Eligibility Check page	1. Navigate to the Admission Round Eligibility Check page 2. Select Educational Information 3. Select Graduated Program 4. Enter GPA (4 semesters) 5. Enter Combined GPA (Mathematics, Chemistry, Physics, English) 6. Enter Combined GPA (Mathematics, Chemistry, Physics) 7. Enter Preparatory Engineering courses 8. Select English	1. Program = Quota 2. Educational Information = Currently studying in Grade 12 3. Graduated Program = Thai standard curriculum, School = School Ladprao 4. GPA (4 semesters) = 2.50 5. Combined GPA (Mathematics, Chemistry,	The system successfully displays the Eligibility Result page. Quota round shows Not Eligible status for all conditions with Your Score < Requirement. Difference column displays negative values for each condition. Calculate Score button is disabled.	User remains on the Eligibility Result page. No navigation to Calculate Score page occurs. User can return to the Score Entry page to re-enter information	The Quota Round page UI has not been implemented yet. Unable to verify whether the system displays Not Eligible result for Quota round.	Fail

				Proficiency Test and enter score 9.Select Netsat and enter scores for Mathematics, Physics, English, and T-PATs 10.Select A-Level and enter scores for Mathematics, Physics, and English 11.Select GCE A-Level Group A (2 subjects) and Group B (1 subject) and enter grades 12.Select Academic Achievements (if any) 13.Click Check Eligibility button 14.Verify that the Eligibility Result page is displayed with Not Eligible status for Admission round	Physics, English) = 2.00 6.Combined GPA (Mathematics, Chemistry, Physics) = 1.90 7.EN 001203 Computer Programming = D, EN 001202 Engineering Drawing = D 8.English Proficiency Test = TOEIC, Score = 300 9.Netsat Mathematics = 30, Physics = 25, English = 20, T-PATs = 28 10.A-Level Mathematics = 30, Physics = 25, English = 20 11.GCE A-Level Group A = Mathematics (Grade E), Chemistry (Grade E) / Group B = Digital Media and Design (Grade E) 12.Academic Achievements = None	Suggestion section is displayed recommending scores that need to be improved.			
TS0003 : Verify user	TC06	Verify eligible	1.System is running	1.Navigate to the Admission Round	1.Program = Admission	The system successfully	User is navigated	The Admission	Fail

can check eligibility for admission rounds (Portfolio, Quota, Admission)		applicant in Admission round can proceed to calculate score	2.User is on Eligibility Check page	Eligibility Check page 2.Select Educational Information 3.Select Graduated Program 4.Enter GPA (4 semesters) 5.Enter Combined GPA (Mathematics, Chemistry, Physics, English) 6.Enter Combined GPA (Mathematics, Chemistry, Physics) 7.Enter Preparatory Engineering courses 8.Select English Proficiency Test and enter score 9.Select Netsat and enter scores for Mathematics, Physics, English, and T-PATs 10.Select A-Level and enter scores for Mathematics, Physics, and English 11.Select GCE A-Level Group A (2 subjects) and Group B (1 subject) and enter grades 12.Select Academic Achievements (if any) 13.Click Check Eligibility button 14.Verify that the Eligibility Result page is displayed with Eligible status for Admission round 15.Click Calculate Score button	2.Educationa l Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School = School Ladprao 4.GPA (4 semesters) = 3.50 5.Combined GPA (Mathematics, Chemistry, Physics, English) = 3.20 6.Combined GPA (Mathematics, Chemistry, Physics) = 3.10 7.EN 001203 Computer Programmin g = B+, EN 001202 Engineering Drawing = B 8.English Proficiency Test = TOEIC, Score = 700 9.Netsat Mathematics = 80, Physics = 75, English = 70, T-PATs = 78 10.A-Level Mathematics = 100, Physics = 90, English	displays the Eligibility Result page. Admission round shows Eligible status with Your Score $\geq$ Requirement for all conditions. Calculate Score button is enabled and user can proceed to the Calculate Score page	to the Calculate Score page for the Admission round. The entered score information is retained and pre-filled for score calculation .	Round page is not available. UI for Admission round has not been implemented yet.	
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					= 80 11.GCE A-Level Group A = Mathematics (Grade A), Chemistry (Grade B) / Group B = Digital Media and Design (Grade B) 12.Academi c Achievemen ts = Participated in science project at provincial level				
TS0003 : Verify user can check eligibility for admission rounds (Portfolio, Quota, Admission)	TC07	Verify system shows not eligible message for Admission round	1.System is running 2.User is on Eligibility Check page	1.Navigate to the Admission Round Eligibility Check page 2.Select Educational Information 3.Select Graduated Program 4.Enter GPA (4 semesters) 5.Enter Combined GPA (Mathematics, Chemistry, Physics, English) 6.Enter Combined GPA (Mathematics, Chemistry, Physics) 7.Enter Preparatory Engineering courses 8.Select English Proficiency Test and enter score 9.Select Netsat and enter scores for Mathematics, Physics, English, and T-PATs 10.Select A-Level and enter scores for	1.Program = High-Potenti al School Student Admission Program 2.Educationa l Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School = School Ladprao 4.GPA (4 semesters) = 2.50 5.Combined GPA (Mathematic s, Chemistry, Physics, English) = 2.00 6.Combined GPA	The system successfully displays the Eligibility Result page. Admission round shows Not Eligible status for all conditions with Your Score < Requirement. Difference column displays negative values for each condition. Calculate Score button is disabled. Suggestion section is displayed recommending scores that need to be improved.	User remains on the Eligibility Result page. No navigation to Calculate Score page occurs. User can return to the Score Entry page to re-enter informatio n	The Admission Round page is not available. UI for Admission round has not been implement ed yet.	Fail

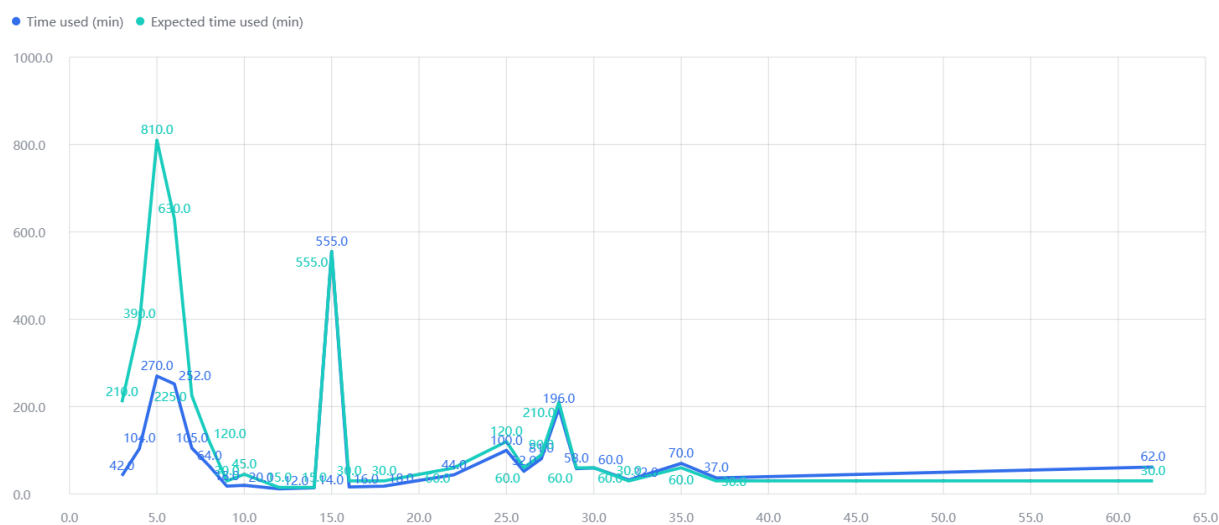
				Mathematics, Physics, and English 11.Select GCE A-Level Group A (2 subjects) and Group B (1 subject) and enter grades 12.Select Academic Achievements (if any) 13.Click Check Eligibility button 14.Verify that the Eligibility Result page is displayed with Not Eligible status for Admission round	(Mathematics, Chemistry, Physics) = 1.90 7.EN 001203 Computer Programming = D, EN 001202 Engineering Drawing = D 8.English Proficiency Test = TOEIC, Score = 300 9.Netsat Mathematics = 30, Physics = 25, English = 20, T-PATs = 28 10.A-Level Mathematics = 30, Physics = 25, English = 20 11.GCE A-Level Group A = Mathematics (Grade E), Chemistry (Grade E) / Group B = Digital Media and Design (Grade E) 12.Academic Achievements = None				
TS0003 : Verify user can check eligibility for admission rounds	TC08	Verify user selects round but does not perform eligibility check	1.System is running 2.User is on Eligibility Check page	1.Navigate to the Portfolio Round Eligibility Check page 2.Select Educational Information	1.Program = High-Potential School Student Admission Program 2.Educational	The system does not perform any validation or eligibility calculation. No eligibility	User remains on the Score Entry (Fill Information) page. All	The Portfolio Round page UI for this behavior has not	Fail

(Portfolio, Quota, Admission)				3.Select Graduated Program 4.Enter GPA (4 semesters) 5.Enter Combined GPA (Mathematics, Chemistry, Physics, English) 6.Enter Combined GPA (Mathematics, Chemistry, Physics) 7.Enter Preparatory Engineering courses 8.Select English Proficiency Test and enter score 9.Select Netsat and enter scores for Mathematics, Physics, English, and T-PATs 10.Select A-Level and enter scores for Mathematics, Physics, and English 11.Select GCE A-Level Group A (2 subjects) and Group B (1 subject) and enter grades 12.Select Academic Achievements (if any) 13.Do not click Check Eligibility button 14.Verify that no eligibility result is displayed and user remains on the Score Entry page	1 Information = Currently studying in Grade 12 3.Graduated Program = Thai standard curriculum, School = School Ladprao 4.GPA (4 semesters) = 3.50 5.Combined GPA (Mathematics, Chemistry, Physics, English) = 3.20 6.Combined GPA (Mathematics, Chemistry, Physics) = 3.10 7.EN 001203 Computer Programming = B+, EN 001202 Engineering Drawing = B 8.English Proficiency Test = TOEIC, Score = 700 9.Netsat Mathematics = 80, Physics = 75, English = 70, T-PATs = 78 10.A-Level Mathematics = 100, Physics = 90, English = 80	result is displayed. User remains on the Score Entry (Fill Information) page with all entered information retained in the input fields.	entered information is retained. No eligibility result page is displayed.	been implemented yet. Unable to verify whether the system retains entered information without clicking Check Eligibility.	
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					11.GCE A-Level Group A = Mathematics (Grade A), Chemistry (Grade B) / Group B = Digital Media and Design (Grade B) 12.Academi c Achievemen ts = Participated in science project at provincial level				
--	--	--	--	--	---	--	--	--	--

4.3 Sprint Monitoring

4.3.1 Burndown Chart



Sprint 1 consisted of 231 tasks with a total expected time of 3,915 minutes. The team completed all tasks with an actual time of 2,282 minutes, finishing approximately 42% faster than planned. The significant gap between expected and actual time reflects that many individual UI design tasks — particularly the

calculator pass/fail case variations — were completed more efficiently as the team built familiarity with the design patterns. A spike in actual time was observed mid-sprint due to the complexity of handling multiple fail case scenarios for the portfolio round calculator, but the team recovered and closed the sprint on schedule.

4.4 Sprint 1 Evaluation

4.4.1 Sprint Review Discussion

During the Sprint 1 Review, the team presented the initial version of the admission eligibility checker. This sprint focused on developing the core functionality that allows users to input academic scores and receive pass/fail results based on admission criteria.

FeedBack	• The test cases do not strictly follow the defined user story. Some test scenarios are too broad and go beyond the scope of the eligibility checker feature. The team should revise the test cases to ensure they align directly with the intended user requirements. • The team should review and clearly understand the user story before designing system components. Some implemented features appear to deviate from the original requirements. • The ER diagram contains structural issues and does not accurately reflect the data relationships required for the system. It should be corrected to match the actual data flow and functionality. • The current UI requires users to select admission rounds (Round 1, 2, 3) before entering scores, which may confuse first-time users. The interface should allow users to directly input their scores without needing to choose a round. • The UI design lacks consistency across pages. Visual style, layout, and components should be standardized to provide a cohesive user experience. • On the result page, when a student does not meet the criteria, the system should clearly indicate how many additional points are required to pass. This information would help users better understand their chances and next steps.
Decision	• Revise all test cases to ensure they strictly align with the defined user story and remain within the scope of the eligibility checker feature. • Re-evaluate the user story and requirements to ensure a shared understanding among team members before further development. • Redesign and correct the ER diagram to accurately represent system

	data relationships. • Modify the UI to allow users to directly enter their scores without selecting admission rounds beforehand. • Standardize the UI design across all pages to maintain consistent style and usability. • Enhance the result page to display the additional score required when eligibility criteria are not met.
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4.4.2 Sprint Retrospective Discussion

What went well	• Initial Planning Completed: The team successfully established project goals, scope, and basic requirements for the system. • Effective Brainstorming: Communication sheets and idea discussions helped generate clear feature directions. • Task Allocation: Responsibilities were assigned to each member, allowing parallel work on design tasks. • Strong Team Participation: All members contributed ideas and actively engaged in discussions. • Foundational Design Progress: Key system models such as ERD and DFD were initiated.
What went poorly	• Unclear Requirements at the Start: Some features were not fully defined, causing confusion during implementation. • Time Management Issues: Several tasks were completed close to the deadline. • Inconsistent Understanding: Team members had different interpretations of certain requirements and diagrams. • Limited Experience: Lack of familiarity with system modeling tools slowed progress.
What can we improve	• Clearer Requirement Definition: Spend more time clarifying user stories and scope before starting development. • Better Task Scheduling: Distribute workload more evenly throughout the sprint. • More Frequent Communication: Hold short meetings to ensure all members share the same understanding. • Early Review of Work: Check progress regularly to detect issues sooner.
What action will we take	• Conduct Requirement Review Meetings at the

	beginning of the next sprint. ● Create a detailed task timeline with milestones. ● Schedule regular progress check-ins among team members. ● Allocate buffer time for revisions and unexpected issues.
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Chapter 5 Sprint 2 Implementation

5.1 Sprint 2 Planning

5.1.1 Sprint Backlog (Lark)

☐ A Task	A Notes	Status	Task leader	Task start ...	Task due ...	Expecte...	Task start...	Task due ...	Time us...
▼ As a high school student, I wa...									
1 Design UI : Admission criteria main page	UX/UI	Done	Deena	02/18/2026	02/18/2026	30.0	2026/02/18	2026/02/18	18.0
2 Design UI : Fill information page	UX/UI	Done	Orange	02/18/2026	02/18/2026	30.0	2026/02/18	2026/02/18	48.0
3 Fix Design UI : Fill information page	UX/UI	Done	Orange	02/19/2026	02/19/2026	30.0	2026/02/19	2026/02/19	30.0
4 Design UI : Show result page (Fail)	UX/UI	Done	Deena	02/19/2026	02/19/2026	30.0	2026/02/19	2026/02/19	41.0
5 Design DFD : Level 1	Backend	Done	Aumaim	02/18/2026	02/18/2026	30.0	2026/02/18	2026/02/18	40.0
6 Design ER diagram overall	Backend	Done	Proud	02/18/2026	02/18/2026	30.0	2026/02/18	2026/02/18	60.0
7 Design ER diagram data type	Backend	Done	Proud	02/19/2026	02/19/2026	30.0	2026/02/19	2026/02/19	30.0
8 Design UI : Fill information page error ...	UX/UI	Done	Opal	02/18/2026	02/18/2026	30.0	2026/02/18	2026/02/18	28.0
9 Design UI : Show result page (Pass)	UX/UI	Done	Mo	02/18/2026	02/18/2026	30.0	2026/02/18	2026/02/18	36.0
10 > Test scenario 3	Tester	Done	Mo	02/19/2026	02/19/2026	30.0	2026/02/19	2026/02/19	20.0
11 test case Scenario 1	Tester	Done	Aumaim	02/21/2026	02/21/2026	30.0	2026/02/21	2026/02/21	15.0
12 test case Scenario 2	Tester	Done	Opal	02/21/2026	02/21/2026	30.0	2026/02/21	2026/02/21	20.0
13 test case Scenario 3	Tester	Done	Aumaim	02/22/2026	02/22/2026	30.0	2026/02/22	2026/02/22	20.0

[Sprint Backlog 2](#)

5.1.2 Sprint Goal

The goal of Sprint 2 is to refine and improve the admission score checker based on feedback received from Sprint 1. This sprint focuses on addressing usability issues identified during review, such as unclear input labels, inconsistent score calculation results, and missing guidance for first-time users.

By the end of this sprint, the score checker will include improved input validation, clearer error messages when scores are out of range, and a more intuitive layout that guides students through the calculation process step by step. Visual presentation of the pass/fail result will also be enhanced to reduce confusion.

This sprint ensures the feature is reliable and user-friendly before moving toward broader system integration in Sprint 3.

5.2 Sprint 2 Development (Increment)

5.2.1 User Interface (UI) Design

DIGITAL MEDIA ENGINEERING

Home Course **Admission** News About

Admission Calculator

Check your university admission chance instantly.
Enter your scores and see eligible rounds.

Calculate Now

Picture 5.1 Main Page

DIGITAL MEDIA ENGINEERING

Home Course **Admission** News About

Score Entry

Education Information

GPAX (% semesters) Pre-engineering courses

Combined GPA (Mathematics, Chemistry, Physics, English) EN 001 203 Computer Programming

Combined GPA (Mathematics, Chemistry, Physics) EN 001 202 Engineering Drawing

Test Scores

English Proficiency Score

Select Enter your Score

Netsat

Mathematics Physics English T-9AT3

A-LEVEL

Mathematics Physics English

GCE A-Level

Requirements: Applicants must have at least a C grade in 3 GCE A-Level subjects, with 2 subjects from Group A and 1 subject from Group B.

Group A: Select 2 subjects from Group A.

☐ Mathematics ☐ Chemistry

☐ Physics ☐ Computer Science

Group B: Select 1 subject from Group B.

☐ Digital Media and Design ☐ Design and Technology

☐ Mechanics ☐ Information Technology

☐ Statistics ☐ Economics

☐ Geography

☐ Academic Achievements

Do you have academic achievements or awards in science, mathematics, or computer science at provincial, regional, national, or international level?

Check Eligibility

Picture 5.2 Score Entry Page and Error Message Score Entry Page


[Home](#)
[Course](#)
[Admission](#)
[News](#)
[About](#)

Eligibility Result

Student Information

Full Name	Province
Sompong Meechai	Khon Kean
School	Grade
Khon Kaen Wittayayon School	12

Round 1 - Portfolio

Eligible

1 High-Potential School Student Admission Program

Qualifications

Condition	Requirement	Your Score	Status
Curriculum	Thai curriculum	Thai curriculum	Eligible
Partner School	29 schools	Khon Kaen Wittayayon School	Eligible
GPAK (4 Semesters)	≥ 3.25	3.82	Eligible
GPA : Math	≥ 3.25	3.50	Eligible
GPA : Chem	≥ 3.25	3.50	Eligible
GPA : Physics	≥ 3.25	3.50	Eligible
GPA : English	≥ 3.25	3.50	Eligible
EN001203 Computer Programming	$\geq B$	B+	Eligible
EN001202 Engineering Drawing	$\geq B$	B	Eligible
IELTS	≥ 5	5.5	Eligible
Academic Achievement	Science / Math / CS activities at provincial level or above	Yes	Eligible

2 High-Potential Student Program

Qualifications

Condition	Requirement	Your Score	Status
Curriculum	Thai / International / OED (age ≥ 16)	Thai curriculum	Eligible
GPAK (4 Semesters)	≥ 2.75	3.82	Eligible
GPA: Math	≥ 2.75	3.50	Eligible
GPA: Chemistry	≥ 2.75	3.36	Eligible
GPA: Physics	≥ 2.75	3.35	Eligible
IELTS	≥ 5	5.5	Eligible
Academic Achievement	Science / Math / CS activities at provincial level or above, or other achievements per attached document	Yes	Eligible

Round 2 - Quota

Eligible

Qualifications

Condition	Requirement	Your Score	Status
TPAT3	≥ 30	58	Eligible
NETSAT: Math	≥ 30	80	Eligible
NETSAT: Physics	≥ 30	70	Eligible
NETSAT: English	≥ 30	76	Eligible
Domicile	Household registration in the Northeast region ≥ 3 years	Khon Kaen	Eligible

Round 3 - Admission

Eligible

Qualifications

Condition	Requirement	Your Score	Status
TPAT3	≥ 30	58	Eligible
A-Level Mathematics 1	≥ 30	50	Eligible
A-Level Physics	≥ 30	60	Eligible
A-Level English	≥ 30	79	Eligible


[Home](#)
[Course](#)
[Admission](#)
[News](#)
[About](#)

Eligibility Result

Student Information

Full Name	Province
Sombat Charonphanthich	Patumthani
School	Grade
Patumthani Vocational Education College	4

You Are Not Eligible Yet

You are currently below the required criteria for all admission rounds. See how much you are missing in each round and explore possible next steps below.

Round 1 - Portfolio

Not Eligible Yet

1 High-Potential School Student Admission Program

Qualifications

Condition	Requirement	Your Score	Difference	Status
Curriculum	Thai curriculum	vocational curriculum	-	Not Eligible
Partner School	29 schools	Patumthani Vocational Education College	-	Not Eligible
GPAK (4 Semesters)	≥ 3.25	3.15	-0.1	Not Eligible
GPA : Math	≥ 3.25	3.00	-0.25	Not Eligible
GPA : Chem	≥ 3.25	3.17	-0.08	Not Eligible
GPA : Physics	≥ 3.25	3.18	-0.09	Not Eligible
GPA : English	≥ 3.25	3.15	-0.1	Not Eligible
EN001203 Computer Programming	$\geq B$	C	Next 1 Grade Higher	Not Eligible
EN001202 Engineering Drawing	$\geq B$	C	Next 1 Grade Higher	Not Eligible
IELTS	≥ 5	4.5	-0.5	Not Eligible
Academic Achievement	Science / Math / CS activities at provincial level or above	-	-	Not Eligible

2 High-Potential Student Program

Qualifications

Condition	Requirement	Your Score	Difference	Status
Curriculum	Thai / International / OED (age ≥ 16)	vocational curriculum	-	Not Eligible
GPAK (4 Semesters)	≥ 2.75	2.55	-0.2	Not Eligible
GPA: Math	≥ 2.75	2.59	-0.2	Not Eligible
GPA: Chem	≥ 2.75	2.90	-0.25	Not Eligible
GPA: Physics	≥ 2.75	2.70	-0.05	Not Eligible
IELTS	≥ 5	4.5	-0.5	Not Eligible
Academic Achievement	Science / Math / CS activities at provincial level or above, or other achievements per attached document	-	-	Not Eligible

Suggestion

To qualify for this round, you need to improve your academic performance and supporting qualifications.

- 1 Increase your GPAK by at least 0.10 - 0.25 points depending on subject requirements.
- 2 Improve course grades (e.g., rates C or at least B in required subjects).
- 3 Improve your English proficiency (IELTS ≥ 5.0 required).
- 4 Participate in academic competitions or activities related to Science / Math / Computer Science at provincial level or higher.

Round 2 - Quota

Not Eligible Yet

Qualifications

Condition	Requirement	Your Score	Difference	Status
TPAT3	≥ 30	15	-15	Not Eligible
NETSAT: Math	≥ 30	55	-35	Not Eligible
NETSAT: Physics	≥ 30	70	-70	Not Eligible
NETSAT: English	≥ 30	60	-30	Not Eligible
Domicile	Household registration in the Northeast region ≥ 3 years	Patumthani	Not in the Northeast region	Not Eligible

Suggestion

You may consider improving your standardized test scores. If possible, retake the examination in the next academic year.

- 1 TPAT3: Need +15 points
- 2 NETSAT (Maths): Need +35 points
- 3 NETSAT (Physics): Need +70 points
- 4 NETSAT (English): Need +30 points

For domicile requirements, this round is limited to applicants residing in the Northeast region for at least 3 years. You may consider applying through other rounds if you do not meet this condition.

Round 3 - Admission

Not Eligible Yet

Qualifications

Condition	Requirement	Your Score	Difference	Status
TPAT3	≥ 30	15	-15	Not Eligible
A-Level Mathematics 1	≥ 30	20	-10	Not Eligible
A-Level Physics	≥ 30	19	-29	Not Eligible
A-Level English	≥ 30	5	-25	Not Eligible

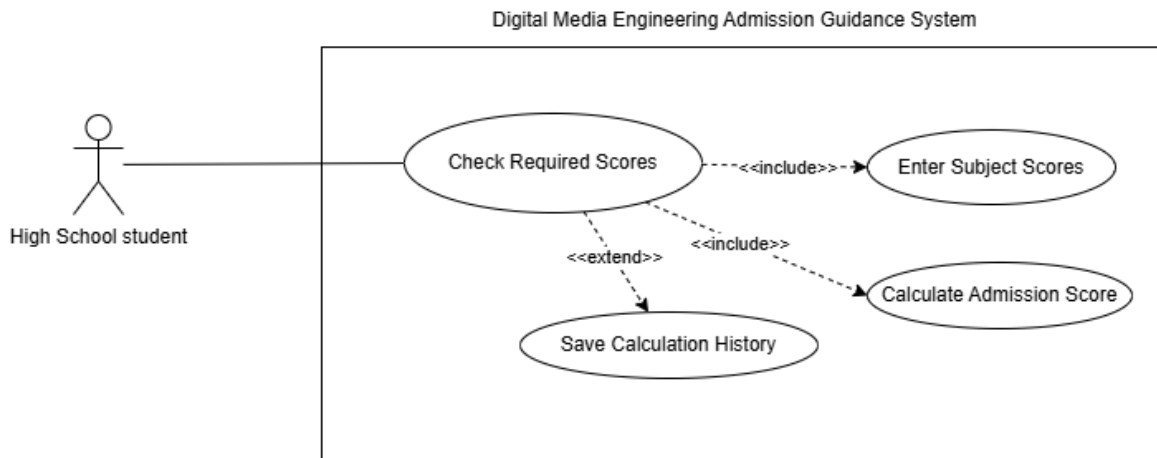
Suggestion

You may consider improving your standardized test scores. If possible, retake the examination in the next academic year.

- 1 TPAT3: Need +15 points
- 2 A-Level Mathematics 1: +10 points
- 3 A-Level Physics: +20 points
- 4 A-Level English: +25 points

Picture 5.3 Show result all pass case Page and Show result all fail case Page

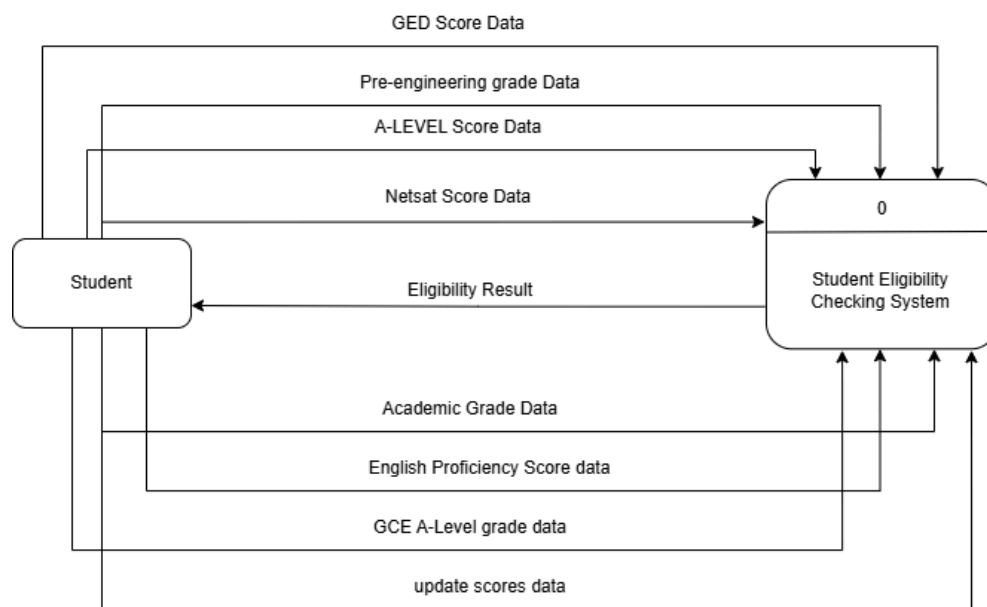
5.2.2 Use Case Diagram



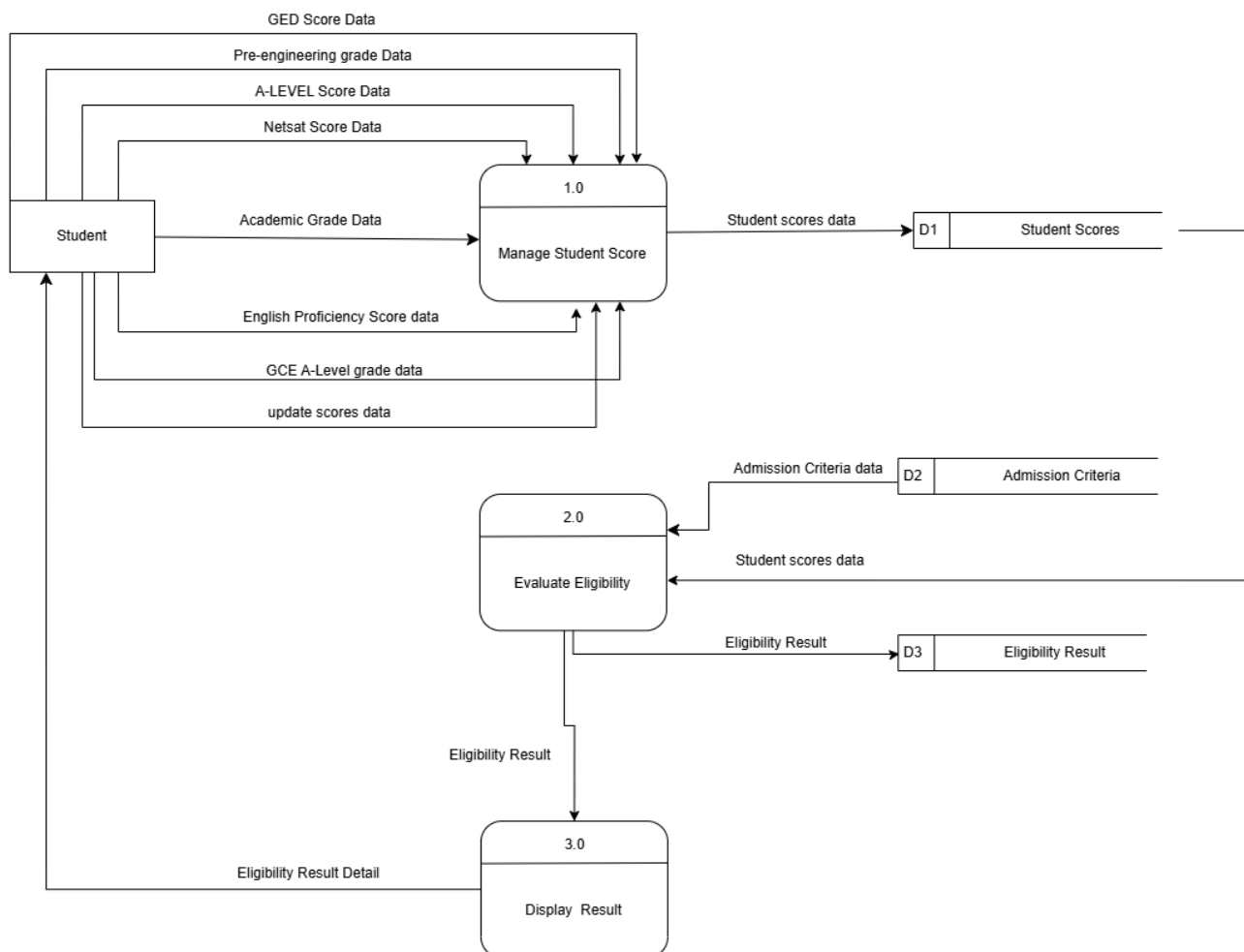
This use case diagram represents the process of checking admission score requirements in the Digital Media Engineering Admission Guidance System. A high school student can enter subject scores and calculate the admission score to determine whether their results meet the admission criteria. The system also provides an optional feature to save the calculation history.

5.2.3 Data Flow Diagram (DFD)

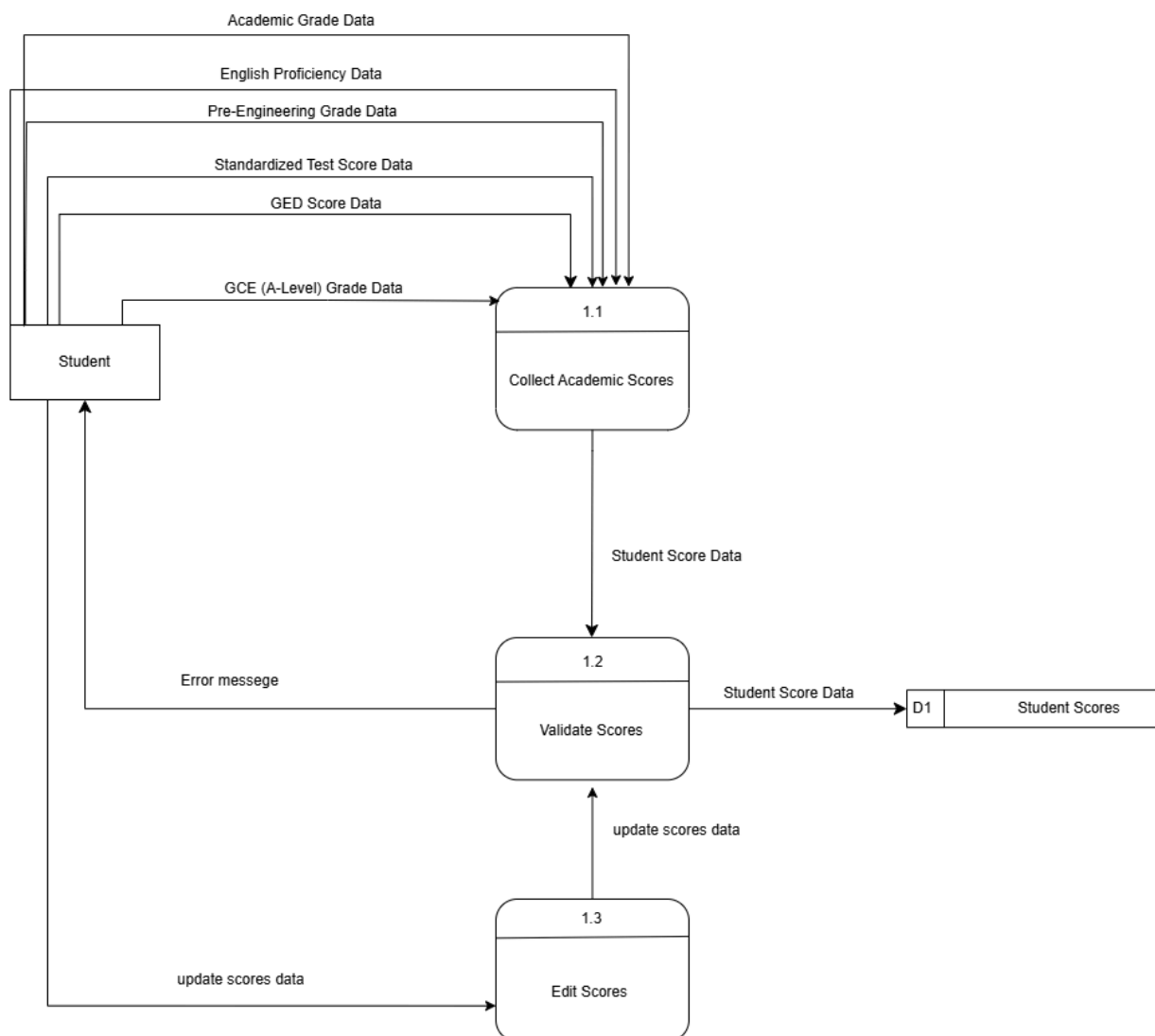
- Context DFD



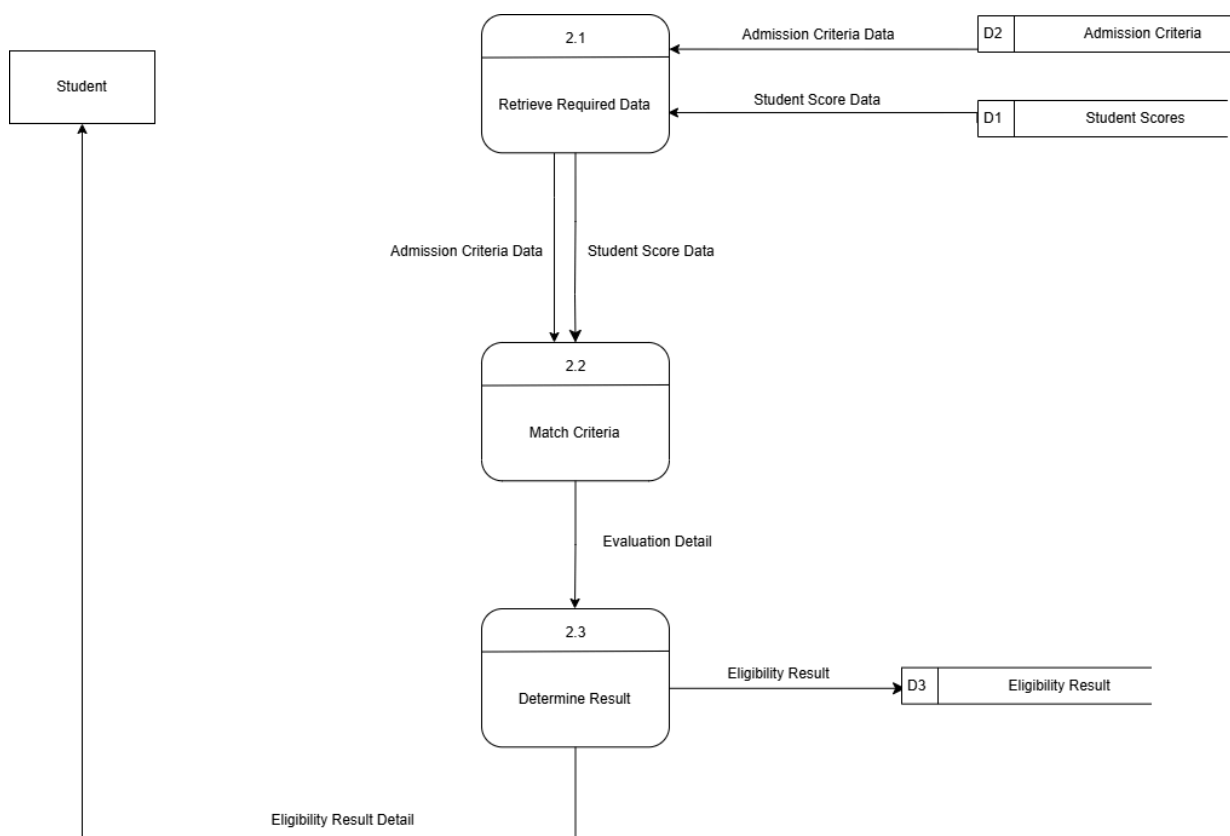
- DFD Level 1



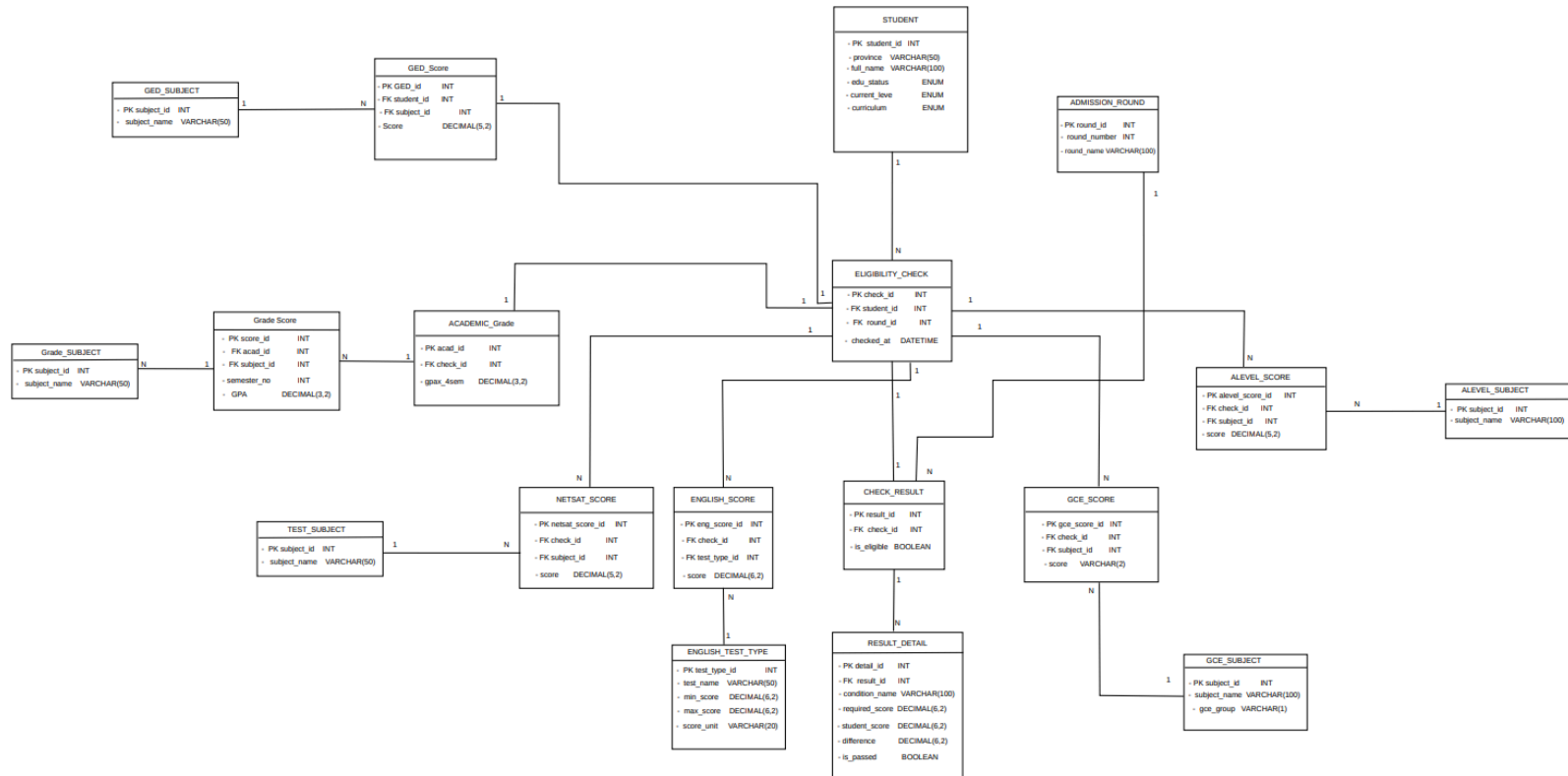
- DFD Level 2
 - DFD Level 2 1.0



- DFD Level 2 2.0



5.2.4 Entity-Relationship Diagram (ER)



5.2.5 Test Cases

5.2.5.1 Test Scenarios

User story ID	Test scenario ID	Test scenario	Number of test cases	Priority
US0001	TS0001	Verify that the system displays error messages when users submit incomplete or invalid score information	6	High
US0001	TS0002	Verify that the system displays all rounds as Eligible when all score criteria are met	3	High

US0001	TS0003	Verify that the system correctly displays mixed eligibility results when scores meet criteria in some rounds but not others	3	High
US0001	TS0004	Verify that the system displays all rounds as Not Eligible with suggestions when no score criteria are met	3	High

5.2.5.2 Test Cases

Test Scenario	Test Case ID	Test Case	Pre-Condition	Test Step	Test Data	Expect Result	Post Condition	Actual Result	Status (Pass/Fail)
TS0001: Verify that the system displays error messages when users submit incomplete or invalid score information	TC01	Verify that the system shows an error when GPAX exceeds the allowed range	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 5.00 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.50 4.Enter Combined GPA (Math, Chem, Physics) = 3.35 5.Enter EN001203 Computer Programming grade = B 6.Enter EN001202 Engineering Drawing grade = B 7.Select English Proficiency = IELTS, Enter score = 5.5 8.Enter NETSAT Mathematics = 80 9.Enter NETSAT Physics = 70 10.Enter NETSAT English = 76 11.Enter TPAT3 = 58 12.Enter A-Level	GPAX (4 Semesters) = 5.00 ✗ (invalid, max allowed: 4.00) Combined GPA (Math, Chem, Physics, English) = 3.50 Combined GPA (Math, Chem, Physics) = 3.35 EN001203 Computer Programming grade = B EN001202 Engineering Drawing grade = B EN001203 Computer Programming = B EN001202 Engineering Drawing = B English Proficiency = IELTS, Score = 5.5 NETSAT Mathematics = 80 NETSAT	System displays error on GPAX field "GPAX value must be between 0.00 and 4.00", form is not submitted	Form remains on Score Entry page GPAX field is highlighted with error message All other entered data remains in the fields No eligibility result is generated	System displayed error message on GPAX field "GPAX value must be between 0.00 and 4.00", form was not submitted	Pass

				Mathematics = 50 13.Enter A-Level Physics = 60 14.Enter A-Level English = 79 15.In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16.In GCE A-Level Group B: check 1 subject (e.g., Digital Media and Design), enter grade = B 17.Select Academic Achievements = Yes 18.Click "Check Eligibility"	Physics = 70 NETSAT English = 76 TPAT3 = 58 A-Level Mathematics = 50 A-Level Physics = 60 A-Level English = 79 GCE Group A: Mathematics = B, Physics = B GCE Group B: Digital Media and Design = B Academic Achievem ts = Yes				
TS0001: Verify that the system displays error messages when users submit incomplete or invalid score information	TC02	Verify that the system shows an error when Combined GPA is entered as a non-numer ic value	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.82 3.Enter Combined GPA (Math, Chem, Physics, English) = "abc" 4.Enter Combined GPA (Math, Chem, Physics) = 3.35 5.Enter EN001203 Computer Programming grade = B 6.Enter EN001202 Engineering Drawing grade = B 7.Select English Proficiency = IELTS, Enter score = 5.5 8.Enter NETSAT Mathematics = 80 9.Enter NETSAT Physics = 70 10.Enter NETSAT English = 76 11.Enter TPAT3 =	GPAX (4 Semesters) = 3.82 Combined GPA (Math, Chem, Physics, English) = "abc" ✗ (invalid, non-numeric) Combined GPA (Math, Chem, Physics) = 3.35 EN001203 Computer Programmin g = B EN001202 Engineering Drawing = B English Proficiency = IELTS, Score = 5.5 NETSAT Mathematics = 80	System displays error on GPAX field "GPAX value must be between 0.00 and 4.00", form is not submitted	Form remains on Score Entry page Combined GPA (Math, Chem, Physics, English) field is highlighte d with error message All other entered data remains in the fields No eligibility result is generated	System displayed error message on Combined GPA field indicating invalid non-numer ic input, form was not submitted	Pass

				58 12.Enter A-Level Mathematics = 50 13.Enter A-Level Physics = 60 14.Enter A-Level English = 79 15.In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16.In GCE A-Level Group B: check 1 subject (e.g., Digital Media and Design), enter grade = B 17.Select Academic Achievements = Yes 18.Click "Check Eligibility"	NETSAT Physics = 70 NETSAT English = 76 TPAT3 = 58 A-Level Mathematics = 50 A-Level Physics = 60 A-Level English = 79 GCE Group A: Mathematics = B, Physics = B GCE Group B: Digital Media and Design = B Academic Achievements = Yes				
TS0001: Verify that the system displays error messages when users submit incomplete or invalid score information	TC03	Verify that the system shows an error when TOEIC score exceeds the maximum allowed value	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.82 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.50 4.Enter Combined GPA (Math, Chem, Physics) = 3.35 5.Enter EN001203 Computer Programming grade = B 6.Enter EN001202 Engineering Drawing grade = B 7.Select English Proficiency = TOEIC, Enter score = 1005 8.Enter NETSAT Mathematics = 80 9.Enter NETSAT Physics = 70 10.Enter NETSAT	GPAX (4 Semesters) = 3.82 Combined GPA (Math, Chem, Physics, English) = 3.50 Combined GPA (Math, Chem, Physics) = 3.35 EN001203 Computer Programming = B EN001202 Engineering Drawing = B English Proficiency = TOEIC, Score = 1005 ✗ (invalid, max allowed:	System displays error on TOEIC field "TOEIC score must be between 0 and 990", form is not submitted	Form remains on Score Entry page TOEIC score field is highlighted with error message All other entered data remains in the fields No eligibility result is generated	System displayed error message on TOEIC field "TOEIC score must be between 0 and 990", form was not submitted	Pass

				English = 76 11.Enter TPAT3 = 58 12.Enter A-Level Mathematics = 50 13.Enter A-Level Physics = 60 14.Enter A-Level English = 79 15.In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16.In GCE A-Level Group B: check 1 subject (e.g., Digital Media and Design), enter grade = B 17.Select Academic Achievements = Yes 18.Click "Check Eligibility"	990) NETSAT Mathematics = 80 NETSAT Physics = 70 NETSAT English = 76 TPAT3 = 58 A-Level Mathematics = 50 A-Level Physics = 60 A-Level English = 79 GCE Group A: Mathematics = B, Physics = B GCE Group B: Digital Media and Design = B				
TS0001: Verify that the system displays error messages when users submit incomplete or invalid score information	TC04	Verify that the system shows an error when fewer than required subjects are selected in GCE Group A	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.82 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.50 4.Enter Combined GPA (Math, Chem, Physics) = 3.35 5.Enter EN001203 Computer Programming grade = B 6.Enter EN001202 Engineering Drawing grade = B 7.Select English Proficiency = IELTS, Enter score = 5.5 8.Enter NETSAT Mathematics = 80 9.Enter NETSAT	GPAX (4 Semesters) = 3.82 Combined GPA (Math, Chem, Physics, English) = 3.50 Combined GPA (Math, Chem, Physics) = 3.35 EN001203 Computer Programming = B EN001202 Engineering Drawing = B English Proficiency = IELTS, Score = 5.5 NETSAT Mathematics	System displays error "Please select at least 2 subjects from Group A", form is not submitted	Form remains on Score Entry page GCE Group A section is highlighted with error message Previously selected subject (Mathematics) and its grade remain checked No eligibility result is generated	System displays error "Please select at least 2 subjects from Group A", form is not submitted	Pass

				Physics = 70 10.Enter NETSAT English = 76 11.Enter TPAT3 = 58 12.Enter A-Level Mathematics = 50 13.Enter A-Level Physics = 60 14.Enter A-Level English = 79 15.In GCE A-Level Group A: check only Mathematics, enter grade = B (do NOT select Physics or any second subject) 16.In GCE A-Level Group B: check 1 subject (e.g., Digital Media and Design), enter grade = B 17.Select Academic Achievements = Yes 18.Click "Check Eligibility"	= 80 NETSAT Physics = 70 NETSAT English = 76 TPAT3 = 58 A-Level Mathematics = 50 A-Level Physics = 60 A-Level English = 79 GCE Group A: Mathematics = B only ✗ (invalid, minimum 2 subjects required, Physics not selected) GCE Group B: Digital Media and Design = B Academic Achievements = Yes				
TS0001: Verify that the system displays error messages when users submit incomplete or invalid score information	TC05	Verify that the system shows an error when no subject is selected in GCE Group B	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.82 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.50 4.Enter Combined GPA (Math, Chem, Physics) = 3.35 5.Enter EN001203 Computer Programming grade = B 6.Enter EN001202 Engineering Drawing grade = B 7.Select English Proficiency = IELTS, Enter score = 5.5	GPAX (4 Semesters) = 3.82 Combined GPA (Math, Chem, Physics, English) = 3.50 Combined GPA (Math, Chem, Physics) = 3.35 EN001203 Computer Programming = B EN001202 Engineering Drawing = B English Proficiency = IELTS,	System displays error "Please select at least 1 subject from Group B", form is not submitted	Form remains on Score Entry page GCE Group B section is highlighted with error message All other entered data remains in the fields No eligibility result is generated	System displayed error message "Please select at least 1 subject from Group B", form was not submitted	Pass

				8.Enter NETSAT Mathematics = 80 9.Enter NETSAT Physics = 70 10.Enter NETSAT English = 76 11.Enter TPAT3 = 58 12.Enter A-Level Mathematics = 50 13.Enter A-Level Physics = 60 14.Enter A-Level English = 79 15.In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16.Leave GCE A-Level Group B: no subject selected 17.Select Academic Achievements = Yes 18.Click "Check Eligibility"	Score = 5.5 NETSAT Mathematics = 80 NETSAT Physics = 70 NETSAT English = 76 TPAT3 = 58 A-Level Mathematics = 50 A-Level Physics = 60 A-Level English = 79 GCE Group A: Mathematics = B, Physics = B GCE Group B: no subject selected ✗ (invalid, minimum 1 subject required) Academic Achievements = Yes				
TS0001: Verify that the system displays error messages when users submit incomplete or invalid score information	TC06	Verify that the system shows errors when submitting the form without entering any required information	User is on Score Entry page, all fields are empty, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Leave GPAX (4 Semesters) = empty 3.Leave Combined GPA (Math, Chem, Physics, English) = empty 4.Leave Combined GPA (Math, Chem, Physics) = empty 5.Leave EN001203 Computer Programming grade = empty 6.Leave EN001202 Engineering Drawing grade = empty 7.Leave English	GPAX (4 Semesters) = empty ✗ Combined GPA (Math, Chem, Physics, English) = empty ✗ Combined GPA (Math, Chem, Physics) = empty ✗ EN001203 Computer Programming grade = empty ✗ EN001202 Engineering Drawing =	System displays error messages on all required fields, all mandatory fields are highlighted, form is not submitted	Form remains on Score Entry page All required fields are highlighted with error messages simultaneously No data is retained in any field No eligibility result is generated	System displayed error messages on all required fields, all mandatory fields were highlighted, form was not submitted	Pass

				Proficiency selection = not selected 8. Leave NETSAT Mathematics = empty 9. Leave NETSAT Physics = empty 10. Leave NETSAT English = empty 11. Leave TPAT3 = empty 12. Leave A-Level Mathematics = empty 13. Leave A-Level Physics = empty 14. Leave A-Level English = empty 15. Leave GCE Group A = no subjects selected 16. Leave GCE Group B = no subjects selected 17. Leave Academic Achievements = not selected 18. Click "Check Eligibility"	empty ✗ English Proficiency = not selected ✗ NETSAT Mathematics = empty ✗ NETSAT Physics = empty ✗ NETSAT English = empty ✗ TPAT3 = empty ✗ A-Level Mathematics = empty ✗ A-Level Physics = empty ✗ A-Level English = empty ✗ GCE Group A = no subjects selected ✗ GCE Group B = no subjects selected ✗ Academic Achievements = not selected ✗				
TS0002: Verify that the system displays all rounds as Eligible when all score criteria are met	TC01	Verify that the system displays Eligible status for all rounds when scores exceed the required criteria	User is on Score Entry page, Browser is open	1. Navigate to the Admission page and click "Calculate Now" 2. Enter GPAX (4 Semesters) = 3.82 3. Enter Combined GPA (Math, Chem, Physics, English) = 3.50 4. Enter Combined GPA (Math, Chem, Physics) = 3.35 5. Enter EN001203 Computer Programming grade = B+	GPAX (4 Semesters) = 3.82 ✓ (pass all rounds ≥ 3.25) Combined GPA (Math, Chem, Physics, English) = 3.50 ✓ (pass Round 1 ≥ 3.25 , Round 1 High-Potential ≥ 2.75)	System navigates to "Eligibility Result" page, displays Round 1 Portfolio – Eligible, Round 2 Quota – Eligible, Round 3 Admission – Eligible, no error messages shown	-System navigates to Eligibility Result page -Student information section displays name, province, school, and grade -Round 1 Portfolio shows	System navigated to "Eligibility Result" page, displayed Round 1 Portfolio – Eligible, Round 2 Quota – Eligible, Round 3 Admission – Eligible, no error	Pass

				6.Enter EN001202 Engineering Drawing grade = B 7.Select English Proficiency = IELTS, Enter score = 5.5 8.Enter NETSAT Mathematics = 80 9.Enter NETSAT Physics = 70 10.Enter NETSAT English = 76 11.Enter TPAT3 = 58 12.Enter A-Level Mathematics = 50 13.Enter A-Level Physics = 60 14.Enter A-Level English = 79 15.In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16.In GCE A-Level Group B: check 1 subject (e.g., Digital Media and Design), enter grade = B 17.Select Academic Achievements = Yes 18.Click "Check Eligibility"	Combined GPA (Math, Chem, Physics) = 3.35 ✓ (pass Round 1 ≥ 3.25 , Round 1 High-Potential ≥ 2.75) EN001203 Computer Programming = B+ ✓ (pass Round 1 $\geq B$) EN001202 Engineering Drawing = B ✓ (pass Round 1 $\geq B$) English Proficiency = IELTS, Score = 5.5 ✓ (pass all rounds ≥ 5.0) NETSAT Mathematics = 80 ✓ (pass Round 2 ≥ 30) NETSAT Physics = 70 ✓ (pass Round 2 ≥ 30) NETSAT English = 76 ✓ (pass Round 2 ≥ 30) TPAT3 = 58 ✓ (pass Round 2 and Round 3 ≥ 30) A-Level Mathematics = 50 ✓ (pass Round 3 ≥ 30) A-Level Physics = 60 ✓ (pass	Eligible with full qualification breakdown -Round 2 Quota shows Eligible with full qualification breakdown -Round 3 Admission shows Eligible with full qualification breakdown -No error messages are present on the page	messages shown	
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					Round 3 ≥ 30) A-Level English = 79 ✓ (pass Round 3 ≥ 30) GCE Group A: Mathematics = B, Physics = B ✓ GCE Group B: Digital Media and Design = B ✓ Academic Achievements = Yes ✓ Domicile = Khon Kaen ✓ (Northeast region ≥ 3 years)				
TS0002: Verify that the system displays all rounds as Eligible when all score criteria are met	TC02	Verify that the system displays Eligible status when scores are exactly at the minimum required boundary values	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.25 (exact minimum) 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.25 (exact minimum) 4.Enter Combined GPA (Math, Chem, Physics) = 2.75 (exact minimum) 5.Enter EN001203 Computer Programming grade = B (exact minimum) 6.Enter EN001202 Engineering Drawing grade = B (exact minimum) 7.Select English Proficiency = IELTS, Enter score	GPAX (4 Semesters) = 3.82 Combined GPA (Math, Chem, Physics, English) = 3.50 Combined GPA (Math, Chem, Physics) = 3.35 EN001203 Computer Programmin g = B+ EN001202 Engineering Drawing = B English Proficiency = IELTS, Score = 5.5 NETSAT Mathematics = 80	System navigates to Eligibility Result page, all rounds display Eligible status, no error messages shown	-System displays Eligibility Result page -All 3 rounds show Eligible status -Full qualificati on breakdow n visible for each round -No error messages present	System navigated to Eligibility Result page, displayed Round 1 Portfolio – Eligible, Round 2 Quota – Eligible, Round 3 Admission – Eligible, no error messages shown	Pass

				= 5.0 (exact minimum) 8.Enter NETSAT Mathematics = 30 (exact minimum) 9.Enter NETSAT Physics = 30 (exact minimum) 10.Enter NETSAT English = 30 (exact minimum) 11.Enter TPAT3 = 30 (exact minimum) 12.Enter A-Level Mathematics = 30 (exact minimum) 13.Enter A-Level Physics = 30 (exact minimum) 14.Enter A-Level English = 30 (exact minimum) 15. In GCE A-Level Group A: check Mathematics, enter grade = C; check Physics, enter grade = C (exact minimum) 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = C (exact minimum) 17.Select Academic Achievements = Yes 18.Click "Check Eligibility"	NETSAT Physics = 70 NETSAT English = 76 TPAT3 = 58 A-Level Mathematics = 50 A-Level Physics = 60 A-Level English = 79 GCE Group A: Mathematics = B, Physics = B GCE Group B: Digital Media and Design = B Academic Achievements = Yes Domicile = Khon Kaen (Northeast region) Full Name = Sompong Meechai Province = Khon Kaen School = Khon Kaen Wittayayon School Grade = 12				
TS0002: Verify that the system displays all rounds as Eligible when all score criteria are met	TC03	Verify that the system displays student information correctly on the result page	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.82 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.50 4.Enter Combined	GPAX (4 Semesters) = 3.82 ✓ (pass Round 1 ≥ 3.25) Combined GPA (Math, Chem, Physics, English) = 3.50 ✓	Eligibility Result page displays Student Information section with correct Full Name, Province, School, and Grade	-Student Information section is visible on result page -All student details are displayed accurately	Eligibility Result page displayed Student Information section with correct Full Name, Province,	Pass

				GPA (Math, Chem, Physics) = 3.35 5. Enter EN001203 Computer Programming grade = B+ 6. Enter EN001202 Engineering Drawing grade = B 7. Select English Proficiency = IELTS, Enter score = 5.5 8. Enter NETSAT Mathematics = 80 9. Enter NETSAT Physics = 70 10. Enter NETSAT English = 76 11. Enter TPAT3 = 58 12. Enter A-Level Mathematics = 50 13. Enter A-Level Physics = 60 14. Enter A-Level English = 79 15. In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = B 17. Select Academic Achievements = Yes 18. Click "Check Eligibility" 19. Verify Student Information section on Eligibility Result page shows correct Full Name, Province, School, and Grade	(pass Round 1 ≥ 3.25) Combined GPA (Math, Chem, Physics) = 3.35 ✓ (pass Round 1 ≥ 3.25) EN001203 Computer Programming = B+ ✓ (pass Round 1 $\geq B$) EN001202 Engineering Drawing = B ✓ (pass Round 1 $\geq B$) English Proficiency = IELTS, Score = 5.5 ✓ (pass Round 1 ≥ 5.0) NETSAT Mathematics = 20 ✗ (fail Round 2 minimum 30) NETSAT Physics = 20 ✗ (fail Round 2 minimum 30) NETSAT English = 20 ✗ (fail Round 2 minimum 30) TPAT3 = 15 ✗ (fail Round 2 and Round 3 minimum 30) A-Level Mathematics = 20 ✗ (fail Round 3)	matching what was entered	-Rounds eligibility results are shown below student information	School, and Grade	
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					minimum 30) A-Level Physics = 10 ✗ (fail Round 3 minimum 30) A-Level English = 5 ✗ (fail Round 3 minimum 30) GCE Group A: Mathematics = B, Physics = B ✓ GCE Group B: Digital Media and Design = B ✓ Academic Achievements = Yes ✓ Domicile = Patumthani ✗ (fail Round 2 domicile requirement – not Northeast region)				
TS0003: Verify that the system correctly displays mixed eligibility results when scores meet criteria in some rounds but not others	TC01	Verify that the system displays Eligible status for Portfolio round and Not Eligible status for Quota and Admission rounds	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.82 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.50 4.Enter Combined GPA (Math, Chem, Physics) = 3.35 5.Enter EN001203 Computer Programming grade = B+ 6.Enter EN001202	GPAX (4 Semesters) = 2.00 ✗ (fail all rounds, diff = -1.25 from Round 1 minimum) Combined GPA (Math, Chem, Physics, English) = 2.00 ✗ (fail all rounds, diff = -1.25) Combined GPA (Math, Chem,	System displays Round 1 Portfolio – Eligible, Round 2 Quota – Not Eligible, Round 3 Admission – Not Eligible, suggestions for improvement shown for Round 2 and Round 3	-Eligibility Result page displays mixed results -Round 1 shows Eligible with qualifications breakdown -Round 2 and Round 3 show Not	System displayed Round 1 Portfolio – Eligible, Round 2 Quota – Not Eligible, Round 3 Admission – Not Eligible, suggestions shown for Round 2 and Round 3	Pass

				Engineering Drawing grade = B 7. Select English Proficiency = IELTS, Enter score = 5.5 8. Enter NETSAT Mathematics = 20 (below minimum 30) 9. Enter NETSAT Physics = 20 (below minimum 30) 10. Enter NETSAT English = 20 (below minimum 30) 11. Enter TPAT3 = 15 (below minimum 30) 12. Enter A-Level Mathematics = 20 (below minimum 30) 13. Enter A-Level Physics = 10 (below minimum 30) 14. Enter A-Level English = 5 (below minimum 30) 15. In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = B 17. Select Academic Achievements = Yes 18. Click "Check Eligibility"	Physics) = 2.00 ✗ (fail minimum 2.75, diff = -0.75) EN001203 Computer Programming = D ✗ (fail minimum B, need 2 grades higher) EN001202 Engineering Drawing = D ✗ (fail minimum B, need 2 grades higher) English Proficiency = IELTS, Score = 3.0 ✗ (fail minimum 5.0, diff = -2.0) NETSAT Mathematics = 10 ✗ (fail minimum 30, diff = -20) NETSAT Physics = 10 ✗ (fail minimum 30, diff = -20) NETSAT English = 10 ✗ (fail minimum 30, diff = -20) TPAT3 = 10 ✗ (fail minimum 30, diff = -20) A-Level Mathematics	Eligible with difference values and suggestions		
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					= 10 ✗ (fail minimum 30, diff = -20) A-Level Physics = 10 ✗ (fail minimum 30, diff = -20) A-Level English = 10 ✗ (fail minimum 30, diff = -20) GCE Group A: Mathematics = D, Physics = D ✗ (fail minimum C) GCE Group B: Digital Media and Design = D ✗ (fail minimum C) Academic Achievements = No ✗ Domicile = Patumthani ✗ (not Northeast region)				
TS0003: Verify that the system correctly displays mixed eligibility results when scores meet criteria in some rounds but not others	TC02	Verify that the system displays mixed Eligible and Not Eligible status within the same round	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.15 (below Round 1 minimum 3.25) 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.00 (below Round 1 minimum 3.25) 4.Enter Combined GPA (Math, Chem, Physics) = 2.55 (below Round 1 minimum 2.75)	GPAX (4 Semesters) = 3.15 ✗ (fail minimum 3.25, diff = -0.10) Combined GPA (Math, Chem, Physics, English) = 3.00 ✗ (fail minimum 3.25, diff = -0.25) Combined	System displays Round 1 Portfolio – Not Eligible (some conditions fail), Round 2 Quota – Not Eligible (TPAT3 passes but domicile may fail), Round 3 Admission – Not Eligible, difference values shown	-Eligibility Result page shows Not Eligible for all rounds -Each round displays condition-level breakdown showing which conditions pass and which fail	System displayed mixed Eligible/Not Eligible per condition within each round, difference values shown for failed conditions, suggestions displayed	Pass

				5. Enter EN001203 Computer Programming grade = C (below minimum B) 6. Enter EN001202 Engineering Drawing grade = C (below minimum B) 7. Select English Proficiency = IELTS, Enter score = 4.5 (below minimum 5.0) 8. Enter NETSAT Mathematics = 80 (pass Round 2 minimum 30) 9. Enter NETSAT Physics = 70 (pass Round 2 minimum 30) 10. Enter NETSAT English = 76 (pass Round 2 minimum 30) 11. Enter TPAT3 = 58 (pass Round 2 minimum 30) 12. Enter A-Level Mathematics = 50 (pass Round 3 minimum 30) 13. Enter A-Level Physics = 60 (pass Round 3 minimum 30) 14. Enter A-Level English = 79 (pass Round 3 minimum 30) 15. In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = B 17. Select	GPA (Math, Chem, Physics) = 2.55 ✗ (fail Round 1 minimum 2.75, diff = -0.20) EN001203 Computer Programming = C ✗ (fail Round 1 minimum B, need 1 grade higher) EN001202 Engineering Drawing = C ✗ (fail Round 1 minimum B, need 1 grade higher) English Proficiency = IELTS, Score = 4.5 ✗ (fail minimum 5.0, diff = -0.5) NETSAT Mathematics = 20 ✗ (fail Round 2 minimum 30, diff = -10) NETSAT Physics = 20 ✗ (fail Round 2 minimum 30, diff = -10) NETSAT English = 20 ✗ (fail Round 2 minimum 30, diff = -10) TPAT3 = 15	for each failed condition, suggestions shown for all rounds	-Suggestion section visible for every round		
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				Academic Achievements = Yes 18.Click "Check Eligibility"	✗ (fail Round 2 and Round 3 minimum 30, diff = -15) A-Level Mathematics = 20 ✗ (fail Round 3 minimum 30, diff = -10) A-Level Physics = 10 ✗ (fail Round 3 minimum 30, diff = -20) A-Level English = 5 ✗ (fail Round 3 minimum 30, diff = -25) GCE Group A: Mathematics = B, Physics = B ✓ GCE Group B: Digital Media and Design = B ✓ Academic Achievements = Yes ✓ Domicile = Patumthani ✗ (fail Round 2 domicile – not Northeast region)				
TS0003: Verify that the system correctly displays mixed	TC03	Verify that the system displays suggestions for improvem	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.15	GPAX (4 Semesters) = 3.24 ✗ (fail Round 1 minimum 3.25, diff =	Each Not Eligible round displays a Suggestion section listing specific	-Eligibility Result page displays Not Eligible	Each Not Eligible round displayed Suggestion section	Pass

eligibility results when scores meet criteria in some rounds but not others		ent in rounds that are Not Eligible		3. Enter Combined GPA (Math, Chem, Physics, English) = 3.00 4. Enter Combined GPA (Math, Chem, Physics) = 2.55 5. Enter EN001203 Computer Programming grade = C 6. Enter EN001202 Engineering Drawing grade = C 7. Select English Proficiency = IELTS, Enter score = 4.5 8. Enter NETSAT Mathematics = 20 9. Enter NETSAT Physics = 20 10. Enter NETSAT English = 20 11. Enter TPAT3 = 15 12. Enter A-Level Mathematics = 20 13. Enter A-Level Physics = 10 14. Enter A-Level English = 5 15. In GCE A-Level Group A: check Mathematics, enter grade = B; check Physics, enter grade = B 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = B 17. Select Academic Achievements = Yes 18. Click "Check Eligibility" 19. Scroll down to verify Suggestion section appears under each Not Eligible round	-0.01) Combined GPA (Math, Chem, Physics, English) = 3.24 ✗ (fail minimum 3.25, diff = -0.01) Combined GPA (Math, Chem, Physics) = 2.74 ✗ (fail minimum 2.75, diff = -0.01) EN001203 Computer Programming = C ✗ (fail minimum B, need 1 grade higher) EN001202 Engineering Drawing = C ✗ (fail minimum B, need 1 grade higher) English Proficiency = IELTS, Score = 4.9 ✗ (fail minimum 5.0, diff = -0.1) NETSAT Mathematics = 29 ✗ (fail minimum 30, diff = -1) NETSAT Physics = 29 ✗ (fail minimum 30, diff = -1) NETSAT English = 29 ✗ (fail	conditions that failed with the exact points/grades needed to meet the requirement	for all rounds -Suggestion section is visible under every Not Eligible round -Each suggestion specifies exact improvement needed per condition	with specific improvement points listed for each failed condition	
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				20. Verify each suggestion lists the specific scores that need improvement and by how much	minimum 30, diff = -1) TPAT3 = 29 ✗ (fail minimum 30, diff = -1) A-Level Mathematics = 29 ✗ (fail minimum 30, diff = -1) A-Level Physics = 29 ✗ (fail minimum 30, diff = -1) A-Level English = 29 ✗ (fail minimum 30, diff = -1) GCE Group A: Mathematics = D, Physics = D ✗ (fail minimum C, need 1 grade higher) GCE Group B: Digital Media and Design = D ✗ (fail minimum C, need 1 grade higher) Academic Achievements = No ✗ Domicile = Patumthani ✗ (not Northeast region)				
TS0004: Verify that the system displays all rounds as Not Eligible with suggestions	TC01	Verify that the system displays a "Not Eligible" result when the scores are below the	User is on Score Entry page, Browser is open	1. Navigate to the Admission page and click "Calculate Now" 2. Enter GPAX (4 Semesters) = 2.00 3. Enter Combined GPA (Math, Chem, Physics, English) =	GPAX (4 Semesters) = 3.24 ✗ (fail Round 1 minimum 3.25, diff = -0.01) Combined GPA (Math,	System displays "You Are Not Eligible Yet" message, all rounds show Not Eligible status, difference	-Eligibility Result page displays "You Are Not Eligible Yet" banner	System displayed "You Are Not Eligible Yet" message, all rounds showed	Pass

when no score criteria are met		required criteria in all categories for every round.		2.00 4.Enter Combined GPA (Math, Chem, Physics) = 2.00 5.Enter EN001203 Computer Programming grade = D 6.Enter EN001202 Engineering Drawing grade = D 7.Select English Proficiency = IELTS, Enter score = 3.0 8.Enter NETSAT Mathematics = 10 9.Enter NETSAT Physics = 10 10.Enter NETSAT English = 10 11.Enter TPAT3 = 10 12.Enter A-Level Mathematics = 10 13.Enter A-Level Physics = 10 14.Enter A-Level English = 10 15. In GCE A-Level Group A: check Mathematics, enter grade = D; check Physics, enter grade = D 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = D 17.Select Academic Achievements = No 18.Click "Check Eligibility"	Chem, Physics, English) = 3.24 ✗ (fail Round 1 minimum 3.25, diff = -0.01) Combined GPA (Math, Chem, Physics) = 2.74 ✗ (fail minimum 2.75, diff = -0.01) EN001203 Computer Programmin g = C ✗ (fail minimum B, need 1 grade higher) EN001202 Engineering Drawing = C ✗ (fail minimum B, need 1 grade higher) English Proficiency = IELTS, Score = 4.9 ✗ (fail minimum 5.0, diff = -0.1) NETSAT Mathematics = 29 ✗ (fail minimum 30, diff = -1) NETSAT Physics = 29 ✗ (fail minimum 30, diff = -1) NETSAT English = 29 ✗ (fail minimum 30, diff = -1) TPAT3 = 29	values shown for every failed condition, suggestions shown for all rounds	-All rounds show Not Eligible with full condition breakdown -Suggestion section visible under every round	Not Eligible status, difference values and suggestions displayed for all rounds	
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					✗ (fail minimum 30, diff = -1) A-Level Mathematics = 29 ✗ (fail minimum 30, diff = -1) A-Level Physics = 29 ✗ (fail minimum 30, diff = -1) A-Level English = 29 ✗ (fail minimum 30, diff = -1) GCE Group A: Mathematics = D, Physics = D ✗ (fail minimum C, need 1 grade higher) GCE Group B: Digital Media and Design = D ✗ (fail minimum C, need 1 grade higher) Academic Achievements = No ✗ Domicile = Patumthani ✗ (not Northeast region)				
TS0004: Verify that the system displays all rounds as Not Eligible with suggestions when no score criteria are	TC02	Verify that the system displays a "Not Eligible" result when the scores are below the required criteria by exactly 1	User is on Score Entry page, Browser is open	1.Navigate to the Admission page and click "Calculate Now" 2.Enter GPAX (4 Semesters) = 3.24 (1 below Round 1 minimum 3.25) 3.Enter Combined GPA (Math, Chem, Physics, English) = 3.24 (1 below	GPAX (4 Semesters) = 2.00 ✗ (fail all rounds, diff = -1.25) Combined GPA (Math, Chem, Physics, English) = 2.00 ✗ (fail all rounds,	System displays "You Are Not Eligible Yet" message, all rounds show Not Eligible status, difference values show -1 for every condition,	-Eligibility Result page displays "You Are Not Eligible Yet" banner -All rounds show Not	System displayed "You Are Not Eligible Yet" message, all rounds showed Not Eligible, difference	Pass

met		point (boundary case).		Round 1 minimum 3.25) 4.Enter Combined GPA (Math, Chem, Physics) = 2.74 (1 below minimum 2.75) 5.Enter EN001203 Computer Programming grade = C (1 grade below minimum B) 6.Enter EN001202 Engineering Drawing grade = C (1 grade below minimum B) 7.Select English Proficiency = IELTS, Enter score = 4.9 (0.1 below minimum 5.0) 8.Enter NETSAT Mathematics = 29 (1 below minimum 30) 9.Enter NETSAT Physics = 29 (1 below minimum 30) 10.Enter NETSAT English = 29 (1 below minimum 30) 11.Enter TPAT3 = 29 (1 below minimum 30) 12.Enter A-Level Mathematics = 29 (1 below minimum 30) 13.Enter A-Level Physics = 29 (1 below minimum 30) 14.Enter A-Level English = 29 (1 below minimum 30) 15. In GCE A-Level Group A: check Mathematics, enter grade = D; check Physics, enter	diff = -1.25) Combined GPA (Math, Chem, Physics) = 2.00 ✗ (fail minimum 2.75, diff = -0.75) EN001203 Computer Programming = D ✗ (fail minimum B, need 2 grades higher) EN001202 Engineering Drawing = D ✗ (fail minimum B, need 2 grades higher) English Proficiency = IELTS, Score = 3.0 ✗ (fail minimum 5.0, diff = -2.0) NETSAT Mathematics = 10 ✗ (fail minimum 30, diff = -20) NETSAT Physics = 10 ✗ (fail minimum 30, diff = -20) NETSAT English = 10 ✗ (fail minimum 30, diff = -20) TPAT3 = 10 ✗ (fail minimum	suggestions shown for all rounds	Eligible -Difference column shows -1 (or equivalent) for every condition -Suggestion section visible under every round	values showed -1 for every condition, suggestions displayed	
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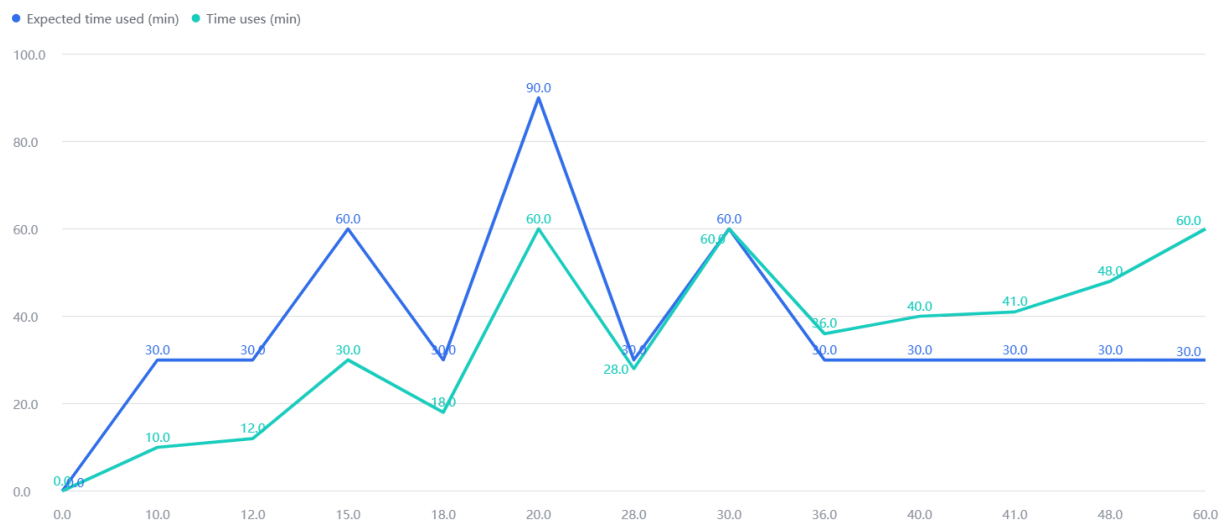
				grade = D (1 grade below minimum C) 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = D (1 grade below minimum C) 17. Select Academic Achievements = No 18. Click "Check Eligibility"	30, diff = -20) A-Level Mathematics = 10 ✗ (fail minimum 30, diff = -20) A-Level Physics = 10 ✗ (fail minimum 30, diff = -20) A-Level English = 10 ✗ (fail minimum 30, diff = -20) GCE Group A: Mathematics = D, Physics = D ✗ (fail minimum C) GCE Group B: Digital Media and Design = D ✗ (fail minimum C) Academic Achievements = No ✗ Domicile = Patumthani ✗ (not Northeast region)				
TS0004: Verify that the system displays all rounds as Not Eligible with suggestions when no score criteria are met	TC03	Verify that the system displays suggestions for every round when all criteria are not met.	User is on Score Entry page, Browser is open	1. Navigate to the Admission page and click "Calculate Now" 2. Enter GPAX (4 Semesters) = 2.00 3. Enter Combined GPA (Math, Chem, Physics, English) = 2.00 4. Enter Combined GPA (Math, Chem, Physics) = 2.00 5. Enter EN001203	GPAX (4 Semesters) = 2.00 ✗ (fail all rounds, diff = -1.25) Combined GPA (Math, Chem, Physics, English) = 2.00 ✗ (fail all rounds, diff = -1.25) Combined	Suggestion section appears under every round, each suggestion lists all failed conditions with specific points/grades needed to meet each requirement	-Eligibility Result page displays Not Eligible for all rounds -Suggestion section present and visible under Round 1,	Suggestion section appeared under every round, each suggestion listed all failed conditions with specific improveme	Pass

				Computer Programming grade = D 6.Enter EN001202 Engineering Drawing grade = D 7.Select English Proficiency = IELTS, Enter score = 3.0 8.Enter NETSAT Mathematics = 10 9.Enter NETSAT Physics = 10 10.Enter NETSAT English = 10 11.Enter TPAT3 = 10 12.Enter A-Level Mathematics = 10 13.Enter A-Level Physics = 10 14.Enter A-Level English = 10 15. In GCE A-Level Group A: check Mathematics, enter grade = D; check Physics, enter grade = D 16. In GCE A-Level Group B: check Digital Media and Design, enter grade = D 17.Select Academic Achievements = No 18.Click "Check Eligibility" 19.Scroll down to verify Suggestion section appears under Round 1 Portfolio 20.Verify Suggestion section appears under Round 2 Quota 21.Verify Suggestion section appears under Round 3	GPA (Math, Chem, Physics) = 2.00 ✗ (fail minimum 2.75, diff = -0.75) EN001203 Computer Programming = D ✗ (fail minimum B, need 2 grades higher) EN001202 Engineering Drawing = D ✗ (fail minimum B, need 2 grades higher) English Proficiency = IELTS, Score = 3.0 ✗ (fail minimum 5.0, diff = -2.0) NETSAT Mathematics = 10 ✗ (fail minimum 30, diff = -20) NETSAT Physics = 10 ✗ (fail minimum 30, diff = -20) NETSAT English = 10 ✗ (fail minimum 30, diff = -20) TPAT3 = 10 ✗ (fail minimum 30, diff = -20)		Round 2, and Round 3 -Each suggestion details exact score improvements needed per failed condition	nt points required	
--	--	--	--	--	---	--	--	--------------------	--

				Admission 22. Verify each suggestion lists specific conditions with exact improvement points required	A-Level Mathematics = 10 ✗ (fail minimum 30, diff = -20) A-Level Physics = 10 ✗ (fail minimum 30, diff = -20) A-Level English = 10 ✗ (fail minimum 30, diff = -20) GCE Group A: Mathematics = D, Physics = D ✗ (fail minimum C) GCE Group B: Digital Media and Design = D ✗ (fail minimum C) Academic Achievements = No ✗ Domicile = Patumthani ✗ (not Northeast region)				
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5.3 Sprint Monitoring

5.3.1 Burndown Chart



Sprint 2 consisted of 13 tasks with a total expected time of 390 minutes. The team completed all tasks with an actual time of 406 minutes, finishing approximately 4% over the planned estimate. This sprint focused on refining the admission score checker UI based on Sprint 1 feedback — including redesigning the fill information page, adding error message handling, and completing pass/fail result pages. Most tasks were completed close to their estimates, however two tasks exceeded the planned time notably: the Fill information page (48 min vs 30 min estimated) and the ER diagram overall (60 min vs 30 min estimated), reflecting the additional complexity encountered when revising the design and data structure from Sprint 1.

5.4 Sprint 2 Evaluation

5.4.1 Sprint Review Discussion

During the Sprint 2 Review, the team demonstrated the refined version of the admission eligibility checker based on feedback received from Sprint 1. This sprint focused on improving usability, input validation, and overall user experience.

FeedBack	• The system requires students to calculate their total credits and grades manually before entering data. This may lead to errors and inconvenience. The interface should provide input fields for both credits and grades so that calculations can be performed automatically by the system. • GED applicants should be identified during the registration process. This would allow the system to hide irrelevant score fields and display only the inputs applicable to that curriculum, reducing user confusion. • The score input fields currently allow free-text entry, which increases the risk of incorrect or invalid data. These fields should be replaced with dropdown selections where appropriate to minimize input errors. • The overall UI layout appears too empty and lacks visual structure. The interface should be enhanced with better spacing, grouping, and visual elements to improve usability and user engagement. • Some parts of the Data Flow Diagram (DFD) do not fully reflect the updated system behavior and should be revised accordingly. • The test cases still appear broader than the scope defined in the user story. They should be refined to focus only on scenarios relevant to the eligibility checking feature.
Decision	• Add input fields for both credits and grades so that the system can automatically calculate results without requiring manual computation by users. • Modify the registration process to allow users to specify their curriculum type (e.g., GED), enabling the system to display only relevant input fields. • Replace free-text score inputs with dropdown menus where possible to reduce data entry errors. • Improve the UI layout by adding structured sections, visual hierarchy, and consistent design elements to make the interface more user-friendly. • Update the DFD to accurately reflect the revised system processes and data flow. • Revise the test cases to ensure they align strictly with the defined user story and remain within its scope.

5.4.2 Sprint Retrospective Discussion

What went well	● Improved Coordination: Team members communicated more effectively compared to Sprint 1. ● Clearer Task Assignments: Responsibilities were better defined, reducing confusion. ● Steady Progress: Most planned tasks were completed within the sprint timeframe. ● Better Understanding of Requirements: The team gained clearer insight into user needs and system functionality. ● Enhanced Work Quality: Deliverables showed improvement due to experience gained from the previous sprint.
What went poorly	● Integration Challenges: Combining individual components required additional time and adjustments. ● Design Inconsistencies: Some UI elements and system diagrams lacked uniformity. ● Minor Communication Gaps: Occasional misunderstandings still occurred during complex tasks. ● Rework Needed: Some completed work had to be revised based on stakeholder feedback.
What can we improve	● Standardize Design Guidelines: Establish consistent UI and modeling standards across the system. ● Earlier Integration Testing: Combine components sooner to identify issues early. ● Strengthen Team Communication: Encourage more frequent discussions for complex tasks. ● Allocate Time for Revisions: Plan buffer time for improvements after feedback.
What action will we take	● Define and document design standards for UI and system models. ● Schedule intermediate integration checkpoints during development. ● Conduct regular short meetings to review progress and resolve issues. ● Reserve dedicated time at the end of the sprint for revisions and refinements.

Chapter 6 Sprint 3 Implementation

6.1 Sprint 3 Planning

6.1.1 Sprint Backlog (Lark)

☐ Task	position	Status	Task leader	Task start ti...	Task due d...	Expecte...	Task start ...	Task due d...	Time us...
▼ As a high school student...									
1 GED	backend	Done	Opal	02/25/2026	02/25/2026	30.0	2026/02/25	2026/02/25	15.0
2 GCE	backend	Done	Deena	02/25/2026	02/25/2026	30.0	2026/02/25	2026/02/25	15.0
3 Netsat	backend	Done	Aumaim	02/25/2026	02/25/2026	30.0	2026/02/25	2026/02/25	15.0
4 Alevel	backend	Done	Mo	02/25/2026	02/25/2026	30.0	2026/02/25	2026/02/25	15.0
5 English score ** คำขั้วมวักวัก 1 รั...	backend	Done	Orange	02/25/2026	02/25/2026	30.0	2026/02/25	2026/02/25	12.0
6 Grade high school + pre engineer...	backend	Done	Proud	02/25/2026	02/25/2026	30.0	2026/02/25	2026/02/25	30.0
7 Design Flow 1	backend	Done	Deena	02/26/2026	02/26/2026	30.0	2026/02/26	2026/02/26	30.0
8 Design Flow 2	backend	Done	Orange	02/26/2026	02/26/2026	30.0	2026/02/26	2026/02/26	30.0
9 Design Flow 3	backend	Done	Opal	02/26/2026	02/26/2026	30.0	2026/02/26	2026/02/26	30.0
10 Design Flow 4	backend	Done	Proud	02/26/2026	02/26/2026	30.0	2026/02/26	2026/02/26	30.0
11 Design Flow 5	backend	Done	Aumaim	02/26/2026	02/26/2026	30.0	2026/02/26	2026/02/26	30.0
12 Design Flow 6	backend	Done	Mo	02/26/2026	02/26/2026	30.0	2026/02/26	2026/02/26	30.0
13 Design UI : Fill score page(GED)	UX/UI	Done	Proud	02/27/2026	02/27/2026	30.0	2026/02/27	2026/02/27	25.0
14 Design UI : Fill score page(GCE)	UX/UI	Done	Mo	02/27/2026	02/27/2026	30.0	2026/02/27	2026/02/27	29.0
15 Design UI : Fill score page(Normal)	UX/UI	Done	Orange	02/27/2026	02/27/2026	30.0	2026/02/27	2026/02/27	25.0
16 Design UI : Show result(Round1)	UX/UI	Done	Opal	02/27/2026	02/27/2026	30.0	2026/02/27	2026/02/27	28.0
17 Design UI : Show result(Round2)	UX/UI	Done	Deena	02/27/2026	02/27/2026	30.0	2026/02/27	2026/02/27	24.0
18 Design UI : Show result(Round3)	UX/UI	Done	Aumaim	02/27/2026	02/27/2026	30.0	2026/02/27	2026/02/27	20.0
19 Design dfd	backend	Done	Aumaim	02/28/2026	02/28/2026	30.0	2026/02/28	2026/02/28	180.0
20 Design ER	backend	Done	Proud	03/28/2026	02/28/2026	30.0	2026/02/28	2026/02/28	30.0
21 > Test scenario ๓ 3	tester	Done	Deena	03/28/2026	02/28/2026	30.0	2026/02/28	2026/02/28	15.0
22 Test case	tester	Done	Opal	03/28/2026	02/28/2026	30.0	2026/02/28	2026/02/28	30.0
23 Test case	tester	Done	Mo	03/28/2026	02/28/2026	30.0	2026/02/28	2026/02/28	30.0
24 Fix figma	UX/UI	Done	Orange	03/28/2026	02/28/2026	30.0	2026/02/28	2026/02/28	22.0
25 Drop down page 3 + Add more s...	UX/UI	Done	Orange	03/01/2026	03/01/2026	30.0	2026/03/01	2026/03/01	26.0
26 Drop down page 2	UX/UI	Done	Opal	03/01/2026	03/01/2026	30.0	2026/03/01	2026/03/01	28.0
27 Drop down page 1	UX/UI	Done	Mo	03/01/2026	03/01/2026	30.0	2026/03/01	2026/03/01	25.0
28 Error message	UX/UI	Done	Deena	03/01/2026	03/01/2026	30.0	2026/03/01	2026/03/01	27.0
29 DFD Level 1	backend	Done	Aumaim	03/01/2026	03/01/2026	30.0	2026/03/02	2026/03/02	30.0
30 DFD Level 1	backend	Done	Proud	03/01/2026	03/01/2026	30.0	2026/03/01	2026/02/01	30.0
31 DFD Level 2	backend	Done	Proud	03/01/2026	03/01/2026	30.0	2026/03/02	2026/03/02	30.0
32 Show result (Not pass)	UX/UI	Done	Orange	03/02/2026	03/02/2026	30.0	2026/03/02	2026/03/02	22.0
33 DFD Level context	backend	Done	Aumaim	03/02/2026	03/02/2026	30.0	2026/03/02	2026/03/02	20.0

Sprint Backlog 3

6.1.2 Sprint Goal

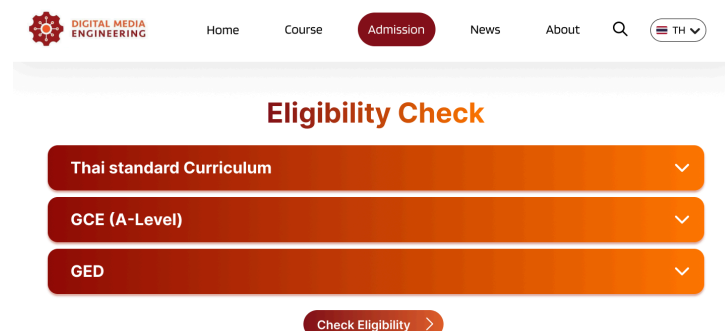
The goal of Sprint 3 is to complete and stabilize the admission score checker by resolving all remaining issues identified in Sprint 2 and preparing the feature for final delivery. This sprint focuses on edge case handling, performance consistency, and ensuring the calculator functions correctly across all supported devices and browsers.

By the end of this sprint, the feature will be fully tested, with all known bugs resolved and the interface polished to meet the project's visual design standards. Score calculation history will be saved to the user profile for logged-in users, and the feature will be ready for integration with the broader admission information system.

This sprint marks the completion of the score checker as a production-ready feature, closing out the core deliverable introduced in Sprint 1.

6.2 Sprint 3 Development (Increment)

6.2.1 User Interface (UI) Design



The screenshot shows a web application interface for an 'Eligibility Check'. At the top, there is a navigation bar with a logo on the left and links for 'Home', 'Course', 'Admission' (which is highlighted with a dark red background), 'News', 'About', a search icon, and a language selector set to 'TH'. Below the navigation bar, the main heading 'Eligibility Check' is centered in a bold, orange font. Underneath this heading, there are three stacked orange buttons with white text and downward-pointing chevrons, representing dropdown menus. The first button is labeled 'Thai standard Curriculum', the second is 'GCE (A-Level)', and the third is 'GED'. At the bottom of this section, there is a dark red button with the text 'Check Eligibility' followed by a white right-pointing arrow.

Picture 6.1 Main Page

DIGITAL MEDIA ENGINEERING Home Course **Admission** News About  

Eligibility Check

Thai standard Curriculum

- Grade**

Grade 10 semester 1

Physics

Chemistry

Grade 10 semester 2

Physics

Chemistry

Grade 11 semester 1

Physics

Chemistry

Grade 11 semester 2

Physics

Chemistry

GRADE (4 semesters)

Combine Mathematics Grade

Combine English Grade
- English Proficiency Score**
- Netsat**

Mathematics (100)

/100

Physics (204)

/100

English (1002)

/100
- A-LEVEL**

Mathematics

/100

Physics

/100

English

/100

T-PWT3

/100
- Pre-engineering**

EN 001 203 Computer Programming

EN 001 202 Engineering Drawing

GCE (A-Level)

GED

[Check Eligibility](#)

DIGITAL MEDIA ENGINEERING Home Course **Admission** News About  

Eligibility Check

Thai standard Curriculum

- Grade**

Grade 10 semester 1

Physics

Chemistry

Grade 10 semester 2

Physics

Chemistry

Grade 11 semester 1

Physics

Chemistry

Grade 11 semester 2

Physics

Chemistry

GRADE (4 semesters)

Combine Mathematics Grade

Combine English Grade
- English Proficiency Score**
- Netsat**

Mathematics (100)

/100

Physics (204)

/100

English (1002)

/100
- A-LEVEL**

Mathematics

/100

Physics

/100

English

/100

T-PWT3

/100
- Pre-engineering**

EN 001 203 Computer Programming

EN 001 202 Engineering Drawing

/100

GCE (A-Level)

GED

[Check Eligibility](#)

Picture 6.2 Thai standard Curriculum Eligibility Check Page

DIGITAL MEDIA ENGINEERING Home Course **Admission** News About  

Eligibility Check

Thai standard Curriculum

GCE (A-Level)

- Subjects**

Group A : Select 2 subjects from Group A.

Group B : Select 1 subject from Group B.
- English Proficiency Score**
- A-LEVEL**

Mathematics

/100

Physics

/100

English

/100

T-PWT3

/100
- Pre-engineering**

EN 001 203 Computer Programming

EN 001 202 Engineering Drawing

GED

[Check Eligibility](#)

DIGITAL MEDIA ENGINEERING Home Course **Admission** News About  

Eligibility Check

Thai standard Curriculum

GCE (A-Level)

- Subjects**

Group A : Select 2 subjects from Group A.

Group B : Select 1 subject from Group B.
- English Proficiency Score**
- A-LEVEL**

Mathematics

/100

Physics

/100

English

/100

T-PWT3

/100
- Pre-engineering**

EN 001 203 Computer Programming

/100

EN 001 202 Engineering Drawing

GED

[Check Eligibility](#)

Picture 6.3 GCE (A-Level) Eligibility Check Page

Eligibility Check

Thai standard Curriculum

GCE (A-Level)

GED

• GED Score

Reasoning Through Language Arts (RLA)

Social Studies

Mathematical Reasoning

Science

• English Proficiency Score

Select your English test Enter your Score

• A-LEVEL

Mathematics Physics English TM023

Enter your Score Enter your Score Enter your Score Enter your Score

• Pre-engineering

EN 001 203 Computer Programming

EN 001 203 Engineering Drawing

Select your grade Select your grade

[Check Eligibility](#)

Picture 6.4 GED Eligibility Check Page

Eligibility Result

Student Information

Full Name: Sompong Meechal Province: Khon Kean

School: Khonkaen Wittayayon School Grade: 12

• Thai standard Curriculum

Round 1 - Portfolio [Eligible for 2 Programs](#)

High-Potential School Student Admission Program [Eligible](#)

Condition	Requirement	Your Score	Status
Curriculum	Thai curriculum	Thai curriculum	✓
Partner School	20 schools ID	Khonkaen Wittayayon School	✓
GPA (3 semester)	≥ 2.5	3.82	✓
GPA - Math	≥ 2.5	3.50	✓
GPA - Chem	≥ 2.5	3.50	✓
GPA - Physics	≥ 2.5	3.50	✓
GPA - English	≥ 2.5	3.50	✓
EN00100 Computer Programming	≥ 8	8+	✓
EN00200 Engineering Drawing	≥ 8	8	✓
IELTS	≥ 5	5.5	✓
Academic Achievement	Science / Math / CI activities at practical level or above	Yes	✓

High-Potential Student Program [Eligible](#)

Condition	Requirement	Your Score	Status
Curriculum	Thai International (IB) / A-level	Thai curriculum	✓
GPA (3 Semester)	≥ 2.75	3.82	✓
GPA - Math	≥ 2.75	3.50	✓
GPA - Chemistry	≥ 2.75	3.50	✓
GPA - Physics	≥ 2.75	3.50	✓
IELTS	≥ 5	5.5	✓
Academic Achievement	Science / Math / CI activities at practical level or above	Yes	✓

Round 2 - Quota [Eligible](#)

Round 3 - Admission [Eligible](#)

• Thai standard Curriculum

Round 1 - Portfolio [Eligible for 2 Programs](#)

Round 2 - Quota [Eligible](#)

Round 3 - Admission [Eligible](#)

Condition	Requirement	Your Score	Status
TM023	≥ 50	58	✓
A-Level Mathematics 1	≥ 50	50	✓
A-Level Physics	≥ 50	60	✓
A-Level English	≥ 50	79	✓

Picture 6.5 Show All Pass result Round 1 Page, Show All Pass result Round 2 Page, Show All Pass result Round 3 Page


**DIGITAL MEDIA
ENGINEERING**

Home
Course
Admission
News
About




TH
VN

Eligibility Result

Student Information

Full Name

Sombat Chareonphanitch

Province

Patumthani

School

Patumthani Vocational Education College

Grade

4

Thai standard Curriculum

You Are Not Eligible Yet

You are currently below the required criteria for all admission rounds.
See how much you are missing in each round and explore possible next steps below.

Round 1 - Portfolio

High-Potential School Student Admission Program

Not Eligible Yet

Condition	Requirement	Your Score	Status
Curriculum	Thai curriculum	Vocational Curriculum	✗
Partner School	29 schools	Patumthani Vocational Education College	✗
GPAX (4 Semesters)	≥ 2.25	3.15 < 3.1 grade required	✗
GPA : Math	≥ 2.25	3.00 < 3.0 grade required	✗
GPA : Chem	≥ 2.25	3.17 < 3.08 grade required	✗
GPA : Physics	≥ 2.25	3.16 < 3.07 grade required	✗
GPA : English	≥ 2.25	3.15 < 3.1 grade required	✗
EN001203 Computer Programming	≥ B	C < 1 grade higher	✗
EN001202 Engineering Drawing	≥ B	C < 1 grade higher	✗
IELTS	≥ 5	4.5 < 5.0 band required	✗
Academic Achievement	Science / Math / CS activities at provincial level or above	-	✗

High-Potential Student Program

Not Eligible Yet

Round 2 - Quota

Not Eligible Yet

Round 3 - Admission

Not Eligible Yet


**DIGITAL MEDIA
ENGINEERING**

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TH
VN

Eligibility Result

Student Information

Full Name

Sombat Chareonphanitch

Province

Patumthani

School

Patumthani Vocational Education College

Grade

4

Thai standard Curriculum

You Are Not Eligible Yet

You are currently below the required criteria for all admission rounds.
See how much you are missing in each round and explore possible next steps below.

Round 1 - Portfolio

High-Potential School Student Admission Program

Not Eligible Yet

Condition	Requirement	Your Score	Status
Curriculum	Thai / International / CED (age ≥ 14)	Vocational Curriculum	✗
GPAX (4 Semesters)	≥ 2.75	2.55 < 2.25 grade required	✗
GPA : Math	≥ 2.75	2.55 < 2.2 grade required	✗
GPA : Chem	≥ 2.75	2.5 < 2.25 grade required	✗
GPA : Physics	≥ 2.75	2.7 < 2.50 grade required	✗
IELTS	≥ 5	4.5 < 5.0 band required	✗
Academic Achievement	Science / Math / CS activities at provincial level or above, or other achievements per attached document	-	✗

High-Potential Student Program

Not Eligible Yet

Round 2 - Quota

Not Eligible Yet

Round 3 - Admission

Not Eligible Yet



DIGITAL MEDIA ENGINEERING

Home Course Admission News About



DIGITAL MEDIA ENGINEERING

Home Course Admission News About

Eligibility Result

Student Information

Full Name	Province
Sombat Chareonphanthich	Patumthani
School	Grade
Patumthani Vocational Education College	4

⚠ You Are Not Eligible Yet

You are currently below the required criteria for all admission rounds. See how much you are missing in each round and explore possible next steps below.

Round 1 - Portfolio

⚠ Not Eligible for 2 Programs Yet

Round 2 - Quota

⚠ Not Eligible Yet

Round 3 - Admission

⚠ Not Eligible Yet

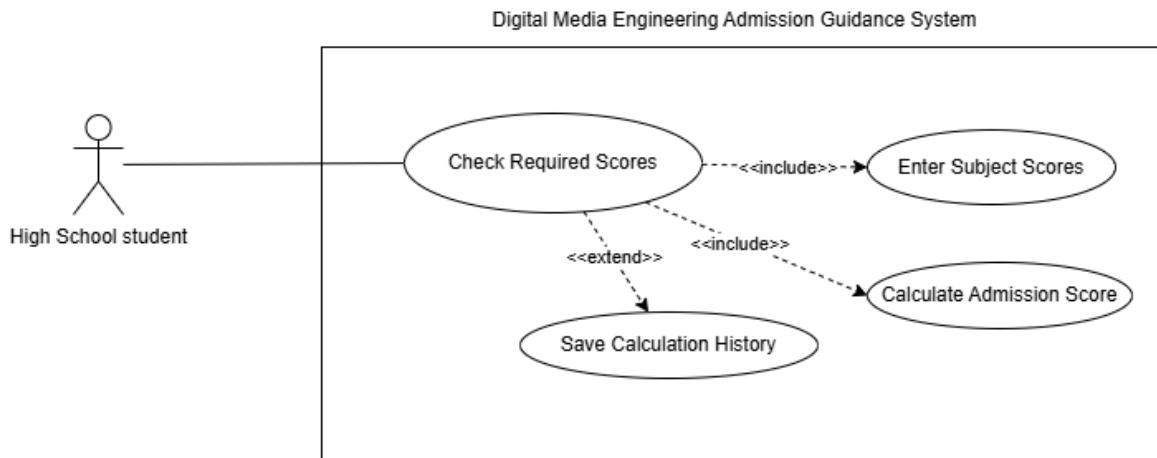
Condition	Requirement	Your Score	Status
Average score @ (INTAK 1 - INTAK 2)	>30	20 <10 points required	✗
Domestic	Household registration in the Northeast region > 3 years	Patumthani	✗

Round 3 - Admission

⚠ Not Eligible Yet

Picture 6.6 Show All Fail result Round 1 Page, Show All Fail result Round 2 Page, Show All Fail result Round 3 Page

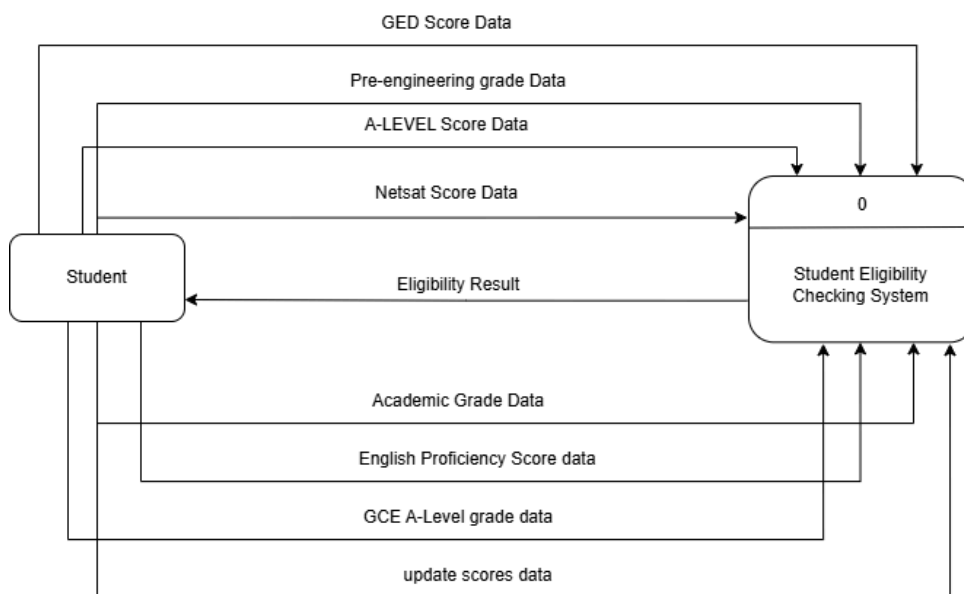
6.2.2 Use Case Diagram



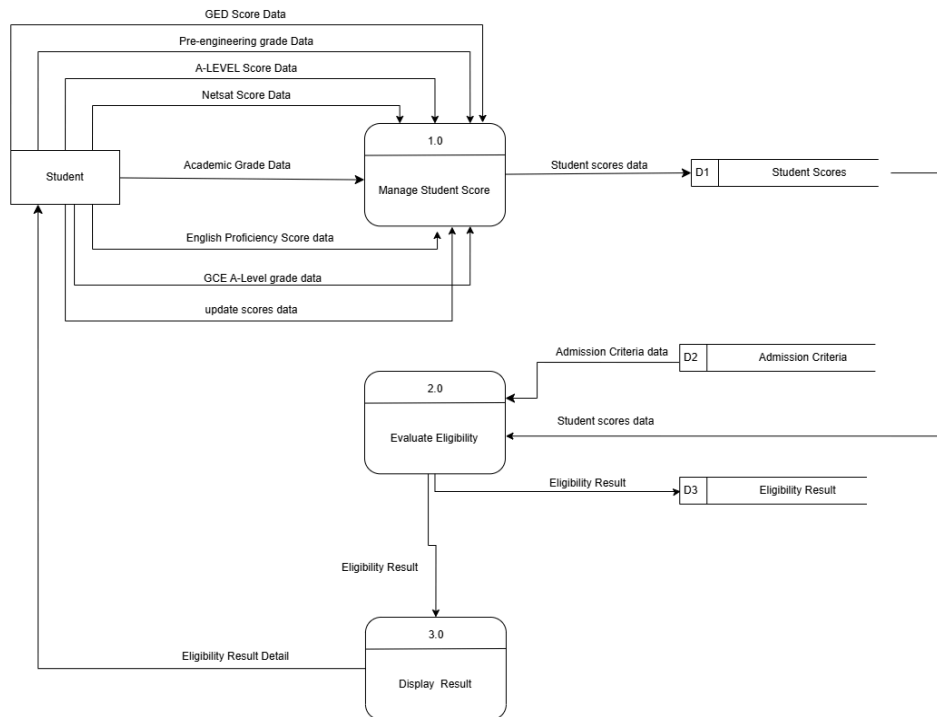
This use case diagram represents the process of checking admission score requirements in the Digital Media Engineering Admission Guidance System. A high school student can enter subject scores and calculate the admission score to determine whether their results meet the admission criteria. The system also provides an optional feature to save the calculation history.

6.2.3 Data Flow Diagram (DFD)

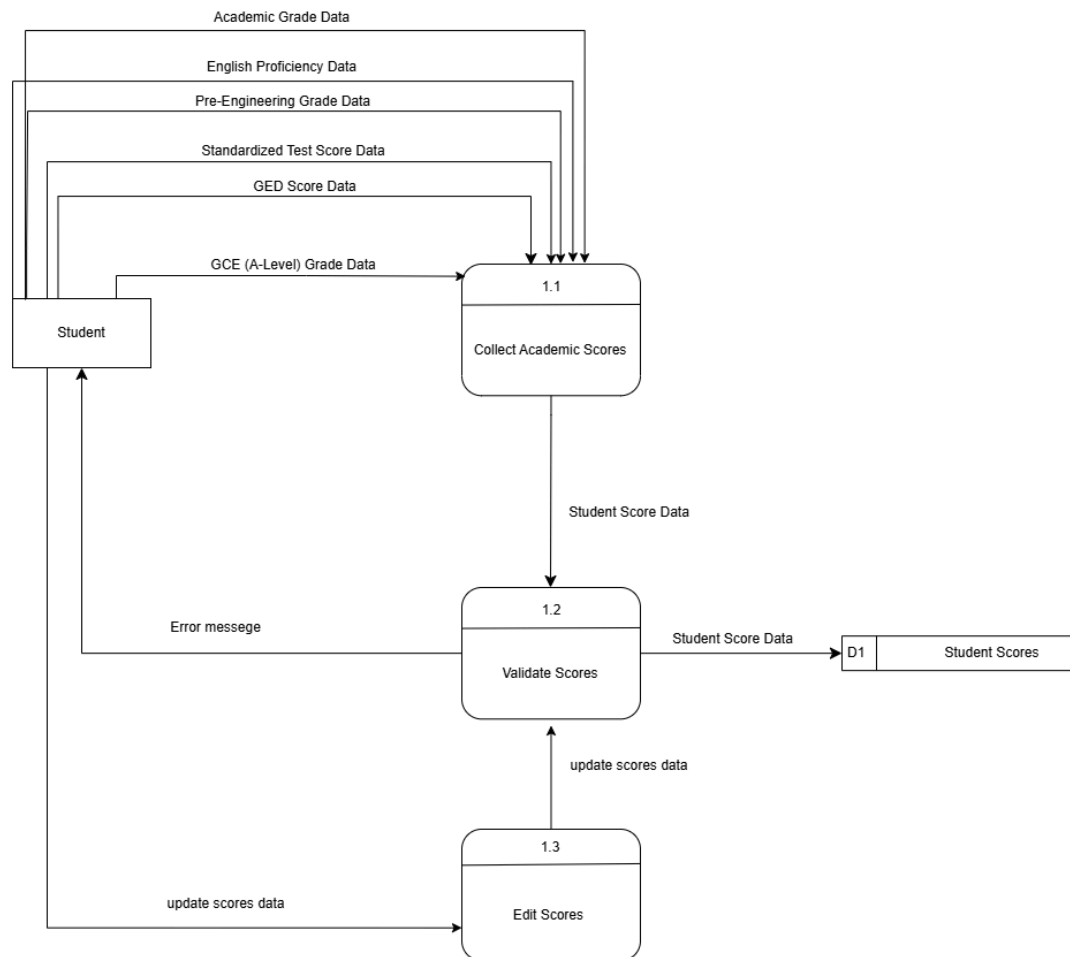
- Context DFD



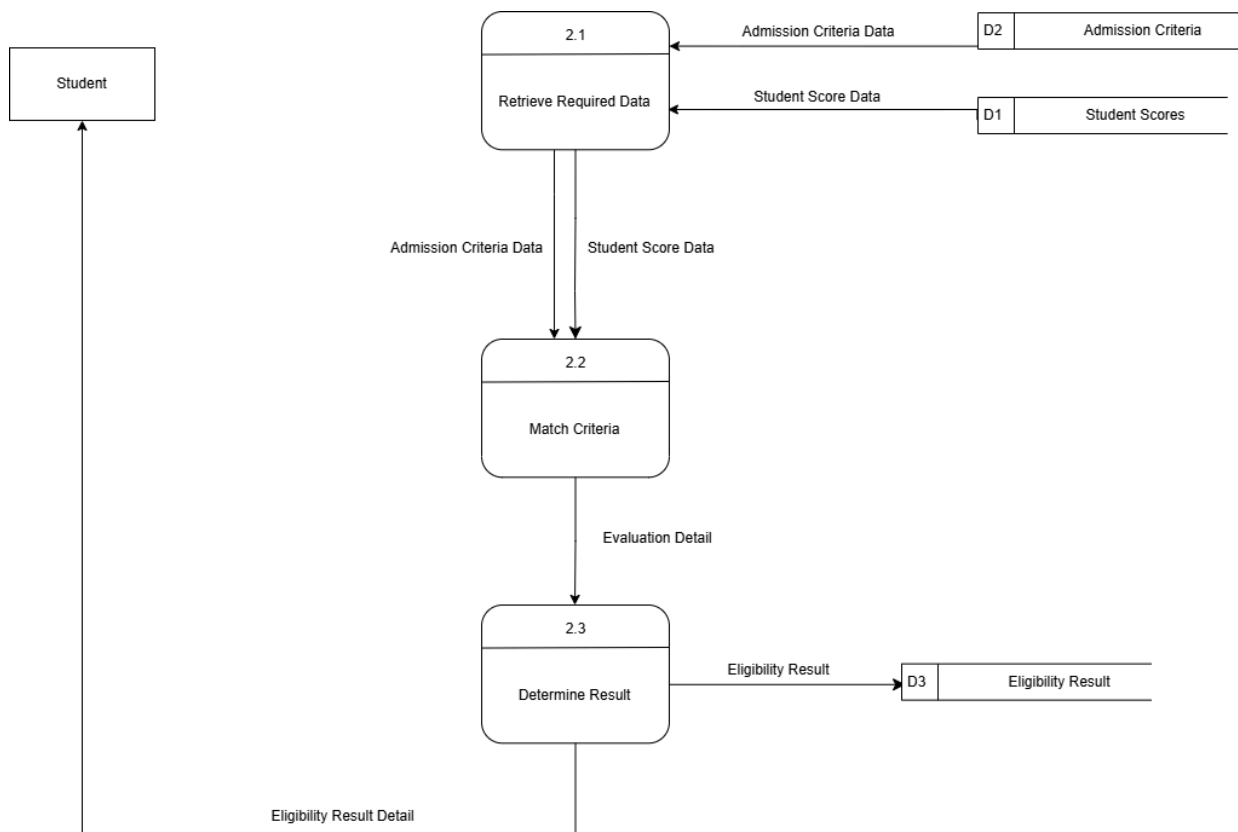
- DFD Level 1



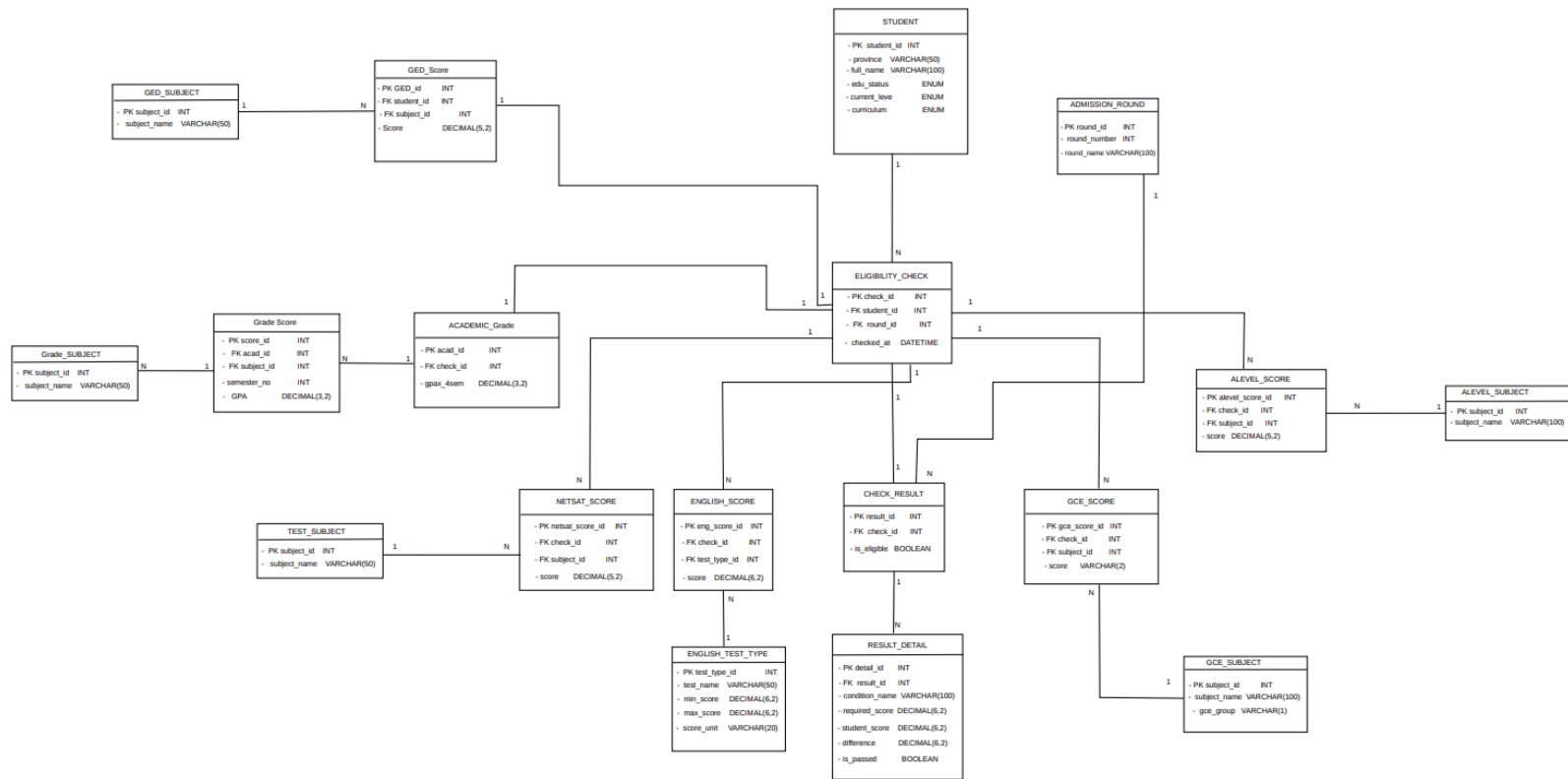
- DFD Level 2
 - DFD Level 2 1.0



- DFD Level 2 2.0



6.2.4 Entity-Relationship Diagram (ER)



6.2.5 Test Cases

6.2.5.1 Test Scenarios

User story ID	Test scenario ID	Test scenario	Number of test cases	Priority
US0003	TS0001	Verify that user can fill in the information page with valid inputs	3	High
US0003	TS0002	Verify that the system handles invalid inputs on the information page correctly	13	High
US0003	TS0003	Verify that the system displays eligibility result correctly after processing	16	High

6.2.5.2 Test Cases

Test Scenario	Test Case ID	Test Case	Pre-Condition	Test Step	Test Data	Expect Result	Post Condition	Actual Result	Status (Pass/Fail)
TS0001: Verify that user can fill in the information page with valid inputs	TC01	Verify that the system displays eligibility results when all Thai Standard Curriculum inputs are valid	User is on Eligibility Check page	1.Open Thai Standard Curriculum section 2.Select Grade and Credits for all Physics and Chemistry 3.Enter GPAX 4.Enter Combined Mathematics Grade 5.Enter Combined English Grade 6.Select English Proficiency test and enter score 7.Enter Netsat scores 8.Enter A-LEVEL scores 9.Select Pre-engineering grades 10.Click Check Eligibility button	1.Grade = A, Credit = 1.0 2.GPAX = 4.00 3.Combine d Mathematics Grade = 3.50 4.Combine d English Grade = 3.50 5.English Proficiency Test = IELTS, Score = 5.5 6.Netsat Mathematics = 80 7.Netsat Physics = 75 8.Netsat English = 70 9.A-LEVEL Mathematics = 80 10.A-LEVEL Physics = 75 11.A-LEVEL English = 70 12.A-LEVEL TPAT3 = 60 13.EN001203 Computer Programming = B+ 14.EN001202	All fields accept valid inputs correctly and the system redirects to the Eligibility Result page	The system successfully displays the Eligibility Result page	All fields accept valid inputs correctly and the system displays the Eligibility Result page	Pass

					Engineering Drawing = B				
TS0001: Verify that user can fill in the information page with valid inputs	TC02	Verify that the system displays eligibility results when all GCE (A-Level) inputs are valid	User is on Eligibility Check page	1.Open GCE (A-Level) section 2.Select 2 subjects and grades for Group A 3.Select 1 subject and grade for Group B 4.Select English Proficiency test and enter score 5.Enter A-LEVEL scores 6.Select Pre-engineering grades 7.Click Check Eligibility	1.Group A Subject 1 = Mathematics, Grade = A 2.Group A Subject 2 = Physics, Grade = A 3.Group B Subject = Statistics, Grade = A 4.English Proficiency Test = IELTS, Scores = 5.5 5.A-LEVEL Mathematics = 80 6.A-LEVEL Physics = 75 7.A-LEVEL English = 70 8.A-LEVEL TPAT3 = 60 9.EN001203 Computer Programming = B+ 10.EN001202 Engineering Drawing = B	All fields accept valid inputs correctly and the system redirects to the Eligibility Result page	The system successfully displays the Eligibility Result page	All fields accept valid inputs correctly and the system displays the Eligibility Result page	Pass
TS0001: Verify that user can fill in the information page with valid inputs	TC03	Verify that the system displays eligibility results when GED score input is valid	User is on Eligibility Check page	1.Open GED section 2.Enter GED Reasoning Through Language	1.GED Reasoning Through Language Arts (RLA) = 170 2.GED Social	All fields accept valid inputs correctly and the system redirects to the	The system successfully displays the Eligibility Result page	All fields accept valid inputs correctly and the system displays the Eligibility	Pass

				Arts (RLA) score 3.Enter GED Social Studies score 4.Enter GED Mathematical Reasoning score 5.Enter GED Science score 6.Select English Proficiency test and enter score 7.Enter A-LEVEL scores 8.Select Pre-engineering grades 9.Click Check Eligibility	Studies = 165 3.GED Mathematical Reasoning = 175 4.GED Science = 160 5.English Proficiency Test = IELTS, Scores = 5.5 6.A-LEVEL Mathematics = 80 7.A-LEVEL Physics = 75 8.A-LEVEL English = 70 9.A-LEVEL TPAT3 = 60 10.EN001203 Computer Programming = B+ 11.EN001202 Engineering Drawing = B	Eligibility Result page		Result page	
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC01	Verify that the system shows an error when grade is selected but credit is not selected	User is on Eligibility Check page	1.Open Thai Standard Curriculum section 2.Select grade for Physics Grade 3.Leave credit field empty 4.Enter GPAX 5.Enter Combined	1.Grade = 3.5, Credits = Not selected 2.GPAX = 3.82 3.Combine d Mathematics Grade = 3.50 4.Combine d English Grade = 3.50	1.System shows error "Please select a credits" only on Chemistry Credits Grade 2.All other fields show no error	System remains on Eligibility Check page	System shows error "Please select a credits" only on Credits Grade	Pass

				Mathematic s Grade 6.Enter Combined English Grade 7.Select English Proficiency test and enter score 8.Enter Netsat scores 9.Enter A-LEVEL scores 10.Select Pre-enginee ring grades 11.Click Check Eligibility button	5.English Proficiency Test = IELTS, Score = 5.5 6.Netsat Mathemati cs = 80 7.Netsat Physics = 75 8.Netsat English = 70 9.A-LEVE L Mathemati cs = 80 10.A-LEV EL Physics = 75 11.A-LEV EL English = 70 12.A-LEV EL TPAT3 = 60 13.EN0012 03 Computer Programmi ng = B+ 14.EN0012 02 Engineerin g Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC02	Verify that the system shows an error when credit is selected but grade is not selected	User is on Eligibility Check page	1.Open Thai Standard Curriculum section 2.Leave grade field empty 3.Select grade for Chemistry Credit 4.Enter GPAX 5.Enter Combined Mathematic s Grade	1.Grade = Not select, Credits = 1.5 2.GPAX = 3.82 3.Combine d Mathemati cs Grade = 3.50 4.Combine d English Grade = 3.50 5.English Proficiency	1.System shows error "Please select a grade" only on Physics Grade 2.All other fields show no error	System remains on Eligibility Check page	System shows error "Please select a grade" only on Physics Grade	Pass

				6.Enter Combined English Grade 7.Select English Proficiency test and enter score 8.Enter Netsat scores 9.Enter A-LEVEL scores 10.Select Pre-engineering grades 11.Click Check Eligibility button	Test = IELTS, Score = 5.5 6.Netsat Mathematics = 80 7.Netsat Physics = 75 8.Netsat English = 70 9.A-LEVEL Mathematics = 80 10.A-LEVEL Physics = 75 11.A-LEVEL English = 70 12.A-LEVEL TPAT3 = 60 13.EN001203 Computer Programming = B+ 14.EN001202 Engineering Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC03	Verify that the system shows an error when English subject is selected but score is not entered	User is on Eligibility Check page	1.Open Thai Standard Curriculum section 2.Select grade 3.Select credit 4.Enter GPAX 5.Enter Combined Mathematics Grade 6.Enter Combined English Grade 7.Select	1.Grade = 3.5, Credits = 1.5 2.GPAX = 3.82 3.Combine d Mathematics Grade = 3.50 4.Combine d English Grade = 3.50 5.English Proficiency Test = IELTS, Score =	System shows error "Please enter your score" only on English Proficiency Score field. All other fields show no error	System remains on Eligibility Check page	System shows error "Please enter your score" only on English Proficiency Score field	Pass

				English Proficiency test 8.Leave English Proficiency Score empty 9.Enter Netsat scores 10.Enter A-LEVEL scores 11.Select Pre-engineering grades 12.Click Check Eligibility button	empty 6.Netsat Mathematics = 80 7.Netsat Physics = 75 8.Netsat English = 70 9.A-LEVEL Mathematics = 80 10.A-LEVEL Physics = 75 11.A-LEVEL English = 70 12.A-LEVEL TPAT3 = 60 13.EN001203 Computer Programming = B+ 14.EN001202 Engineering Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC04	Verify that the system shows an error when score is entered but English subject is not selected	User is on Eligibility Check page	1.Open Thai Standard Curriculum section 2.Select grade 3.Select credit 4.Enter GPAX 5.Enter Combined Mathematics Grade 6.Enter Combined English Grade 7.Leave English Proficiency	1.Grade = 3.5, Credits = 1.5 2.GPAX = 3.82 3.Combine d Mathematics Grade = 3.50 4.Combine d English Grade = 3.50 5.English Proficiency Test = empty, Score = 5.5 6.Netsat Mathemati	System shows error on English Proficiency Test field that was not selected. All other fields show no error	System remains on Eligibility Check page	System does not show any error on English Proficiency Test field when score is entered but test is not selected	Fail

				test empty 8.Enter English Proficiency Score 9.Enter Netsat scores 10.Enter A-LEVEL scores 11.Select Pre-enginee ring grades 12.Click Check Eligibility button	cs = 80 7.Netsat Physics = 75 8.Netsat English = 70 9.A-LEVE L Mathemati cs = 80 10.A-LEV EL Physics = 75 11.A-LEV EL English = 70 12.A-LEV EL TPAT3 = 60 13.EN0012 03 Computer Programmi ng = B+ 14.EN0012 02 Engineerin g Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC05	Verify that the system shows an error when score is entered as text	User is on Eligibility Check page	1.Open GED section 2.Enter GED Reasoning Through Language Arts (RLA) score 3.Enter GED Social Studies score 4.Enter GED Mathematic al Reasoning score 5.Enter GED Science score wrong	1.GED Reasoning Through Language Arts (RLA) = 170 2.GED Social Studies = 165 3.GED Mathemati cal Reasoning = 175 4.GED Science = A 5.English Proficiency Test = IELTS, Scores = 5.5	System shows error "Invalid format. Please enter a numeric value." only on GED Science field. All other fields show no error	System remains on Eligibility Check page	System shows error "Invalid format. Please enter a numeric value." only on GED Science field	Pass

				format 6.Select English Proficiency test and enter score 7.Enter A-LEVEL scores 8.Select Pre-engineering grades 9.Click Check Eligibility	6.A-LEVEL Mathematics = 80 7.A-LEVEL Physics = 75 8.A-LEVEL English = 70 9.A-LEVEL TPAT3 = 60 10.EN001203 Computer Programming = B+ 11.EN001202 Engineering Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC06	Verify that the system shows an error when score is outside the allowed range	User is on Eligibility Check page	1.Open Thai Standard Curriculum section 2.Select grade 3.Select credit 4.Enter GPAX 5.Enter Combined Mathematics Grade 6.Enter Combined English Grade 7.Select English Proficiency test and enter score 8.Enter Netsat scores 9.Enter A-LEVEL scores outside the allowed	1.Grade = 3.5, Credits = 1.5 2.GPAX = 3.82 3.Combined Mathematics Grade = 3.50 4.Combined English Grade = 3.50 5.English Proficiency Test = IELTS, Score = 5.5 6.Netsat Mathematics = 80 7.Netsat Physics = 75 8.Netsat English = 70 9.A-LEVEL Mathematics	System shows error "This score exceeds the maximum value. Please enter a score between 0 and 100." on A-LEVEL Mathematics field only. and System shows error "This score less than the minimum value. Please enter a score between 0 and 100." on A-LEVEL Physics field only.	System remains on Eligibility Check page	System shows error "This score exceeds the maximum value. Please enter a score between 0 and 100." only on A-LEVEL Mathematics field. and and System shows error "This score less than the minimum value. Please enter a score between 0 and 100." only on A-LEVEL Physics field.	Pass

				range 10.Select Pre-enginee ring grades 11.Click Check Eligibility button	cs = 1008 10.A-LEV EL Physics = -75 11.A-LEV EL English = 70 12.A-LEV EL TPAT3 = 60 13.EN0012 03 Computer Programmi ng = B+ 14.EN0012 02 Engineerin g Drawing = B	All other fields show no error.			
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC07	Verify that the system shows an error when GPAX is outside the allowed range (0–4)	User is on Eligibility Check page	1.Open Thai Standard Curriculum section 2.Select grade 3.Select credit 4.Enter GPAX outside the allowed range 5.Enter Combined Mathematic s Grade 6.Enter Combined English Grade 7.Select English Proficiency test and enter score 8.Enter Netsat scores 9.Enter A-LEVEL scores 10.Select Pre-enginee	1.Grade = 3.5, Credits = 1.5 2.GPAX = 4.5 3.Combine d Mathemati cs Grade = 3.50 4.Combine d English Grade = 3.50 5.English Proficiency Test = IELTS, Score = 5.5 6.Netsat Mathemati cs = 80 7.Netsat Physics = 75 8.Netsat English = 70 9.A-LEVE L Mathemati cs = 80 10.A-LEV EL Physics	System shows error "GPAX must be between 0 and 4" on GPAX field only.	System remains on Eligibility Check page	System shows error "GPAX must be between 0 and 4" on GPAX field.	Pass

				ring grades 11.Click Check Eligibility button	= 75 11.A-LEV EL English = 70 12.A-LEV EL TPAT3 = 60 13.EN0012 03 Computer Programmi ng = B+ 14.EN0012 02 Engineerin g Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC08	Verify that the system shows an error when Group A/B subject is selected but grade is not selected	User is on Eligibility Check page	1.Open GCE (A-Level) section Select Group A subject 2.Leave grade field empty 3.Open GCE (A-Level) section Select Group B subject 4.Leave grade field empty 5.Enter English Proficiency test and score 6.Enter other A-LEVEL scores 7.Select Pre-enginee ring grades 8.Click Check Eligibility button	1.Subject (Group A) = Math 2.Grade (Group A) = empty 3.Subject (Group A) = Chemistry 4.Grade (Group A) = empty 5.Subject (Group B) = Economics 6.Grade (Group B) = empty 7.English Proficiency Test = IELTS, Score = 5.5 8.A-LEVE L Mathemati cs = 80 9.A-LEVE L Physics = 75 10.A-LEV EL English = 70 11.A-LEV EL TPAT3	System shows error "Please select your grade" on the subject grade field.	System remains on Eligibility Check page	System shows error "Please select your grade" on subject grade field.	Pass

					= 60 12.EN0012 03 Computer Programmi ng = B+ 13.EN0012 02 Engineerin g Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC09	Verify that the system shows an error when grade is selected but Group A/B subject is not selected	User is on Eligibility Check page	1.Open GCE (A-Level) section Leave Group A subject field empty 2.Select grade 3.Open GCE (A-Level) section Leave Group B subject field empty 4.Select grade 5.Enter English Proficiency test and score 6.Enter other A-LEVEL scores 7.Select Pre-enginee ring grades 8.Click Check Eligibility button	1.Subject (Group A) = empty 2.Grade (Group A) = B 3.Subject (Group A) = empty 4.Grade (Group A) = C 5.Subject (Group B) = empty 6.Grade (Group B) = A 7.English Proficiency Test = IELTS, Score = 5.5 8.A-LEVE L Mathemati cs = 80 9.A-LEVE L Physics = 75 10.A-LEV EL English = 70 11.A-LEV EL TPAT3 = 60 12.EN0012 03 Computer Programmi ng = B+ 13.EN0012	System shows error "Please select a subject" on subject field.	System remains on Eligibility Check page	System shows error "Please select a subject" on subject field.	Pass

					02 Engineerin g Drawing = B				
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC10	Verify that the system shows an error when score is entered as text	User is on Eligibility Check page	1.Open GCE (A-Level) section Select Group A subject 2.Select Grade 3.Open GCE (A-Level) section Select Group B subject 4.Select Grade 5.Enter English Proficiency test and score 6.Enter A-LEVEL scores as text 7.Select Pre-enginee ring grades 8.Click Check Eligibility button	1.Subject (Group A) = Math 2.Grade (Group A) = B 3.Subject (Group A) = Chemistry 4.Grade (Group A) = C 5.Subject (Group B) = Economics 6.Grade (Group B) = B+ 7.English Proficiency Test = IELTS, Score = 5.5 8.A-LEVE L Mathemati cs = ABC 9.A-LEVE L Physics = 70 10.A-LEV EL English = 65 11.A-LEV EL TPAT3 = 80 12.EN0012 03 Computer Programmi ng = B+ 13.EN0012 02 Engineerin g Drawing = B	System shows error "Invalid format. Please enter a numeric value." on A-LEVEL Mathematic s field only. All other fields show no error.	System remains on Eligibility Check page	System shows error "Invalid format. Please enter a numeric value." only on A-LEVEL Mathematic s field.	Pass

TS0002: Verify that the system handles invalid inputs on the information page correctly	TC11	Verify that the system shows an error when score is outside the allowed range	User is on Eligibility Check page	1.Open GCE (A-Level) section Select Group A subject 2.Select Grade 3.Open GCE (A-Level) section Select Group B subject 4.Select Grade 5.Enter English Proficiency test and score 6.Enter A-LEVEL scores outside range 7.Select Pre-engineering grades 8.Click Check Eligibility button	1.Subject (Group A) = Math 2.Grade (Group A) = B 3.Subject (Group A) = Chemistry 4.Grade (Group A) = C 5.Subject (Group B) = Economics 6.Grade (Group B) = B+ 7.English Proficiency Test = IELTS, Score = 5.5 8.A-LEVEL Mathematics = 1008 9.A-LEVEL Physics = -80 10.A-LEVEL English = 70 11.A-LEVEL TPAT3 = 85 12.EN001203 Computer Programming = B+ 13.EN001202 Engineering Drawing = B	System shows error "This score exceeds the maximum value. Please enter a score between 0 and 100." on A-LEVEL Mathematics field only. and System shows error "This score less than the minimum value. Please enter a score between 0 and 100." on A-LEVEL Physics field only. All other fields show no error.	System remains on Eligibility Check page	System shows error "This score exceeds the maximum value. Please enter a score between 0 and 100." only on A-LEVEL Mathematics field. and System shows error "This score less than the minimum value. Please enter a score between 0 and 100." only on A-LEVEL Physics field.	Pass
TS0002: Verify that the system handles invalid	TC12	Verify that the system shows an error when score is	User is on Eligibility Check page	1.Open GED section 2.Enter GED	1.Subject (Group A) = Math 2.Grade (Group A)	System shows error "This score exceeds the maximum	System remains on Eligibility Check page	System shows error "This score exceeds the maximum	Pass

inputs on the information page correctly		entered as text		Reasoning Through Language Arts (RLA) score 3.Enter GED Social Studies score 4.Enter GED Mathematical Reasoning score 5.Enter GED Science score wrong format 6.Select English Proficiency test and enter score 7.Enter A-LEVEL scores 8.Select Pre-engineering grades 9.Click Check Eligibility	= B 3.Subject (Group A) = Chemistry 4.Grade (Group A) = C 5.Subject (Group B) = Economics 6.Grade (Group B) = B+ 7.English Proficiency Test = IELTS, Score = 5.5 8.A-LEVEL Mathematics = 1008 9.A-LEVEL Physics = -80 10.A-LEVEL English = 70 11.A-LEVEL TPAT3 = 85 12.EN001203 Computer Programming = B+ 13.EN001202 Engineering Drawing = B	value. Please enter a score between 0 and 100." on A-LEVEL Mathematics field only. and System shows error "This score less than the minimum value. Please enter a score between 0 and 100." on A-LEVEL Physics field only. All other fields show no error.		value. Please enter a score between 0 and 200." only on GED Reasoning Through Language Arts (RLA) field. and and System shows error "This score less than the minimum value. Please enter a score between 0 and 200." on GED Social Studies field only.	
TS0002: Verify that the system handles invalid inputs on the information page correctly	TC13	Verify that the system shows an error when GED score is outside the allowed range (0–200)	User is on Eligibility Check page	1.Open GED section 2.Enter GED Reasoning Through Language Arts (RLA) score	1.GED Reasoning Through Language Arts (RLA) = 245 2.GED Social Studies = -175	System shows error "This score exceeds the maximum value. Please enter a score between 0	System remains on Eligibility Check page	System shows error "This score exceeds the maximum value. Please enter a score between 0 and 200."	Pass

				outside allowed range 3.Enter GED Social Studies score outside allowed range 4.Enter GED Mathematical Reasoning score 5.Enter GED Science score 6.Select English Proficiency test and enter score 7.Enter A-LEVEL scores 8.Select Pre-engineering grades 9.Click Check Eligibility	3.GED Mathematical Reasoning = 175 4.GED Science = 175 5.English Proficiency Test = IELTS, Scores = 5.5 6.A-LEVEL Mathematics = 80 7.A-LEVEL Physics = 75 8.A-LEVEL English = 70 9.A-LEVEL TPAT3 = 60 10.EN001203 Computer Programming = B+ 11.EN001202 Engineering Drawing = B	and 200." on GED Reasoning Through Language Arts (RLA) field only. and System shows error "This score less than the minimum value. Please enter a score between 0 and 200." on GED Social Studies field only. All other fields show no error.		only on GED Reasoning Through Language Arts (RLA) field. and and System shows error "This score less than the minimum value. Please enter a score between 0 and 200." on GED Social Studies field only.	
TS0003: Verify that the system displays eligibility result correctly after processing	TC01	Verify result displays Thai Standard only	1.System is running 2.User is on Eligibility Check page	1.Open Eligibility Check page 2.Open Thai Standard Curriculum section 3.Fill in all required Thai Standard Curriculum fields 4.Ensure GED section is	1.Province = Khon Kaen 2.School = Khon Kaen Wittayayon School 3.Grade = 12 4.Curriculum = Thai Standard 5.Physics Grade = 3.00 6.Chemistr	System displays Thai Standard Curriculum eligibility result only	Eligibility Result page displays Thai Standard Curriculum section only	Thai Standard Curriculum result is displayed correctly	Pass

				empty 5.Ensure GCE section is empty 6.Click Check Eligibility 7.Verify that Eligibility Result page is displayed 8.Verify that Student Information section is displayed correctly 9.Verify that Thai Standard Curriculum result section is displayed 10.Verify that Round 1 – Portfolio is displayed 11.Verify that Round 2 – Quota is displayed 12.Verify that Round 3 – Admission is displayed 13.Verify that no GED Curriculum result section is displayed 14.Verify that no GCE Curriculum result section is displayed	y Grade = 3.00 7.GPAX = 3.60 8.Combine Mathemati cs Grade = 3.50 9.Combine English Grade = 3.40 10.English Proficiency English Proficiency Test = IELTS 11.English Proficiency Score = 5.5 12.Netsat Mathemati cs = 45 13.Netsat Physics = 42 14.Netsat English = 40 15.A-LEV EL Mathemati cs = 75 16.A-LEV EL Physics = 72 17.A-LEV EL English = 70 18.A-LEV EL TPAT3 = 65 19.EN0012 03 Computer Programmi ng = B+ 20.EN0012 02 Engineerin g Drawing = B				
TS0003:	TC02	Verify	1.System is	1.Open	1.Province	System	Eligibility	GED	Pass

Verify that the system displays eligibility result correctly after processing		result displays GED only	running 2.User is on Eligibility Check page	Eligibility Check page 2.Open GED section 3.Fill in all required GED fields 4.Ensure Thai Standard section is empty 5.Ensure GCE section is empty 6.Click Check Eligibility 7.Verify that Eligibility Result page is displayed 8.Verify that GED Curriculum result section is displayed 9.Verify that Round 1 – Portfolio is displayed 10.Verify that Round 2 – Quota is displayed 11.Verify that Round 3 – Admission is displayed 12.Verify that Thai Standard Curriculum section is not displayed 13.Verify that GCE Curriculum section is not displayed	= Bangkok 2.School = Bangkok GED Center 3.Grade = 12 4.Curriculum = GED 5.GED Reasoning Through Language Arts (RLA) = 170 6.GED Social Studies = 165 7.GED Mathematical Reasoning = 175 8.GED Science = 168 9.English Proficiency English Proficiency Test = IELTS 10.English Proficiency Score = 5.5 11.A-LEVEL Mathematics = 75 12.A-LEVEL Physics = 70 13.A-LEVEL English = 70 14.A-LEVEL TPAT3 = 65 15.EN001203 Computer Programming = B 16.EN0012	displays GED Curriculum eligibility result only	Result page displays GED Curriculum section only	Curriculum result is displayed correctly	
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					02 Engineerin g Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC03	Verify result displays GCE only	1.System is running 2.User is on Eligibility Check page	1.Open Eligibility Check page 2.Open GCE section 3.Fill in all required GCE fields 4.Ensure Thai Standard section is empty 5.Ensure GED section is empty 6.Click Check Eligibility 7.Verify that Eligibility Result page is displayed 8.Verify that GCE Curriculum result section is displayed 9.Verify that Round 1 – Portfolio is displayed 10.Verify that Round 2 – Quota is displayed 11.Verify that Round 3 – Admission is displayed 12.Verify that Thai Standard section is not displayed 13.Verify	1.Province = Bangkok 2.School = Shrewsbury Internation al School 3.Grade = 12 4.Curriculu m = GCE 5.Group A Subject 1 = Mathemati cs (Grade A) 6.Group A Subject 2 = Computer Science (Grade B) 7.Group B Subject Group B Subject = Geography (Grade A) 8.English Proficiency English Proficiency Test = IELTS 9.English Proficiency Score = 6.0 10.A-LEV EL Mathemati cs = 80 11.A-LEV EL Physics = 78 12.A-LEV EL English = 75 13.A-LEV EL TPAT3 = 70 14.EN0012	System displays GCE Curriculum eligibility result only	Eligibility Result page displays GCE Curriculum section only	GCE Curriculum result is displayed correctly	Pass

				that GED section is not displayed	03 Computer Programming = B+ 15.EN0012 02 Engineering Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC04	Verify result displays Thai Standard and GED	1.System is running 2.User is on Eligibility Check page	1.Open Eligibility Check page 2.Open Thai Standard Curriculum section 3.Fill in all required Thai Standard Curriculum fields 4.Open GED section 5.Fill in all required GED fields 6.Ensure that GCE section is not filled 7.Click Check Eligibility 8.Verify that Eligibility Result page is displayed 9.Verify that Thai Standard Curriculum result section is displayed 10.Verify that GED Curriculum result section is displayed 11.Verify that GCE	1.Province = Pathum Thani 2.School = Pathumthani Vocational Education College 3.Grade = 12 Curriculum = Thai Standard + GED 4.GPAX = 3.40 5.Combine Mathematics Grade = 3.30 6.Physics Grade = 3.00 7.Chemistry Grade = 3.00 8.Combine English Grade = 3.30 9.GED Reasoning Through Language Arts (RLA) = 168 10.GED Social Studies = 165 11.GED Mathematical Reasoning	System displays Thai Standard Curriculum and GED Curriculum eligibility results, and GCE Curriculum result is not displayed	Eligibility Result page displays Thai Standard and GED curriculum sections	System displays Thai Standard Curriculum and GED Curriculum results correctly	Pass

				Curriculum result section is not displayed	= 170 12.GED Science = 166 13.English Proficiency English Proficiency Test = IELTS 14.English Proficiency Score = 5.5 15.A-LEVEL Mathematics = 72 16.A-LEVEL Physics = 70 17.A-LEVEL English = 68 18.A-LEVEL TPAT3 = 65 19.EN001203 Computer Programming = B 20.EN001202 Engineering Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC05	Verify result displays Thai Standard and GCE	1.System is running 2.User is on Eligibility Check page	1.Open Eligibility Check page 2.Open Thai Standard Curriculum section 3.Fill in all required Thai Standard Curriculum fields 4.Open GCE section 3.Fill in all required GCE fields	1.Province = Chiang Mai 2.School = Montfort College 3.Grade = 12 4.Curriculum = Thai Standard + GCE 5.GPAX = 3.55 6.Combine Mathematics Grade = 3.50	System displays Thai Standard Curriculum and GCE Curriculum eligibility results, and GED Curriculum result is not displayed	Eligibility Result page displays Thai Standard and GCE curriculum sections	System displays Thai Standard Curriculum and GCE Curriculum results correctly	Pass

				8.Ensure that GED section is not filled 9.Click Check Eligibility 10.Verify that Eligibility Result page is displayed 11.Verify that Thai Standard Curriculum result section is displayed 12.Verify that GCE Curriculum result section is displayed 13.Verify that GED Curriculum result section is not displayed	7.Physics Grade = 3.00 8.Chemistry Grade = 3.00 9.Combine English Grade = 3.45 10.Group A Subject 1 = Mathematics (Grade A) 11.Group A Subject 2 = Computer Science (Grade B) 12.Group B Subject = Geography (Grade A) 13.English Proficiency English Proficiency Test = IELTS 14.English Proficiency Score = 6.0 15.A-LEVEL Mathematics = 78 16.A-LEVEL Physics = 75 17.A-LEVEL English = 74 18.A-LEVEL TPAT3 = 70 19.EN001203 Computer Programming = B+ 20.EN001202				
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					Engineering Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC06	Verify result displays GED and GCE	1.System is running 2.User is on Eligibility Check page Test Steps	1.Open Eligibility Check page 2.Open GED section 3.Fill in all required GED fields 4.Open GCE section 5.Fill in all required GCE fields 6.Ensure that Thai Standard section is not filled 7.Click Check Eligibility 8.Verify that Eligibility Result page is displayed 9.Verify that GED Curriculum result section is displayed 10.Verify that GCE Curriculum result section is displayed 11.Verify that Thai Standard Curriculum result section is not displayed	1.Province = Bangkok 2.School = International Community School 3.Grade = 12 4.Curriculum = GED + GCE 5.GED RLA = 170 6.GED Social Studies = 168 7.GED Mathematical Reasoning = 175 8.GED Science = 172 9.Group A Subject 1 = Mathematics (Grade A) 10.Group A Subject 2 = Computer Science (Grade B) 11.Group B Subject = Geography (Grade A) 12.English Proficiency English Proficiency Test = IELTS 13.English Proficiency Score = 5.5 14.A-LEV	System displays GED Curriculum and GCE Curriculum eligibility results, and Thai Standard Curriculum result is not displayed	Eligibility Result page displays GED and GCE curriculum sections	System displays GED Curriculum and GCE Curriculum results correctly	Pass

					EL Mathematics = 80 15.A-LEV EL Physics = 75 16.A-LEV EL English = 72 17.A-LEV EL TPAT3 = 68 18.EN0012 03 Computer Programming = B+ 19.EN0012 02 Engineering Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC07	Verify result displays all curriculums	1.System is running 2.User is on Eligibility Check page	1.Open Eligibility Check page 2.Open Thai Standard Curriculum section 3.Fill in all required Thai Standard Curriculum fields 4.Open GED section 5.Fill in all required GED fields 6.Open GCE section 7.Fill in all required GCE fields 8.Click Check Eligibility 9.Verify that Eligibility Result page is displayed 10.Verify	1.Province = Bangkok 2.School = Shrewsbury Internation al School 3.Grade = 12 4.Curriculu m = Thai Standard + GED + GCE 5.GPAX = 3.70 6.Combine Mathemati cs Grade = 3.60 7.Physics Grade = 3.00 8.Chemistr y Grade = 3.00 9.Combine English Grade = 3.50 10.GED RLA = 175	System displays Thai Standard Curriculum , GED Curriculum , and GCE Curriculum eligibility results	Eligibility Result page displays all curriculum sections	System displays all curriculum results correctly	Pass

				that Thai Standard Curriculum result section is displayed 11. Verify that GED Curriculum result section is displayed 12. Verify that GCE Curriculum result section is displayed	11. GED Social Studies = 170 12. GED Mathematical Reasoning = 178 13. GED Science = 172 14. Group A Subject 1 = Mathematics (Grade A) 15. Group A Subject 2 = Computer Science (Grade B) 16. Group B Subject = Geography (Grade A) 17. English Proficiency English Proficiency Test = IELTS 18. English Proficiency Score = 6.0 19. A-LEVEL Mathematics = 85 20. A-LEVEL Physics = 80 21. A-LEVEL English = 78 22. A-LEVEL TPAT3 = 72 23. EN001203 Computer Programming = A				
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					24.EN0012 02 Engineerin g Drawing = B+				
TS0003: Verify that the system displays eligibility result correctly after processing	TC08	Verify all rounds show Eligible	1.System is running 2.User has filled curriculum data with scores meeting all criteria 3.User has clicked Check Eligibility	1.Open Eligibility Check page 2.Open Thai Standard Curriculum section 3.Fill in curriculum data with scores above required criteria 4.Click Check Eligibility 5.Verify that "You Are Eligible." banner is displayed 6.Verify that Round 1 - Portfolio displays Eligible status 7.Verify that Round 2 - Quota displays Eligible status 8.Verify that Round 3 - Admission displays Eligible status	1.Province = Khon Kaen 2.School = Khonkaen Wittayayon School 3.Grade = 12 4.Physics Grade= 4.00 5.Chemistr y Grade = 4.00 6.GPAX = 4.00 7.Combine d Mathemati cs Grade = 4.00 8.Combine d English Grade = 4.00 9.English Proficiency Test = IELTS 10.English Proficiency Score = 6.0 11.Netsat Mathemati cs = 80 12.Netsat Physics = 80 13.Netsat English = 80 14.A-LEV EL Mathemati cs = 80 15.A-LEV EL Physics = 80	System displays "You Are Eligible" banner and all rounds show Eligible status	Eligibility Result page displays Eligible status for all rounds	System displays "You Are Eligible" banner. Round 1 shows "Eligible for 2 Programs", Round 2 and Round 3 show " Eligible"	Pass

					16.A-LEV EL English = 80 17.A-LEV EL TPAT3 = 80 18.EN0012 03 Computer Programmi ng = B 19.EN0012 02 Engineerin g Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC09	Verify all rounds show Not Eligible	1. System is running 2.User has filled in Thai Standard Curriculum with scores below criteria 3.User has clicked Check Eligibility	1.Open Eligibility Check page 2.Open Thai Standard Curriculum section 3.Fill in all fields with scores below criteria 4.Click Check Eligibility 5.Verify that "You Are Not Eligible Yet" banner is displayed 6.Verify that Round 1 - Portfolio displays Not Eligible status 7.Verify that Round 2 - Quota displays Not Eligible status 8.Verify that Round 3 - Admission displays Not Eligible status	1.Province = Bangkok 2.School = Shrewsbury Internation al School 3.Grade = 12 4.Curriculu m = GED + GCE 5.GED Reasoning Through Language Arts (RLA) scores = 170 6.GED Social Studies scores = 165 7.GED Mathemati cal Reasoning scores = 175 8.GED Science scores = 160 9.Group A Subject 1 = Math ,Grade = A	System displays "You Are Not Eligible Yet" banner and all rounds show Not Eligible status	Eligibility Result page displays Not Eligible status for all rounds	System displays "You Are Not Eligible Yet" banner. Round 1 shows "Not Eligible for 2 Programs Yet", Round 2 and Round 3 show "Not Eligible Yet"	Pass

					10.Group A Subject 2 = Computer Science ,Grade = B 11.Group B Subject = Geography ,Grade = A 12.English Proficiency Test = IELTS 13.English Proficiency Score = 5.5 14.-LEVE L Mathemat ics = 80 15.A-LEV EL Physics = 75 16.A-LEV EL English = 70 17.A-LEV EL TPAT3 = 60 18.EN0012 03 Computer Programmi ng = B+ 19.EN0012 02 Engineerin g Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC10	Verify mixed Eligible and Not Eligible across rounds	1. System is running 2.User is on Eligibility Check page	1.Open Eligibility Check page 2.Open GED section and fill in all GED scores 3.Open GCE section and select subjects for Group A and Group	1.Province = Bangkok 2.School = Shrewsbury International School 3.Grade = 12 4.Curriculum = GED + GCE 5.GED Reasoning	GED Curriculum shows Eligible for all rounds. GCE Round 1 shows "Eligible for 1 Programs" where High-Potential School	Eligibility Result page displays mixed status correctly. High-Potential School Student Admission Program shows Not Eligible Yet due to	GED Curriculum shows Eligible for all rounds. GCE Round 1 shows "Eligible for 1 Programs" with High-Potential School	Pass

				B 4.Select English Proficiency test and enter score 5.Enter A-LEVEL scores 6.Select Pre-engineering grades 7.Click Check Eligibility 8.Verify that GED Curriculum section displays all rounds as Eligible 9.Verify that GCE Round 1 displays "Eligible for 1 Programs" 10.Verify that GCE Round 1 High-Potential School Student Admission Program displays Not Eligible Yet because curriculum is not Thai Curriculum 11.Verify that GCE Round 1 High-Potential Student Program displays Eligible 12.Verify that GCE Round 2 and Round 3 display Eligible	Through Language Arts (RLA) scores = 170 6.GED Social Studies scores = 165 7.GED Mathematical Reasoning scores = 175 8.GED Science scores = 160 9.Group A Subject 1 = Math ,Grade = A 10.Group A Subject 2 = Computer Science ,Grade = B 11.Group B Subject = Geography ,Grade = A 12.English Proficiency Test = IELTS 13.English Proficiency Score = 5.5 14.-LEVEL Mathematics = 80 15.A-LEVEL Physics = 75 16.A-LEVEL English = 70 17.A-LEVEL TPAT3 = 60 18.EN0012	Student Admission Program is Not Eligible Yet because student's curriculum is not Thai Curriculum , and High-Potential Student Program is Eligible. GCE Round 2 and Round 3 show Eligible	curriculum requirement not met	Student Admission Program as Not Eligible Yet due to curriculum mismatch and High-Potential Student Program as Eligible. GCE Round 2 and Round 3 show Eligible	
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					03 Computer Programmi ng = B+ 19.EN0012 02 Engineerin g Drawing = B				
TS0003: Verify that the system displays eligibility result correctly after processing	TC11	Verify Round 1 dropdown shows 2 programs collapsed	1.System is running 2.User has filled in all curriculum fields and clicked Check Eligibility 3. Eligibility Result page is displayed	1.Verify that Eligibility Result page is displayed 2.Verify that Student Information displays correctly 3.Verify that Round 1 - Portfolio is expanded and shows 2 programs in collapsed state 4.Click dropdown of High-Poten tial School Student Admission Program 5.Verify that detail table displays Condition column correctly 6.Verify that detail table displays Requiremen t column correctly 7.Verify that detail table displays Your Score column correctly 8.Verify that detail table	1.Province = Khon Kean 2.School = Khonkaen Wittayayon School 3.Grade = 12 4.Curriculu m = Thai Standard 5.All scores = Pass criteria for all rounds 6.Round 1 - Portfolio Status = Eligible for 2 Programs 7.Program 1 = High-Poten tial School Student Admission Program 8.Program 1 Status = Eligible 9.Program 1 Condition = Curriculum , Partner School, GPAX, GPA-Math, GPA-Chem , GPA-Physi cs,	High-Poten tial School Student Admission Program dropdown expands and displays detail table with Condition, Requireme nt, Your Score and Status correctly	High-Poten tial School Student Admission Program dropdown expands and displays detail table with Condition, Requireme nt, Your Score and Status correctly	High-Poten tial School Student Admission Program dropdown expands and displays detail table correctly	Pass

				displays Status column correctly 9. Verify that High-Potential Student Program remains in collapsed state	GPA-English, EN001203, EN001202, IELTS, Academic Achievement 10. Program 1 Requirement = Thai curriculum, 29 schools, ≥ 3.25, ≥ 3.25, ≥ 3.25, ≥ 3.25, ≥ 3.25, $\geq B$, $\geq B$, ≥ 5, Science/Math/CS at provincial level or above 11. Program 1 Your Score = Thai curriculum, Khon Kaen Wittayayon School, 3.82, 3.50, 3.50, 3.50, 3.50, B+, B, 5.5, Yes 12. Program 1 All Status = Pass all conditions 13. Program 2 = High-Potential Student Program 14. Program 2 Status = Eligible (collapsed) Total programs in Round 1 = 2 programs				
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TS0003: Verify that the system displays eligibility result correctly after processing	TC12	Verify Round 1 program 1 dropdown shows detail	1.System is running 2.User has filled in all fields and clicked Check Eligibility 3.Eligibility Result page is displayed 4. Round 1 dropdown is expanded	1.Verify that Eligibility Result page is displayed 2.Verify that Student Information displays correctly 3.Verify that Round 1 - Portfolio is expanded and shows 2 programs 4.Click dropdown of High-Potential Student Program 5.Verify that detail table displays Condition column correctly 6.Verify that detail table displays Requirement column correctly 7.Verify that detail table displays Your Score column correctly 8.Verify that detail table displays Status column correctly 9.Verify that High-Potential School Student Admission Program remains in collapsed	1.Province = Khon Kean 2.School = Khonkaen Wittayayon School 3.Grade = 12 4.Curriculum = Thai Standard 5.All scores = Pass criteria for all rounds 6.Round 1 - Portfolio Status = Eligible for 2 Programs 7.Program 1 = High-Potential School Student Admission Program 8.Program 1 Status = Eligible 9.Program 1 Condition = Curriculum , Partner School, GPAX, GPA-Math, GPA-Chem , GPA-Physics, GPA-English, EN001203, EN001202, IELTS, Academic Achievement	High-Potential School Student Admission Program dropdown expands and displays detail table with Condition, Requirement, Your Score and Status correctly	High-Potential School Student Admission Program is expanded showing detail table	High-Potential School Student Admission Program dropdown expands and displays detail table correctly	Pass

				state	10.Program 1 Requireme nt = Thai curriculum, 29 schools, ≥3.25, ≥3.25, ≥3.25, ≥3.25, ≥3.25, ≥B, ≥B, ≥5, Science/Ma th/CS at provincial level or above 11.Program 1 Your Score = Thai curriculum, Khon Kaen Wittayayon School, 3.82, 3.50, 3.50, 3.50, 3.50, B+, B, 5.5, Yes 12.Program 1 All Status = Pass all conditions 13.Program 2 = High-Poten tial Student Program 14.Program 2 Status = Eligible (collapsed) Total programs in Round 1 = 2 programs				
TS0003: Verify that the system displays eligibility result	TC13	Verify Round 1 program 2 dropdown shows detail	1. System is running 2.Eligibility Result page is displayed 3. Round 1	1.Verify that Eligibility Result page is displayed 2.Verify that Student	1.Province = Khon Kean 2.School = Khonkaen Wittayayon	High-Poten tial Student Program dropdown expands and	High-Poten tial Student Program is expanded showing detail table	High-Poten tial Student Program dropdown expands and	Pass

correctly after processing			dropdown is expanded	Information displays correctly 3. Verify that Round 1 - Portfolio is expanded and shows 2 programs 4. Click dropdown of High-Potential Student Program 5. Verify that detail table displays Condition column correctly 6. Verify that detail table displays Requirement column correctly 7. Verify that detail table displays Your Score column correctly 8. Verify that detail table displays Status column correctly 9. Verify that High-Potential School Student Admission Program remains in collapsed state	School 3. Grade = 12 4. Curriculum = Thai Standard 5. All scores = Pass criteria for all rounds 6. Round 1 - Portfolio Status = Eligible for 2 Programs 7. Program 1 = High-Potential School Student Admission Program 8. Program 1 Status = Eligible (collapsed) 9. Program 2 = High-Potential Student Program 10. Program 2 Status = Eligible 11. Program 2 Condition = Curriculum, GPAX, GPA-Math, GPA-Chemistry, GPA-Physics, IELTS, Academic Achievement 12. Program 2 Requirement = Thai/International/GE	displays detail table with Condition, Requirement, Your Score and Status correctly		displays detail table correctly	
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					D (age $\geq$ 16), ≥ 2.75 , ≥ 2.75 , ≥ 2.75 , ≥ 2.75 , ≥ 5 , Science/Math/CS at provincial level or above 13. Program 2 Your Score = Thai curriculum, 3.82, 3.50, 3.36, 3.25, 5.5, Yes 14. Program 2 All Status = Pass all conditions 15. Total programs in Round 1 = 2 programs				
TS0003: Verify that the system displays eligibility result correctly after processing	TC14	Verify Round 2 dropdown shows detail	1. System is running 2. Eligibility Result page is displayed	1. Verify that Eligibility Result page is displayed 2. Verify that Student Information displays correctly 3. Verify that Round 2 - Quota is displayed in collapsed state before clicking 4. Verify that Round 2 - Quota shows Eligible status 5. Click dropdown of Round 2 -	1. Province = Khon Kean 2. School = Khonkaen Wittayayon School 3. Grade = 12 4. Curriculum = Thai Standard 5. All scores = Pass criteria for all rounds 6. Round 1 - Portfolio Status = Eligible for 2 Programs (collapsed) 7. Round 2 - Quota	Round 2 - Quota dropdown expands and displays detail table with Average score and Domicile conditions, Requirement, Your Score and Status correctly	Round 2 - Quota is expanded showing detail table	Round 2 - Quota dropdown expands and displays detail table with all conditions correctly	Pass

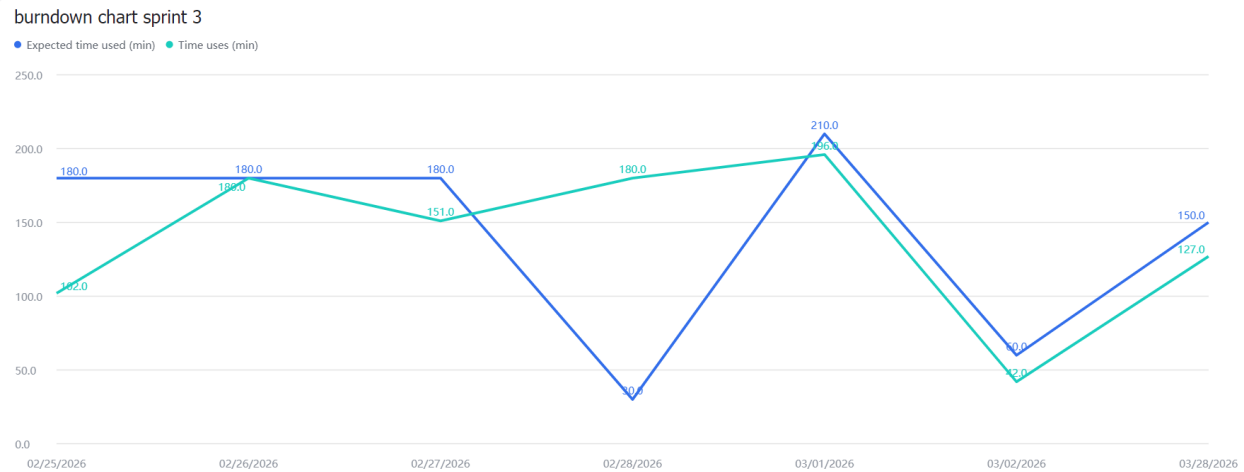
				Quota 6. Verify that detail table displays Condition column correctly 7. Verify that detail table displays Requirement column correctly 8. Verify that detail table displays Your Score column correctly 9. Verify that detail table displays Status column correctly	Status = Eligible 8. Condition 1 = Average score (NETSAT + T-PAT 3) 9. Requirement 1 ≥ 30 10. Your Score 1 = 45.17 11. Status 1 = Pass 12. Condition 2 = Domicile 13. Requirement 2 = Household registration in the Northeast region ≥ 3 years 14. Your Score 2 = Khon Kaen 15. Status 2 = Pass 16. Round 3 - Admission Status = Eligible (collapsed)				
TS0003: Verify that the system displays eligibility result correctly after processing	TC15	Verify Round 3 dropdown shows detail	1. System is running 2. Eligibility Result page is displayed	1. Verify that Eligibility Result page is displayed 2. Verify that Student Information displays correctly 3. Verify that Round 3 - Admission is displayed in collapsed state before clicking	1. Province = Khon Kean 2. School = Khonkaen Wittayayon School 3. Grade = 12 4. Curriculum = Thai Standard 5. All scores = Pass criteria for	Round 3 - Admission dropdown expands and displays detail table with T-PAT3, A-Level Mathematics 1, A-Level Physics and A-Level English	Round 3 - Admission is expanded showing detail table	Round 3 - Admission dropdown expands and displays detail table with all conditions correctly	Pass

				4. Verify that Round 3 - Admission shows Eligible status 5. Click dropdown of Round 3 - Admission 6. Verify that detail table displays Condition column correctly 7. Verify that detail table displays Requirement column correctly 8. Verify that detail table displays Your Score column correctly 9. Verify that detail table displays Status column correctly	all rounds 6. Round 1 - Portfolio Status = Eligible for 2 Programs (collapsed) 7. Round 2 - Quota Status = Eligible (collapsed) 8. Round 3 - Admission Status = Eligible 9. Total conditions in Round 3 = T-PAT3, A-Level Mathematics 1, A-Level Physics, A-Level English	conditions, Requirement, Your Score and Status correctly			
TS0003: Verify that the system displays eligibility result correctly after processing	TC16	Verify detail popup displays correctly when clicking info button	1. System is running 2. Eligibility Result page is displayed	1. Verify that Eligibility Result page is displayed 2. Verify that Student Information displays correctly 3. Click dropdown Round 1 and open High-Potential School Student Admission Program 4. Click info	1. Province = Khon Kean 2. School = Khonkaen Wittayayon School 3. Grade = 12 4. Curriculum = Thai Standard 5. All scores = Pass criteria for all rounds 6. Round 1 Partner	Info button on Partner School displays list of partner schools correctly. Info button on Average Score displays Your Score Breakdown with each subject score, calculation and Total correctly	Popup is displayed correctly when clicking info button	Both info buttons display popup correctly as expected	Pass

				button next to Partner School 5. Verify that popup displays partner school list correctly 6. Click dropdown Round 2 - Quota 7. Click info button next to Average Score 8. Verify that popup displays Your Score Breakdown correctly	School info button = Displays list of 29 partner schools 7. Round 2 Average Score info button = Displays Your Score Breakdown (English, Mathematics, Physics, T-PAT3 $\times$ % = score, Total)				
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6.3 Sprint Monitoring

6.3.1 Burndown Chart



Sprint 3 consisted of 33 tasks with a total expected time of 990 minutes. The team completed all tasks with an actual time of 978 minutes, nearly matching the plan at only 1% variance. This sprint focused on expanding score type support (GED, GCE, Netsat, A-level) and designing result pages for all three admission rounds. The near-perfect alignment between expected and actual time reflects the team's improved estimation accuracy over the course of the project. One notable exception was the DFD design task, which took 180 minutes against an estimate of 30 minutes, but was offset by efficiency gains in other tasks.

6.4 Sprint 3 Evaluation

6.4.1 Sprint Review Discussion

During the Sprint 3 Review, the team presented the final version of the admission eligibility checker. This sprint focused on completing remaining features, handling edge cases, and preparing the system for final delivery.

Feedback	- Minor revisions are required for the test cases to ensure they accurately reflect the final system behavior and fully align with the user story. Some test steps and expected results need clarification. - Small adjustments are needed in Data Flow Diagrams (DFD) Level 1
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	and Level 2 to improve consistency and accuracy with the implemented system processes. No major structural issues were identified.
Decision	• Update and refine the test cases to match the finalized functionality of the system and ensure completeness and clarity. • Revise DFD Level 1 and Level 2 to reflect the final system design while maintaining consistency across all diagrams.

6.4.2 Sprint Retrospective Discussion

What went well	• Strong Collaboration: Team members worked closely to complete the final features and refinements. • Timely Completion: Most tasks were finished according to the planned schedule. • Improved Efficiency: Experience from previous sprints helped the team work faster and more confidently. • System Stability Achieved: Final testing ensured the system functioned correctly across scenarios. • Quality Improvements: UI, diagrams, and documentation were refined to meet project standards.
What went poorly	• Increased Workload Near Deadline: Final adjustments and fixes created pressure toward the end of the sprint. • Minor Revisions Required: Some diagrams and test cases needed last-minute corrections. • Limited Time for Extensive Testing: Not all edge cases could be tested due to time constraints.
What can we improve	• Allocate More Time for Final Testing: Ensure sufficient time for validation and quality assurance. • Plan Earlier Completion of Core Tasks: Finish major components sooner to allow buffer for refinements. • Improve Documentation Workflow: Update documents continuously instead of at the end.

What action will we take	● Schedule a dedicated testing phase before project submission. ● Complete critical tasks earlier in future projects. ● Maintain ongoing documentation updates throughout development. ● Conduct a final team review session before submission.
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